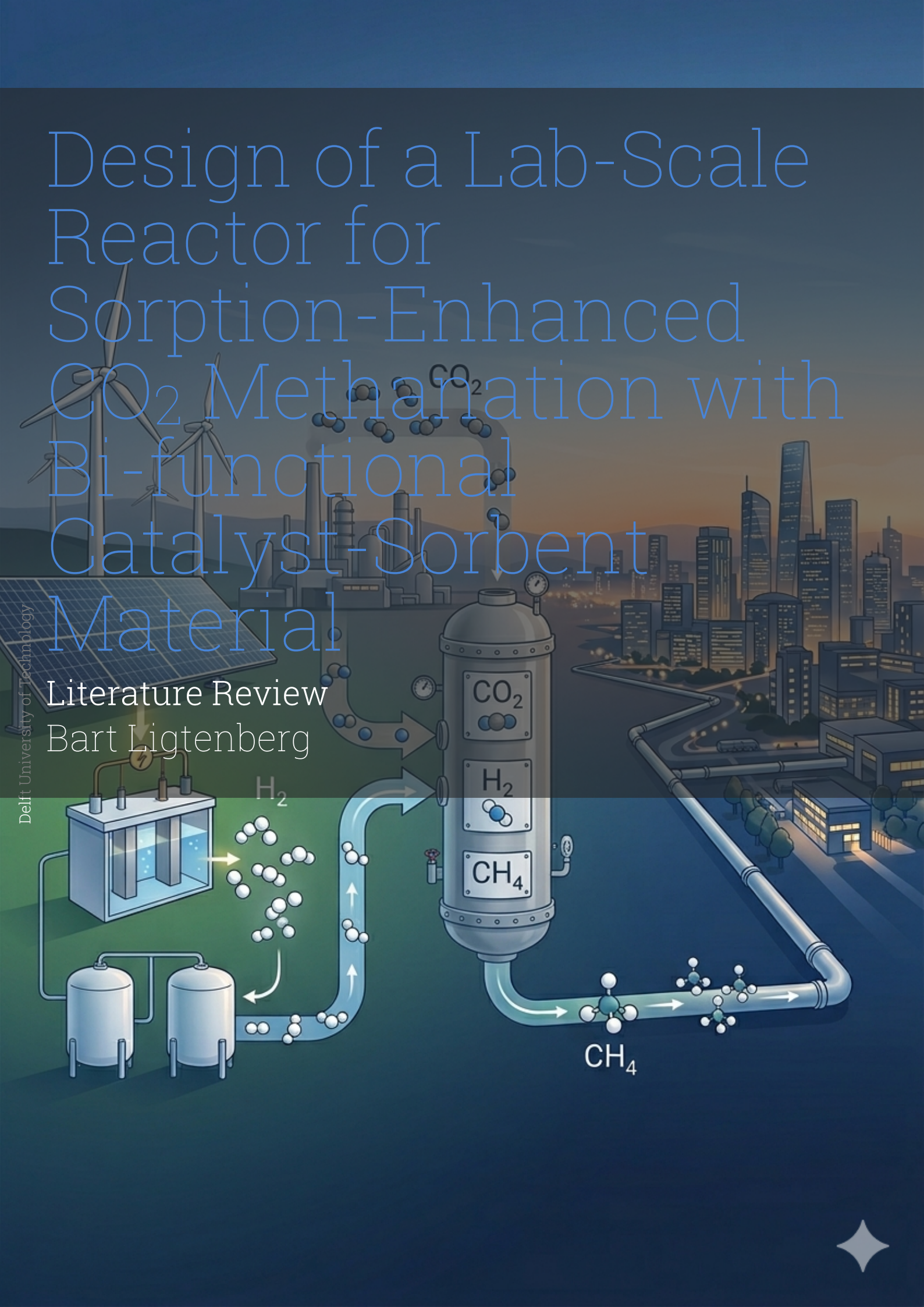
Design of a Lab-Scale Reactor for Sorption-Enhanced CO_2 Methanation with Bi-functional Catalyst-Sorbent Material

Literature Review

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by

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[0] Generative AI tools were used for language editing and stylistic improvements. No AI tools were used to generate technical content, data, or analysis.

Summary

The transition to a fully renewable energy system requires large-scale, long-duration storage of electrical energy that batteries cannot provide, an energy-dense fuel for sectors that resist direct electrification such as shipping, aviation, and high-temperature industrial heat, and relief for a Dutch transmission grid that is approaching its capacity limit. Converting surplus renewable electricity into hydrogen and reacting it with captured CO_2 into a storable, infrastructure-compatible fuel or bulk chemical addresses all three constraints simultaneously. The four candidate products considered are methane (CH_4), dimethyl ether (CH_3OCH_3), methanol (CH_3OH), and carbon monoxide (CO). All four reactions are limited by chemical equilibrium and produce water as a by-product, so removing that water in-situ with a solid sorbent shifts the equilibrium toward the product, raises single-pass conversion, and relaxes the temperature and pressure window required to achieve acceptable reaction rates. Within this sorption-enhanced approach, integrating the catalytic and water-adsorbing functions into a single bi-functional particle places the two sites within nanometre-scale proximity and produces a more efficient equilibrium shift than a physically mixed bed of separate particles and is the focus of this thesis. Because the renewable electricity supply is intermittent by nature, dynamic operability across a wide load range is a central design requirement for the reactor.

Chapter 2 selected the target product through a weighted multi-criteria assessment of methane, dimethyl ether, methanol, and carbon monoxide against five criteria: downstream process complexity, infrastructure readiness, bi-functional material maturity, market value, and operating conditions. Methane was selected as the target product. The decisive advantage is downstream process complexity: sorption enhancement applied to methanation delivers a stream that is already close to the Dutch gas grid injection specification with only water removal required, while the three alternatives all require multi-step downstream separation regardless of the single-pass conversion achieved. Methane is also directly compatible with the existing gas grid, operates within a moderate temperature window, and is the only candidate for which a stable bi-functional material has been demonstrated at pilot scale.

Chapter 3 reviewed the SEM literature in detail, covering reaction thermodynamics and kinetics, catalyst and sorbent selection, feed composition, and regeneration strategies. The bi-functional catalyst-sorbent developed by Wei, consisting of 5 wt% Ni and 2.5 wt% Ce supported on zeolite 13X, was identified as the most promising option, with full conversion shown over 100 cycles. The review also revealed a structural gap in the experimental record: the literature is dominated by fixed-bed reactors, even though non-fixed-bed configurations are essential for scaling beyond batch operation and for managing the heat released by the strongly exothermic methanation reaction. No demonstration or model combining a non-fixed-bed reactor with a bi-functional material has been reported, even though this combination is the most logical route forward, and heat integration between the exothermic reaction stage and the regeneration stage remains unquantified outside fixed-bed geometries. This gap defines the design space addressed in the remainder of the thesis.

Chapter 4 translated these findings into a basis of design for a lab-scale unit. The bi-functional material developed by Wei was selected as the solid inventory. Six reactor configurations were then compared against six weighted criteria: operational bandwidth, continuity of production, attrition resistance, heat management, sorbent capacity utilisation, and start-up time. The moving packed bed was selected because it decouples gas velocity from solid throughput, supports continuous solid circulation, generates only compressive contact forces inside the reactor that minimise attrition risk, and provides a counter-current contact pattern that simultaneously moderates the peak temperature and maximises sorbent utilisation. The reactor is embedded in a two-stage process configuration: a conventional first-stage reactor performs the bulk of the conversion, followed by cooling and water removal, before the moving packed bed second-stage reactor drives conversion to completion, with spent solid continuously transferred to a separate regenerator and returned to the reactor top to close the solid loop.

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Nomenclature

Abbreviations

Abbreviation	Definition
BET	Brunauer–Emmett–Teller (surface area measurement method)
BFB	Bubbling Fluidised Bed
CBAM	Carbon Border Adjustment Mechanism
CCS	Carbon Capture and Storage
CCU	Carbon Capture and Utilisation
CFB	Circulating Fluidised Bed
CFD	Computational Fluid Dynamics
DAC	Direct Air Capture
DA	Dubinin–Astakhov (adsorption model)
DME	Dimethyl Ether
ETS	EU Emissions Trading System
FAU	Faujasite (zeolite structural family)
GHSV	Gas Hourly Space Velocity
LDF	Linear Driving Force
LHHW	Langmuir–Hinshelwood–Hougen–Watson (kinetic model)
LHV	Lower Heating Value
LTA	Linde Type A (zeolite structural family)
MOF	Metal-Organic Framework
MPB	Moving Packed Bed
PSA	Pressure Swing Adsorption
rWGS	Reverse Water–Gas Shift reaction
RF	Radiofrequency
SEM	Sorption-Enhanced Methanation
SEDMES	Sorption Enhanced Dimethyl Ether Synthesis
SNG	Synthetic Natural Gas
STP	Standard Temperature and Pressure
TRL	Technology Readiness Level
TSA	Temperature Swing Adsorption
TPSA	Temperature and Pressure Swing Adsorption
WHSV	Weight Hourly Space Velocity

Roman symbols

Symbol	Definition	Unit
A	Adsorption potential (Dubinin–Astakhov model)	$\mathrm{J\,mol^{-1}}$
A_t	Reactor cross-sectional area	$\mathrm{m^2}$
Ar	Archimedes number	–
c_{ads}	Specific adsorption capacity of sorbent	$\mathrm{mol\,kg^{-1}}$
d^*	Dimensionless particle diameter	–
d_p	Particle diameter	m
D	Water diffusion coefficient in zeolite	$\mathrm{m^2\,s^{-1}}$
E_{char}	Characteristic adsorption energy (DA model)	$\mathrm{J\,mol^{-1}}$
G_s	Solid mass flux	$\mathrm{kg\,m^{-2}\,s^{-1}}$

Symbol	Definition	Unit
k	Reaction rate constant	$\mathrm{mol\,kg_{cat}^{-1}\,s^{-1}\,bar^{-1}}$
K_{eq}	Equilibrium constant of methanation reaction	–
K_{OH}	Adsorption equilibrium constant for hydroxyl groups	$\mathrm{bar}^{-0.5}$
$K_{\mathrm{H_2}}$	Adsorption equilibrium constant for H_2	$\mathrm{bar}^{-0.5}$
K_{mix}	Adsorption equilibrium constant for CO_2	$\mathrm{bar}^{-0.5}$
$M_{\mathrm{H_2O}}$	Molar mass of water	$\mathrm{kg\,mol^{-1}}$
$\dot{n}_{\mathrm{H_2O}}$	Molar flow rate of water	$\mathrm{mol\,s^{-1}}$
n	Structural heterogeneity exponent (DA model)	–
p	Total pressure	bar
p_i	Partial pressure of species i	bar
$p_{\mathrm{CH_4}}$	Partial pressure of methane	bar
$p_{\mathrm{CO_2}}$	Partial pressure of CO_2	bar
$p_{\mathrm{H_2}}$	Partial pressure of hydrogen	bar
$p_{\mathrm{H_2O}}$	Partial pressure of water	bar
p_{sat}	Saturation vapour pressure of water	bar
r	Reaction rate	$\mathrm{mol\,kg_{cat}^{-1}\,s^{-1}}$
R	Universal gas constant	$\mathrm{J\,mol^{-1}\,K^{-1}}$
$S_{\mathrm{CH_4}}$	Methane selectivity	–
T	Temperature	K
t	Time	s
u_0	Superficial gas velocity	$\mathrm{m\,s^{-1}}$
u_t	Terminal velocity of particle	$\mathrm{m\,s^{-1}}$
W	Volumetric adsorption capacity (DA model)	$\mathrm{m^3\,kg^{-1}}$
W_0	Maximum volumetric adsorption capacity (DA model)	$\mathrm{m^3\,kg^{-1}}$
$X_{\mathrm{CO_2}}$	CO_2 conversion	–

Greek symbols

Symbol	Definition	Unit
ΔH°	Standard reaction enthalpy	$\mathrm{kJ\,mol^{-1}}$
$\varepsilon_{\mathrm{se}}$	Solid volume fraction in riser (dilute section)	–
μ	Dynamic viscosity of gas	Pa s
ρ_g	Gas density	$\mathrm{kg\,m^{-3}}$
ρ_s	Solid particle density	$\mathrm{kg\,m^{-3}}$

1

Introduction

1.1. Context and Motivation

Global greenhouse gas emissions must reach net zero by 2050 to limit average warming to 1.5 °C above pre-industrial levels [1]. Meeting this target requires deep decarbonisation of the process industry, which currently relies on fossil fuels not only as an energy source but also as a feedstock for base chemicals and fuels [2]. Electrification based on renewable energy is the primary pathway for reducing the associated CO_2 emissions, but it introduces a fundamental supply-side challenge: solar and wind power are intermittently available, and the periods of peak generation rarely coincide with peak demand. Large-scale, long-duration energy storage is therefore a prerequisite for a fully renewable energy system, yet batteries cannot satisfy this requirement across all sectors and timescales, given their limitations in power capacity, energy density, and raw material availability [3, 2]. Beyond storage, direct electrification is not a viable defossilization route for all sectors: transportation modes such as shipping and aviation, as well as high-temperature industrial processes, require energy-dense fuels rather than direct electrical power [3]. A third structural constraint is that the Dutch high-voltage transmission grid is approaching its capacity limit across multiple regions, meaning that newly generated renewable electricity frequently cannot be transported to where it is needed [4]. Converting surplus electrical energy into a chemical energy carrier therefore addresses all three limitations: it provides long-duration storage, relieves grid transport demand, and produces a fuel for sectors where direct electrification is not feasible.

One route is to convert surplus electricity into hydrogen via water electrolysis. Hydrogen carries a high energy density per unit mass, but its physical properties make large-scale storage and transport technically demanding and costly: at ambient conditions it is a low-density gas, requiring either compression to several hundred bar or liquefaction to below −253 °C for practical storage and pipeline transport, and the existing gas infrastructure is not designed for pure hydrogen at high concentrations [5]. Binding the hydrogen chemically to a carbon-containing molecule to form a fuel that is compatible with existing infrastructure therefore offers a more practical path to large-scale energy storage.

CO_2 is a promising carbon source for this purpose, and concentrated CO_2 streams are available from a range of industrial point sources, including biogas upgrading facilities, fermentation plants, and combustion flue gases from power generation and industrial processes such as steel and cement production. These sources typically supply CO_2 at volume fractions between 15 and 100 % [2]. A complementary route is Direct Air Capture (DAC), in which CO_2 is extracted directly from ambient air at a concentration of approximately 0.04 %. DAC is independent of any particular emission source and could in principle supply CO_2 anywhere that air and renewable electricity are available, but the very low inlet concentration demands larger equipment and higher energy input per unit of CO_2 recovered compared to point-source capture. Both routes are active areas of development: global operational CO_2 capture capacity already stands at approximately 50 Mt per year and is projected to reach 435 Mt per year by 2030 as the project pipeline matures [6]. A further driver is the tightening of European regulatory frameworks: the EU Emissions Trading System is progressively reducing its cap on industrial emissions [7],

while the Carbon Border Adjustment Mechanism, which entered its definitive regime in January 2026, imposes a carbon price on imports of carbon-intensive goods [8]. Together, these instruments create a direct financial incentive to capture CO_2 rather than vent it to the atmosphere. Captured CO_2 can either be permanently stored in deep geological formations (Carbon Capture and Storage, CCS), treating CO_2 purely as a waste stream, or converted into a product of economic value (Carbon Capture and Utilisation, CCU). The present work follows the CCU route, focusing on concentrated point-source CO_2 streams as the nearer-term opportunity given their lower capture cost and higher availability today. Reacting captured CO_2 with electrolytic H_2 into a storable, grid-compatible energy carrier simultaneously addresses the growing regulatory cost of emitting CO_2 and provides a practical route to large-scale renewable energy storage.

To make the CO_2 hydrogenation process economically viable, the dominant cost driver must be addressed: the price of green hydrogen, which accounts for the large majority of operating cost regardless of the target product, since all routes consume hydrogen in large stoichiometric quantities relative to the CO_2 feed. The cost of green hydrogen is in turn dominated by the electricity consumed by the electrolyser, making this the dominant cost driver [9]. This cost is expected to decrease substantially as electrolyser technology matures and renewable electricity capacity expands, but in the near term the process is most competitive when the electrolyser operates primarily during periods of surplus renewable generation, when electricity prices are low. Co-locating the electrolyser with the reactor and coupling their operation to periods of excess grid supply is therefore the primary route to economic viability in the near term. This coupling has a direct consequence for reactor design: the hydrogen supply is intermittent by definition, so the reactor must be capable of frequent start-up and shutdown and must maintain acceptable performance across a wide range of partial loads, making dynamic operability a central design requirement rather than an optional feature, with direct consequences for reactor type selection and process control. Because the plant operates at full capacity only during surplus periods rather than continuously, capital investment is spread over a smaller annual production volume, which strengthens the case for compact, low-cost reactor designs that minimise hardware investment per unit of installed capacity.

Furthermore, CO_2 hydrogenation technology offers a strategic dimension that is increasingly relevant in the current geopolitical climate. Europe's dependence on imported fossil fuels has proven to be a significant vulnerability, as demonstrated by the price volatility and supply disruptions that have characterised European energy markets in recent years [10]. The ability to produce synthetic fuels and chemicals domestically from renewable electricity and captured CO_2 reduces exposure to international commodity markets and the political risks associated with fossil fuel supply chains. This strategic value is difficult to capture in a simple cost comparison, but it underpins much of the policy support that makes power-to-X development attractive at the national and European level. Producing fuels and chemicals from wind and solar power is, in this sense, not only an engineering challenge but also a contribution to energy sovereignty.

The hydrogenation of CO_2 with H_2 can yield several base chemicals and fuels depending on the catalyst and operating conditions, including carbon monoxide, methane, methanol, and dimethyl ether (DME). All four reactions are thermodynamically limited by chemical equilibrium and produce water as a by-product, which constrains the single-pass conversion achievable in a conventional reactor. Removing this water in-situ shifts the equilibrium toward the product, driving conversion beyond the limit that would otherwise be imposed by thermodynamics alone, and is therefore a central strategy for process intensification. Two broad approaches have been pursued in the literature: membrane reactors, in which a water-selective membrane wall allows water to permeate out of the reaction zone, and sorption-enhanced reactors, in which a solid sorbent co-located with the catalyst captures water directly within the bed. Membrane reactors have been demonstrated to improve conversion, but poor selectivity for water over hydrogen causes reactant losses that reduce CO_2 conversion and available inorganic membranes suffer from limited hydrothermal and mechanical stability at the elevated temperatures relevant to CO_2 hydrogenation [2]. Sorption enhancement addresses the equilibrium limitation more directly: by capturing water at the point of formation, the reaction is driven beyond its conventional equilibrium limit without the selectivity and pressure constraints that affect membrane systems. The concept of reactive steam adsorption was first demonstrated experimentally by Carvill et al. in 1996 for the reverse water-gas shift reaction [11]. Subsequent work extended the principle across multiple CO_2 hydrogenation products: Borgschulte et al. demonstrated sorption-enhanced CO_2 methanation in

2013, and contributions by Wei et al., G'omez et al., Van Kampen et al., and Kiefer et al. have progressively advanced the technology from proof of concept to pilot-scale demonstration across methane, DME, and related products [12, 13, 14, 15, 16]. The consequences extend well beyond the conversion benefit alone: near-complete single-pass conversion reduces or eliminates the need for downstream separation and unreacted gas recycle, allows the reactor to operate at milder conditions than its conventional counterpart, and has the potential to lower both capital and operating expenditure significantly, strengthening the economic case for power-to-X processes. Sorption enhancement is therefore the approach pursued in this work.

The sorbent does not, however, have an unlimited capacity to retain water. Once it approaches saturation, its ability to remove water from the gas phase diminishes and the conversion benefit is lost, so it must be periodically regenerated by supplying heat to drive off the accumulated water. Regeneration can be driven by increasing the temperature via a hot purge gas, by reducing the total pressure, or by direct electrical heating of the sorbent, and the choice of method has significant consequences for energy consumption, cycle time, and compatibility with the reactor type. Since the plant is designed to operate on surplus renewable electricity, that same electricity supply is the most practical energy source for driving regeneration, avoiding the need for a separate heat source and limiting the environmental impact of the process. In all cases, regeneration introduces an energy penalty and sets a practical lower bound on cycle time, since the sorbent must be sufficiently dewatered before production can resume. In fixed-bed configurations, this forces the reactor offline, requiring a minimum of two to three parallel vessels operating in staggered phases to maintain a continuous product stream [17, 18], and the total solid inventory must be sized for the offline period rather than for the production phase alone. This batch character becomes increasingly penalising at larger scale, where the cost of duplicated pressure vessels and oversized solid inventories grows in proportion to production capacity, directly undermining the economic case for power-to-X deployment. Non-fixed-bed configurations with continuous solid circulation between a dedicated reaction vessel and a separate regeneration vessel eliminate this penalty by decoupling the two functions in space rather than in time, making the process genuinely continuous without the multiplied hardware and inventory cost of the parallel fixed-bed approach and providing a reactor architecture that can scale toward industrial capacity.

Within sorption enhanced CO_2 hydrogenation, the configuration that has shown the most promising process intensification potential is the bi-functional catalyst-sorbent material, in which the catalytic and water-adsorbing functions are integrated within a single particle rather than being physically mixed as separate solids. Borgschulte et al. first proposed this concept for CO_2 methanation, showing that placing catalytic and adsorption sites in close proximity within the same particle produces a more efficient equilibrium shift than a physically mixed bed [12]. This bi-functional configuration is the primary focus of the CO_2URAGE programme[1], a multi-disciplinary Dutch research effort targeting the development of industrially deployable, electrified reactor concepts for sorption-enhanced CO_2 hydrogenation to base chemicals and fuels. The programme spans the full development chain, from material synthesis and reactor development through to process and system design, dynamic operation and control, and techno-economic assessment, and involves both academic institutions and industrial partners across the value chain.

Despite the potential of using a bi-functional catalyst-sorbent material in a non-fixed bed for continuous CO_2 hydrogenation, it has not been demonstrated or modelled. Across all candidate products, only three non-fixed-bed demonstrations exist in the literature, one for methanation and two for methanol synthesis, all using physically mixed beds of separate catalyst and sorbent particles rather than a bi-functional material, and none were conducted with extended cycling or attrition assessment [19, 20, 21]. The surrounding process that such a configuration requires, including a solid circulation loop and a continuously operating regeneration vessel, is without precedent in the literature. The present thesis addresses this gap by designing and modelling a lab-scale sorption-enhanced CO_2 hydrogenation process with a non-fixed-bed reactor with continuous solid circulation, a bi-functional catalyst-sorbent material, and electrically driven regeneration. The scope includes the selection of the target product, the reactor type, and the surrounding process units.

[1] The full programme proposal is available from the programme coordinator upon request.

1.2. Research Questions

Developing such a process requires closing two compounding gaps in the existing literature simultaneously. First, the experimental and modelling basis for sorption-enhanced CO_2 hydrogenation using a bi-functional material outside fixed-bed geometries is essentially absent: no demonstration or model combining a non-fixed-bed reactor with a bi-functional catalyst-sorbent material has been reported at any scale or for any target product. Second, the process configuration that a non-fixed-bed concept requires, with continuous solid circulation between a reactor and a dedicated regeneration vessel, has never been designed or modelled for any sorption-enhanced CO_2 hydrogenation process. Closing both gaps requires a validated reactor concept and engineering-level model before experimental work can begin; developing and modelling that concept is the primary contribution of this thesis, carried out within the reactor development strand of the CO_2URAGE programme.

The central research question addressed here is:

> How should a lab-scale sorption-enhanced CO_2 hydrogenation process with a bi-functional catalyst-sorbent material, continuous solid circulation between a reactor and a dedicated regeneration vessel, and electrically driven regeneration be designed and modelled?

Five sub-questions address product selection, reactor type, reactor performance, integrated process performance, and mechanical design respectively:

1. Which bulk chemical or fuel is the most suitable target product for a sorption-enhanced CO_2 hydrogenation process with a bi-functional catalyst-sorbent material, based on a weighted multi-criteria assessment of downstream process complexity, product and infrastructure readiness, bi-functional material readiness, and operating conditions?
2. What reactor type is most suitable for a dynamically operable lab-scale sorption-enhanced CO_2 hydrogenation process with a bi-functional catalyst-sorbent material, and provides a clear path toward scale-up?
3. What is the predicted performance of the reactor under varying operating conditions, as described by an engineering-level model?
4. What configuration of the surrounding process units meets the product purity requirements, and what does the energy balance of the integrated system look like under varying operating conditions, accounting for heat integration between the reaction and regeneration stages?
5. What does the mechanical design of the full lab-scale process look like, as expressed in basic construction drawings?

Sub-questions 1 and 2 are addressed in Chapters 2 and 4 respectively as part of the literature review and basis of design; sub-questions 3 through 5 are addressed in the subsequent thesis chapters.

The longer-term programme ambition of replacing conventional thermal regeneration with particle-level electrical heating methods such as microwave or inductive heating requires dedicated material-specific feasibility studies not yet available in the literature and falls outside the scope of this thesis; resistive heating is used here as a practical and well-understood baseline.

Three-dimensional computational fluid dynamics modelling of reactor hydrodynamics and heat and mass transfer, as well as particle-level modelling of intraparticle phenomena, are carried out in parallel by other consortium members at TU Eindhoven and TU Delft and fall outside the scope of this thesis. The modelling work presented here is limited to engineering-level descriptions of the coupled reaction and adsorption process, suitable for process design and parametric studies, and serves as direct input to system-level and control studies carried out elsewhere in the programme. The reactor is designed with sufficient flexibility to support future experimental campaigns on feed compositions and impurities, alternative electric regeneration techniques, and optimisation studies.

2

Selection of Target Reaction Product

The catalytic hydrogenation of CO_2 with H_2 can yield several base chemicals and fuels depending on the catalyst and operating conditions. Four candidate products are considered in this chapter: methane, dimethyl ether (DME), methanol, and carbon monoxide (CO) via the reverse water-gas shift reaction. All four reactions produce water as a by-product, which makes them equilibrium-limited under conventional operating conditions and therefore candidates for sorption enhancement. In all cases, removing the water produced during the reaction shifts the equilibrium beyond its conventional limit, raising the single-pass conversion while at the same time relaxing the temperature and pressure requirements needed to achieve acceptable reaction rates.

Figure 2.1 illustrates this principle for all four candidates simultaneously: it shows the equilibrium product mole fraction as a function of the remaining steam partial pressure in the reactor. As the sorbent removes water from the gas phase and the steam partial pressure decreases, the equilibrium shifts toward the product, with the extent of this shift differing substantially between the candidate reactions. Notably, methanol and DME synthesis show the steepest equilibrium response to water removal, indicating the largest potential yield benefit from sorption enhancement, while the methanation equilibrium is already favourable at higher steam partial pressures. The implications of these differences for process complexity and operability are discussed in the assessment that follows.

The CO_2URAGE programme, within which this thesis is embedded, targets reactor concepts that combine bi-functional catalyst-sorbent materials with electricity-based regeneration. In a bi-functional material, the catalytic and water-adsorbing functions are integrated within a single particle, placing reaction and adsorption sites in close proximity and maximising the equilibrium shift per unit of sorbent inventory. This configuration is the primary scope of the present work. A physically mixed bed of separate catalyst and sorbent particles remains a fallback option within the programme.

The four candidate products are evaluated against five criteria that are developed in Section 2.1: downstream process complexity, infrastructure readiness, technology readiness for bi-functional material configurations, product market value, and operating conditions. A weighted multi-criteria assessment is used to arrive at a justified and transparent product selection, which forms the basis for the remainder of the thesis.

2.1. Assessment Criteria

Five criteria are used to compare the candidate products. Each criterion is defined below, together with its assigned weight and the justification for that weight. The weights reflect the priorities established in Chapter 1: dynamic operability under a variable renewable electricity supply and low capital expenditure are identified there as the central design requirements for a power-to-X process, and the weighting is constructed accordingly.

Downstream process complexity is assigned the highest weight of 35%. This criterion captures the single-pass conversion achievable with sorption enhancement, the need for a recycle loop, and the

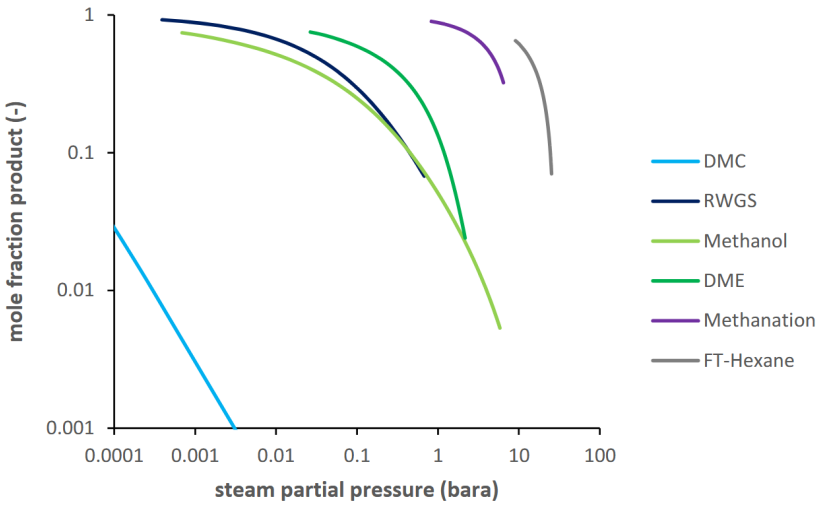


Figure 2.1: Equilibrium product mole fraction as a function of remaining steam partial pressure for the four candidate reactions considered in this chapter, based on minimisation of the Gibbs free energy of reaction. Conditions: stoichiometric H_2/CO_2 feed for all reactions; reverse water-gas shift at 300 °C and 10 bar; methanol synthesis at 250 °C and 30 bar; DME synthesis at 275 °C and 30 bar; methanation at 300 °C and 10 bar. Graph from [22].

number of downstream separation steps required to meet product purity specifications. These factors are closely linked: a low single-pass conversion requires a recycle loop, which increases capital expenditure and fundamentally limits dynamic operability. When feed flow rates change, the entire recycle loop must reach a new stable operating point before the reactor output follows, making fast load-following impossible. A process that achieves near-complete single-pass conversion eliminates this constraint and allows the reactor output to respond directly to changes in the feed.

Infrastructure readiness is assigned a weight of 20%. This criterion covers the standalone value of the product, the availability of an existing market, and the compatibility of the product with existing storage, transport, and distribution infrastructure. A product that is directly compatible with existing infrastructure avoids additional capital investment and accelerates the route to market, while a product that depends on dedicated infrastructure that does not yet exist adds cost and system complexity that undermines the economic case for the power-to-X process.

Bi-functional material maturity is assigned a weight of 20%. This criterion evaluates whether a working bi-functional particle exists for each candidate reaction, and if so, how well characterised it is in terms of material stability and available process-level data. A score of 1 indicates that no bi-functional material has been reported; a score of 2 that a bi-functional effect has been demonstrated but the material is not yet viable for process design; and a score of 3 that a stable material exists but process-level data remain too limited to fully ground a reactor design without further experimental work. No candidate scores above 3, as no bi-functional configuration has been validated in a reactor geometry representative of a process unit at any scale, which is the gap this thesis addresses. The weight is kept moderate because a physically mixed bed of separate catalyst and sorbent particles remains available as a fallback if the bi-functional reactor concept does not perform as intended.

Market value is assigned a weight of 15 %. This criterion covers the standalone value of the product and the availability of an existing market. A product that requires a further conversion step before it can be used, or that competes in a commodity market at low unit value, limits the revenue that can offset the cost of the power-to-X process.

Operating conditions is assigned the lowest weight of 10%. This criterion covers the required operating pressure and temperature window, which together determine the mechanical complexity and cost of the reactor and surrounding equipment. Regarding pressure, elevated pressure is a disadvantage primarily at lab scale, where it adds hardware cost and safety requirements to the experimental setup. However, at industrial scale, higher pressure reduces equipment size and can reduce hardware costs,

making it a neutral or beneficial condition. The low weight assigned to this criterion therefore reflects this context-dependence: pressure should not be the main decisive argument against a product that is otherwise well suited to the objectives. A candidate that does not require elevated pressure to achieve good conversion retains the freedom to operate at higher pressure if that is beneficial for equipment sizing at larger scale, without being constrained to do so. This operational flexibility is rewarded in the scoring. Regarding temperature, a lower operating temperature is universally preferable: it reduces material and insulation requirements, simplifies heat management, and lowers both capital and operating expenditure regardless of the process or scale.

2.2. Product Descriptions and Sorption Enhancement Potential

With these five criteria established, each candidate product and its associated sorption enhanced process are now briefly reviewed. The scores for each product are assigned and justified in in Section 2.3.

2.2.1. Methanol

Methanol (CH_3OH) synthesis from CO_2 and H_2 proceeds at 200–300 °and 50–100 bar over a copper-based catalyst. The reaction is thermodynamically limited, and conventional single-pass CO_2 conversion is low, which necessitates large recycle-to-feed ratios of around 7 and downstream distillation to meet product purity specifications [23]. Methanol has an established global market of approximately 85 Mt per year as both a chemical feedstock and a direct fuel with particular potential in the maritime sector, and is easily stored and transported as a liquid at ambient conditions [23]. Its product value per unit of energy delivered is higher than that of synthetic natural gas, since it competes in chemical and liquid fuel markets rather than against pipeline gas at commodity prices. This partially offsets the process complexity penalty identified above.

Sorption enhancement increases the single-pass yield by removing water in-situ, with mixed-bed experiments reporting a best-case methanol yield of 28% and a CO_2 conversion of 39% [24]. A complicating factor is that the zeolite sorbent also retains methanol and dimethyl ether alongside water, which distributes the equilibrium shift across multiple adsorbed species rather than concentrating it on water removal alone [25]. Figure 2.1 shows that methanol synthesis has among the steepest equilibrium responses to water removal of all four candidates, suggesting a large theoretical potential for sorption enhancement; however, co-adsorption of methanol reduces the effective sorbent capacity available for water removal, and water removal simultaneously promotes CO formation as a side product, so that experimental single-pass yields remain far below the thermodynamic potential visible in the figure [25, 19]. Furthermore, pressures above 40 bar are required for effective sorption enhancement, as lower pressures disproportionately favour an unwanted side reaction that produces CO rather than methanol [26]. Even at these best-case yields, significant recycle of unconverted feed gases remains necessary, which fundamentally limits dynamic operability under a fluctuating H_2 supply.

The only bi-functional study is that of Terreni et al., who deposited copper particles into zeolite 13X beads and demonstrated methanol yield enhancements of up to 130% at 15 bar and 250°C [25]. However, pressure swing regeneration only partially removed the accumulated species over five cycles, copper incorporation reduced the zeolite surface area by a factor of four, and catalytic activity remained well below commercial standards. A bi-functional effect has therefore been demonstrated, but the material is too compromised to serve as a basis for process design.

2.2.2. Dimethyl Ether

Dimethyl ether (DME, CH_3OCH_3) can be synthesised from CO_2 and H_2 at approximately 250,°C and 25 - 50 bar, proceeding via methanol as an intermediate and requiring two catalyst functions in the same reactor [27, 28]. DME is easily liquefied at modest pressure and commands a higher market price per unit of energy than synthetic methane, competing in the LPG replacement and clean transport fuel markets rather than against relatively cheap fossil natural gas [22].

Sorption enhancement is particularly effective for DME synthesis because water removal drives both equilibrium steps forward simultaneously, raising the single-pass DME yield from approximately 9% in the conventional process to over 80% with sorption enhancement [22]. Figure 2.1 shows that DME synthesis has among the steepest equilibrium responses to water removal of the four candidates, consistent with the high single-pass yield improvements reported experimentally. Pressure swing regener-

ation is the preferred regeneration mode, achieving this yield with a factor of four increase in productivity compared to temperature swing [22]. Operating at elevated pressure is a process requirement rather than a convenience: it improves the equilibrium conversion of both reaction steps and is necessary for pressure swing regeneration to function. Carbon selectivity to DME is strongly pressure-dependent, rising from 50.2 % at 20,bar to 88.5 % at 50,bar in modelling studies validated against the TNO pilot plant [28]. The primary cause of reduced selectivity at lower pressures is the reverse water-gas shift reaction, which diverts reactants toward CO as the sorbent saturates[22, 28].

The high single-pass yield substantially reduces but does not eliminate recycle: unconverted feed gases and CO byproduct must be separated and recycled or vented. The downstream separation train requires a flash drum to remove non-condensables, a cryogenic distillation column to separate CO_2 from DME, and a further distillation step to remove methanol and water [22]. The CO_2/DME separation is the most energy-intensive step in the process and, combined with the residual methanol from incomplete dehydration and DME losses during the blowdown step [28], means that downstream complexity remains substantial even at high single-pass conversion.

All experimental demonstrations to date use physically mixed beds of separate catalyst and sorbent particles. No bi-functional material combining both functions in a single particle has been reported for sorption enhanced DME synthesis (SEDMES). The absence of a bi-functional material reflects a fundamental challenge at the material level: integrating a copper-based methanol synthesis catalyst into a zeolite support while preserving both catalytic activity and water adsorption capacity has not yet been achieved. The mixed-bed configuration has been validated on a multi-column test rig under industrially relevant conditions at TRL 6 [16].

2.2.3. Carbon Monoxide

CO is synthesized from CO_2 and H_2 via the reverse water-gas shift reaction converts. The reaction is equilibrium-limited, with practically relevant conversion requiring temperatures above 800,°C in a conventional reactor [29]. Sorption enhancement shifts this window down to 300–400,°C by continuously removing product water, substantially reducing energy demand and material requirements. The reaction requires elevated operating pressure of approximately 25–30,bar to achieve good selectivity and to enable pressure swing regeneration [30]. CO is not a final product but a valuable intermediate: it is more reactive than CO_2 and its conversion to methanol or synthetic fuels via Fischer-Tropsch synthesis is well established industrially. A two-step route in which CO_2 is first converted to CO and subsequently hydrogenated may therefore offer advantages over direct CO_2 hydrogenation, as each step can be optimised separately. The full process chain must however be designed and operated together, since CO has no standalone product value, is toxic, and cannot be stored practically. Furthermore, unlike the other three candidates whose reactions are exothermic, the reverse water-gas shift reaction is endothermic, meaning that heat must be continuously supplied to the reactor to sustain conversion rather than managed and removed.

Sorption enhancement achieves near-complete CO_2 conversion in the pre-breakthrough period, with a maximum of approximately 90% at atmospheric pressure rising to 93% at 25 bar [29, 30]. Near-complete selectivity to CO was demonstrated at 29 bar at the lower end of the operating temperature window [30], but methane formation increases with temperature, reaching approximately 10% of the carbon product distribution at 390°C [29]. Methane is inert in downstream methanol synthesis and Fischer-Tropsch processes and would accumulate in recycle loops, adding a separation requirement that partially offsets the conversion benefit of sorption enhancement. A small fraction of methanol also accumulates on the sorbent between cycles and is not fully removed during regeneration [29]. Pressure swing regeneration is the preferred regeneration mode, consistent with the approach demonstrated for SEDMES [30]. The dependency on a downstream conversion step, combined with the methane and methanol byproducts requiring separation, means that the downstream process complexity is substantial and that dynamic operation under a variable H_2 supply is particularly challenging.

The bi-functional Pt/zeolite 4A configuration has been experimentally demonstrated at kilogram scale over more than 400 cycles without significant decay [29], confirming material stability. However, a multi-column continuous process has not been demonstrated at any scale and no performance data under conditions representative of a process unit are available, leaving a clear gap between the current material-level evidence and what would be required to ground a full reactor design.

2.2.4. Methane

CO_2 methanation proceeds at 200–300°C and moderate pressure over a nickel-based catalyst, which is the industry standard due to its adequate activity and low cost relative to noble metal alternatives [3]. The reaction is strongly exothermic, releasing 165 kJ per mole of CO_2 converted, which makes thermal management the central engineering challenge when scaling the process. The product, synthetic natural gas, is directly compatible with the existing gas grid infrastructure, requiring no downstream conversion, dedicated storage, or separate distribution infrastructure. Methane carries the highest gravimetric heating value of the four candidates at 50 MJ kg^{-1} (LHV), compared to approximately 29 MJ kg^{-1} for DME and 20 MJ kg^{-1} for methanol [12]. However, synthetic methane competes directly with fossil natural gas at commodity prices, meaning its product value per unit of energy is lower than that of DME or methanol, which compete in liquid fuel and chemical feedstock markets [31].

Sorption enhancement is well suited to methanation. Conventional multi-stage fixed-bed methanation reaches approximately 80% CO_2 conversion [31], while sorption enhanced methanation (SEM) achieves near-complete single-pass conversion, with CO_2 conversions of 100% and CH_4 purities above 99 vol% reported at lab scale [13]. As Figure 2.1 shows, the equilibrium response to water removal for methanation is less steep than for DME or methanol, but the shift to full conversion occurs at relatively high steam partial pressures, where the zeolite operates close to its maximum adsorption capacity and sorbent inventory is therefore well utilised. Unlike the other three candidates, whose reactor outlets contain multiple carbon-bearing species requiring distillation or cryogenic separation regardless of the conversion achieved, the SEM outlet contains only CH_4, water, and small residual quantities of H_2, which are easier to separate. SEM applied to methanation therefore delivers a stream that is already close to pipeline-quality synthetic natural gas, with few extra synthesis steps required. The elimination of the recycle loop has a direct benefit for dynamic operability: Barbaresi et al. demonstrated through modelling that the process can be operated across a load range of 25–100% [32], making it well suited for direct coupling with a variable renewable electricity supply. Fluidized bed reactors are a particularly promising configuration, as their intensive solid mixing provides near-isothermal operation and enables continuous regeneration [3].

SEM has been demonstrated at pilot scale through the Empa move-MEGA project, a 50 kW fixed-bed demonstration plant operating at 50 bar using a bi-functional Ni/zeolite 13X material [33]. The most comprehensive material-level evidence supporting this configuration is provided by Wei, who demonstrated that a 5 wt% Ni with 2.5 wt% Ce on zeolite 13X maintained 100% CO_2 conversion and near-complete CH_4 selectivity over 100 adsorption-regeneration cycles at atmospheric pressure [13]. Bi-functional materials have been shown to outperform physically mixed beds due to the shorter transport distance between reaction and adsorption sites [12, 13], making them the preferred material configuration for this work. While Massa et al. demonstrated SEM in a fluidized bed using a physically mixed bed configuration [20], no demonstration combining a non-fixed-bed reactor with a bi-functional material has been reported at any scale.

2.3. Product Selection

Table 2.1 summarises the scores assigned to each candidate product across the five criteria; the reasoning behind each score is presented below, working through the criteria in descending order of weight.

On downstream process complexity, which carries a weight of 35%, methane is the only candidate for which sorption enhancement delivers a stream that is already close to the final product specification without further separation or synthesis steps. All three alternative candidates require multi-step downstream processing regardless of the single-pass conversion achieved: DME requires a flash drum, a cryogenic CO_2/DME separation column, and a methanol removal step; methanol requires distillation with a large recycle-to-feed ratio; and CO has no standalone product value and requires a complete downstream synthesis train. Methanol and CO are tied at 1, rather than one outranking the other because neither penalty is recoverable through better reactor performance: Methanol's single-pass yield of 28% forces a large recycle loop regardless of sorption enhancement [22], and CO has no standalone product value at any conversion. DME scores 3 rather than 1 because its sorption-enhanced single-pass yield of >80% substantially reduces the recycle burden, even though the separation train itself remains complex. For methane, water removal by condensation and near-complete single-pass CO_2 conversion under SEM conditions are sufficient to meet Dutch gas grid injection specifications

[34]. This distinction is not simply a matter of higher conversion but of a fundamentally simpler process architecture downstream of the reactor.

On infrastructure readiness, which carries a weight of 20%, methane scores highest at 5, as synthetic natural gas is directly compatible with the existing Dutch gas grid and requires no dedicated storage or distribution infrastructure. DME and methanol both score 3, since they are easily liquefied and distributed using conventional equipment, but lack universal grid access and depend on more limited distribution networks. CO scores 1, as it has no standalone distribution infrastructure and cannot be stored or transported practically.

On bi-functional material maturity, which carries a weight of 20%, DME scores 1 as no bi-functional material has been reported. Methanol scores 2 because Terreni et al. demonstrated a bi-functional effect, but material degradation renders it insufficient for process design [25]. CO scores 3, as a stable material exists but process-level data remain limited. Methane scores 4: validated up to TRL 5 in the move-MEGA pilot and over 100 cycles at material level [13, 33], but exclusively in fixed-bed reactors.

On market value, which carries a weight of 15%, DME and methanol score highest at 5, as both compete in higher-value liquid fuel and chemical feedstock markets. Methane scores 3 because synthetic methane competes directly with fossil natural gas at commodity prices, giving it a lower product value per unit of energy than the liquid alternatives [31]. CO scores 3, reflecting its value as an industrial intermediate and sought-after feedstock for methanol synthesis and Fischer-Tropsch fuel production. Although it cannot be sold directly from the reactor outlet, the penalties associated with the required downstream conversion train and the absence of standalone distribution infrastructure are already captured under process complexity and infrastructure readiness respectively, and are therefore excluded from this criteria.

On operating conditions, which carries a weight of 10%, methane scores highest at 5. It operates within a moderate temperature window of 200 to 300°C, is compatible with atmospheric-pressure operation, and retains full flexibility to operate at elevated pressure if beneficial for sizing at larger scale [3]. DME scores 3, as it operates at a similarly mild temperature but requires pressures of 25 to 50 bar, reducing this pressure flexibility. Methanol scores 2: it shares a mild temperature window with methane and DME but requires the highest operating pressure of any candidate, above 50 bar, removing pressure flexibility entirely. CO scores 1, as it combines the highest operating temperature of the four candidates with a uniquely endothermic reaction, meaning heat must be continuously supplied rather than managed and removed, adding a hardware and energy management requirement that none of the other candidates face.

Based on the weighted scores across the five criteria, methane achieves the highest total in Table 2.1 at 4.50, well ahead of DME at 2.90 and methanol at 2.30, and is therefore selected as the target product for the remainder of this thesis. The gap is driven primarily by downstream process complexity, where methane is the only candidate for which sorption enhancement delivers a stream already close to the final product specification without further separation or synthesis steps. The result is robust: if all five criteria are assigned equal weights of 20%, methane still leads by 1.40 points, confirming that the selection is not driven by the weighting choices but reflects a consistent advantage across all criteria. Chapter 3 provides a detailed review of the sorption-enhanced methanation literature, covering the thermodynamic and kinetic basis of the process, material selection, and regeneration strategies, and identifies the gaps that motivate the reactor design developed in this thesis.

Table 2.1: Multi-criteria assessment of candidate products for sorption-enhanced CO_2 hydrogenation. Scores are on a scale of 1 to 5, where 5 is most favourable.

Criterion	Weight	Methane	DME	Methanol	CO
Downstream process complexity	35%	5	3	1	1
Infrastructure readiness	20%	5	3	3	1
Bi-functional material maturity	20%	4	1	2	3
Market value	15%	3	5	5	3
Operating conditions	10%	5	3	2	1
Weighted total		**4.50**	2.90	2.30	1.70

3

Sorption-enhanced CO_2 Methanation: Literature Review

SEM combines two functions in a single reactor: the catalytic conversion of CO_2 and H_2 to methane, and the continuous in-situ capture of the water produced by that reaction. This chapter reviews the state of the literature on SEM, covering the thermodynamic and kinetic basis of the reaction, catalyst and sorbent selection and regeneration strategies, and concludes with an overview of the experimental record and the gaps that remain.

The interaction between reaction and adsorption gives SEM its characteristic cyclic behaviour. During the pre-breakthrough period, water is captured by the sorbent as it forms, shifting the reaction equilibrium beyond its thermodynamic limit. This phase ends when the sorbent approaches saturation. During the breakthrough period the equilibrium shift weakens and methane purity at the outlet declines. In the post-breakthrough period the sorbent is fully saturated, the sorption-enhanced effect is lost, and the reactor approaches conventional methanation steady state [3]. The sorbent must then be regenerated before the next cycle begins. The duration of the pre-breakthrough period, the regeneration rate, and the stability of catalyst and sorbent over many cycles are therefore the central engineering variables in SEM design.

This chapter first establishes the thermodynamic and kinetic basis of the reaction, covering the operating window and the role of equilibrium in setting conversion limits, before reviewing the catalyst and sorbent materials reported in the literature and the trade-offs governing their selection. The feed composition, regeneration strategies, and alternative electrically driven regeneration concepts are then discussed, and the chapter closes with a structured overview of the experimental record and the gaps that remain unresolved.

3.1. Thermodynamics of CO_2 Methanation

CO_2 methanation proceeds via the Sabatier reaction:

$$\mathrm{CO_2} + 4\,\mathrm{H_2} \rightleftharpoons \mathrm{CH_4} + 2\,\mathrm{H_2O} \qquad \Delta H^\circ_{298} = -165 \text{ kJ/mol} \tag{3.1}$$

with competing side reactions:

$$\mathrm{CO_2} + \mathrm{H_2} \rightleftharpoons \mathrm{CO} + \mathrm{H_2O} \qquad \Delta H^\circ = +41 \text{ kJ/mol} \tag{3.2}$$

$$\mathrm{CO} + 3\,\mathrm{H_2} \rightleftharpoons \mathrm{CH_4} + \mathrm{H_2O} \qquad \Delta H^\circ = -206 \text{ kJ/mol} \tag{3.3}$$

The CO_2 methanation reaction (Equation 3.1), commonly referred to as the Sabatier reaction, is highly exothermic ($\Delta H^\circ_{298} = -165.4$,kJ mol^{-1}) and proceeds with a reduction in the total number of moles,

making it thermodynamically favoured at low temperatures and elevated pressures. However, temperatures of at least 200–250 °C are required for practically relevant reaction rates over nickel-based catalysts [3].

A competing side reaction is the reverse water-gas shift reaction (rWGS, Equation 3.2), which is endothermic and becomes increasingly favourable at higher temperatures, leading to CO formation and reduced CH_4 selectivity. Tommasi et al. report that methane selectivity drops at temperatures above 400,°C at ambient pressure, while elevated pressure allows higher temperatures to be maintained at 100% selectivity [31]. In the SEM context, sorbent water uptake capacity declines significantly above 300,°C, so the intended operating window remains well below the temperature at which the rWGS reaction becomes significant under normal conditions. CO formation therefore only becomes a concern locally, in the event of hotspots caused by the strong exothermicity of the reaction. Bareschino et al. modelled axial temperature differences of up to 100 K in a fixed-bed SEM reactor with bi-functional Ni and zeolite 13x material, under adiabatic conditions, illustrating that such excursions are plausible if heat management is inadequate [35]. Hotspots reduce methane selectivity, lower local sorbent capacity, and risk catalyst sintering [3]. Kiefer et al. identified that sorption enhancement acts as a partially self-stabilising thermal mechanism: a local temperature rise reduces the water adsorption capacity in that zone, which in turn dampens the local reaction rate and the associated heat release [15]. This effect has not been verified beyond lab scale, and its significance relative to conventional heat management measures at larger reactor diameters remains an open question.

Elevated operating pressure has a compounding thermodynamic benefit in SEM. Since the Sabatier reaction proceeds with a net reduction in molar volume, higher pressure shifts the equilibrium toward CH_4 formation. It simultaneously increases the partial pressure of water in the product gas, which strengthens the thermodynamic driving force for water adsorption on the zeolite and extends the pre-breakthrough period. Gómez et al. demonstrated that increasing the SEM stage pressure from 1 bar to 10 bar tripled the water uptake and therefore the pre-breakthrough duration quadrupled and the CH_4 purity increased [18]. Elevated pressure also enables pressure swing regeneration, since a meaningful pressure differential between the reaction and regeneration stages can be established without reducing the system to vacuum. However, elevated pressure adds engineering complexity in terms of vessel design, sealing, and instrumentation, and most experiments in the literature are therefore conducted at atmospheric pressure.

The methanation reaction is limited by the thermodynamic equilibrium. At 300 °C and ambient pressure, the maximum is 92% and it decreases when the temperature increases [3]. When a chemical reaction is at equilibrium, Le Chatelier's principle states that removing one of the products will cause the reaction to proceed further in the forward direction to compensate. In CO_2 methanation, water is produced alongside methane; if this water is continuously removed from the reactor, the reaction is driven further towards methane formation than would otherwise be possible. This is the core idea behind sorption-enhanced methanation: a water-adsorbing material, often a zeolite, is placed alongside the catalyst in the reactor, so that as water forms it is immediately captured, pushing the reaction beyond its normal equilibrium limit and achieving higher CO_2 conversion in a single pass, up to 100% [13].

3.2. Reaction Kinetics

The hydrogenation of CO_2 to methane over a nickel catalyst proceeds via multiple proposed surface reaction pathways. Three routes have been identified in literature: a carbide pathway, in which CO_2 is first reduced to a surface-bound CO intermediate before sequential hydrogenation to methane; a formate pathway, in which CO_2 is hydrogenated through oxygenated intermediates; and a carboxylate pathway, which is thermodynamically unfavourable on nickel and has not been observed experimentally [36]. Although the reaction mechanism under sorption-enhanced conditions has not been exhaustively characterised, the carbide route is adopted as the governing mechanism in most kinetic modelling work on nickel-based systems [13].

The reaction rate of CO_2 methanation over nickel depends on both temperature and the partial pressures of the reacting species. Several kinetic models have been proposed in the literature, ranging from empirical power-law expressions [13] to mechanistically derived Langmuir–Hinshelwood–Hougen–Watson (LHHW) models [31]. For SEM modelling, the LHHW model proposed by Koschany, Schlereth, and Hin-

richsen is the most widely adopted, having been validated over a Ni-based catalyst in the temperature range 180–360,°C and at pressures between 1 and 15 bar [15, 17, 31, 35, 37]. The model neglects side reactions, which is a reasonable simplification for the temperature range relevant to SEM [35]. The Xu and Froment model is also widely cited in the CO_2 methanation literature [38], but its kinetic parameters were regressed for steam methane reforming at significantly higher temperatures, limiting its reliability in the 200–400,°C range relevant to SEM [31]. The rate expression takes the form:

$$r = k \cdot \frac{\left(p_{\mathrm{H_2}} p_{\mathrm{CO_2}}\right)^{0.5} \left(1 - \dfrac{p_{\mathrm{CH_4}} p_{\mathrm{H_2O}}^2}{p_{\mathrm{H_2}}^4 p_{\mathrm{CO_2}} K_{\mathsf{eq}}}\right)}{\left(1 + K_{\mathsf{OH}} \dfrac{p_{\mathrm{H_2O}}}{p_{\mathrm{H_2}}^{0.5}} + K_{\mathrm{H_2}} p_{\mathrm{H_2}}^{0.5} + K_{\mathsf{mix}} p_{\mathrm{CO_2}}^{0.5}\right)^2} \tag{3.4}$$

The reaction rate r (mol $\mathrm{kg_{cat}^{-1}\,s^{-1}}$) describes the molar conversion rate of CO_2 per unit catalyst mass. The pre-exponential rate constant k follows an Arrhenius temperature dependence, capturing how the intrinsic activity of the nickel surface increases with temperature. The equilibrium constant K_{eq} governs the thermodynamic driving force: the bracketed term in the numerator equals zero at equilibrium and approaches unity far from it, meaning the rate is highest when the gas-phase composition is furthest from the equilibrium composition. The numerator shows that the rate increases with both H_2 and CO_2 partial pressure. The three adsorption equilibrium constants in the denominator, K_{OH}, $K_{\mathrm{H_2}}$, and K_{mix}, each follow a van 't Hoff temperature dependence and represent the degree to which hydroxyl groups, hydrogen, and CO_2 compete for active surface sites on the nickel catalyst. The denominator contains a water inhibition term, $K_{\mathsf{OH}}\, p_{\mathrm{H_2O}}\, p_{\mathrm{H_2}}^{-0.5}$, which accounts for competitive occupation of active surface sites by hydroxyl groups derived from product water. All partial pressures p_i are in bar.

The practical operating window for CO_2 methanation on nickel is approximately 200–400,°C. Below 200,°C, the reaction rate becomes negligible at practical catalyst loadings [3]. The upper bound is set not by kinetics but by thermodynamics: above roughly 400,°C, equilibrium conversion drops, as discussed in Section 3.1. A temperature of 300,°C represents a practical operating point, balancing sufficient reaction rate against favourable equilibrium [15, 35].

The ratio of feed flow rate to bed inventory, expressed as the gas hourly space velocity (GHSV, $\mathrm{h^{-1}}$) or weight hourly space velocity (WHSV, $\mathrm{mL_{STP}\,g_{cat}^{-1}\,h^{-1}}$), determines the gas residence time in the bed and therefore how much of the available kinetic driving force is utilised per pass. A lower space velocity extends the contact time between the gas phase and the solid inventory, allowing the reaction to approach equilibrium more closely within a single pass, at the cost of a lower throughput per unit of catalyst mass. Conversely, a higher space velocity increases throughput but reduces per-pass conversion, which becomes particularly penalising in SEM because the sorbent then saturates more slowly and the pre-breakthrough period is extended without a proportional gain in methane productivity. The space velocities reported across the SEM literature span roughly two orders of magnitude, reflecting differences in catalyst activity, sorbent inventory, and operating pressure between studies, and are summarised in Table 3.1.

3.3. Catalysts

Catalysts are essential in CO_2 methanation to lower the energy barriers of the reaction and enable practically relevant conversion rates at moderate temperatures. A lower operating temperature is doubly beneficial in SEM: it improves the thermodynamic equilibrium conversion and increases the water adsorption capacity of the zeolite sorbent. A catalyst consists of three functional components: the active metal, which provides the surface sites where the reaction takes place; the support, which disperses the active metal over a high surface area and provides mechanical strength; and promoters, which are added in small quantities to improve activity, selectivity, or long-term stability. The selection of each component influences catalytic activity, selectivity, and long-term stability, and is therefore a key design decision.

3.3.1. Active Metal

Nickel is the most widely used active metal for CO_2 methanation in both conventional and sorption-enhanced configurations, due to its high activity, high CH_4 selectivity, and low cost relative to noble metals [13, 2]. Its dominance in the SEM literature reflects a practical preference: most experimental and modelling studies to date employ Ni-based catalysts, making it the de facto reference material for process development.

Ruthenium and rhodium are more active than Ni and operate effectively at lower temperatures, but their high cost makes them unsuitable for large-scale application [13]. Direct comparisons between metals are complicated by the fact that performance depends strongly on the support, promoter, and sorbent combination used. Goméz et al. found that a Ni-based catalyst outperformed both Ruthenium (Ru) and Rhodium (Rh) in terms of activity and selectivity in the 200–275,°C range when combined with zeolite 4A or 13X [39]. Wei showed that adding Ru to a Ni-based zeolite catalyst does not improve CO_2 conversion; selectivity is influenced by Ru addition, but the direction of the effect depends on the zeolite type and operating temperature, with no consistent improvement observed across support materials [13]. Ruthenium and rhodium are therefore not considered further in this work and the focus will be on Ni based catalysts.

3.3.2. Catalyst Support

The support material determines the surface area available for active metal dispersion, which in turn sets the number of accessible catalytic sites per unit mass. Higher surface area supports allow smaller, more numerous Ni particles, directly increasing catalytic activity [3]. Beyond surface area, the support influences CO_2 adsorption and can therefore increase activity [3]. The preparation method and precursor choice also affect the final metal dispersion; for example, nickel citrate precursors have been shown to yield finer particle distributions than nickel nitrate on the same support [13]. For conventional methanation, Al_2O_3 is the most widely used support, offering adequate surface area and thermal stability. However, zeolites are an interesting in SEM because they serve a dual function: providing a surface for Ni dispersion while simultaneously acting as the water sorbent. This eliminates the need for a separate sorbent material in bi-functional bed configurations. Zeolite supports are discussed in Section 3.4.2.

3.3.3. Catalyst Promoters

Promoters are secondary components added in small quantities to improve activity, selectivity, and long-term stability of Ni-based catalysts, primarily by mitigating sintering and carbon deposition [13]. Among the candidates studied, Cerium (Ce) is generally considered the most effective [13, 40]. It improves Ni dispersion across the support surface, suppresses particle agglomeration at reaction temperatures, and reduces carbon deposition [40]. Gavrilović et al. showed that Ce addition to Ni/Al_2O_3 increased CO_2 conversion across all tested temperatures (250–320,°C), with an optimal loading of 4,wt%, beyond which excessive surface coverage of Ni offsets the benefit [40]. Wei confirmed a consistent improvement for Ce-promoted Ni/13X, with higher activity and a lower apparent activation energy compared to the unpromoted reference [13]. Promoter behaviour is support-dependent and cannot be generalised across catalyst systems: Na, for instance, reduces CO_2 adsorption on Ni/CeO_2 while improving it on Ni/SiO_2 [13]. This context-dependence is relevant when selecting a promoted catalyst for the cyclic operating environment of SEM.

3.3.4. Catalyst Stability

Catalyst stability is a critical consideration for any reactor intended for long-term or cyclic operation. Deactivation mechanisms in CO_2 methanation can be broadly grouped into four categories: poisoning of active sites, fouling by carbon deposition (coking), thermal degradation (sintering) and mechanical degradation (attrition and crushing). All of these ultimately reduce the concentration of catalytically active surface sites, lowering the apparent rate constant for methanation.

The most common deactivation risk for Ni-based catalysts is sintering, the thermally driven agglomeration of Ni nanoparticles into larger crystallites which leads to a reduced surface area. Sintering becomes significant above approximately 500,°C and is accelerated by high water partial pressures [3]. For conventional methanation this is a persistent concern, but in SEM the operating temperature window of 200–350,°C is intentionally mild. Delmelle et al. found that at 320,°C and 360,°C, the nor-

malised CH_4/CO_2 ratio lost only about 10% of its original value after 30 hours of continuous operation, while at 500,°C the same drop was 30%. This suggests that as long as the reactor operates within its intended temperature window, sintering is not a dominant deactivation pathway [41]. However, the highly exothermic nature of the reaction creates a risk of localised hotspots in fixed-bed reactors, particularly near the inlet where fresh reactants contact the catalyst. If sustained, these hotspots can drive local temperatures well above the intended operating window, making temperature management a critical design consideration [15].

Carbon deposition on the catalyst surface, also known as coking, is generally suppressed under the strongly hydrogenating and low-temperature conditions of SEM, as confirmed experimentally by Delmelle et al. and Wei.

Attrition is not a significant concern in fixed-bed operation but becomes relevant in fluidized-bed reactors, where particle-wall and inter-particle collisions cause mechanical degradation over time [31]. This has direct implications for the reactor type considered in this work and is discussed further in Chapter 4.

Poisoning by sulphur compounds is the primary chemical deactivation risk. H_2S, present in biogas-derived feedstocks, reacts with active Ni sites to form inactive NiS, with a tolerable threshold of only 10–100 ppb [13, 31], far below typical biogas concentrations. Upstream desulfurisation is therefore mandatory when biogas is used as feedstock. $Ni(CO)_4$ formation is an additional risk below approximately 230 °C [31], setting a practical lower bound on operating temperature. Notably, sulphur poisoning has not been experimentally characterised under SEM-specific conditions, representing a gap in literature.

Despite these mechanisms, multi-cycle experimental evidence is encouraging. Wei demonstrated that the bi-functional 5%Ni–2.5%Ce/13X material maintained 100% CO_2 conversion and CH_4 selectivity throughout 100 adsorption/desorption cycles (223 hours at 360,°C), with only a slight reduction in water uptake capacity [13]. For practical reactor design, thermal management and zeolite hydrothermal durability are the most critical long-term stability parameters.

3.4. Sorbent Selection

Water removal by an in-situ sorbent is central to SEM, but the choice of sorbent is not straightforward. The operating conditions of a methanation reactor impose simultaneous requirements on adsorption capacity, selectivity, regenerability, and long-term stability that few materials can satisfy together. The following subsections establish these requirements, evaluate candidate sorbent classes, discuss the choice between bi-functional and mixed-bed configurations, and examine the trade-offs that govern material selection for SEM.

3.4.1. Sorbent Requirements

The operating environment of a SEM reactor imposes constraints on the sorbent that are more demanding than those of conventional gas drying. Methanation over Ni-based catalysts requires temperatures above approximately 200 °C to proceed at a practically relevant rate [3, 13]. The sorbent must therefore retain sufficient water uptake capacity within this elevated temperature window.

Beyond temperature compatibility, a viable SEM sorbent must simultaneously satisfy the following requirements [40]: high water adsorption capacity at reaction temperature; high selectivity for water over other species present in the gas phase, particularly CO_2 and CH_4; a sufficiently strong affinity for water at reaction conditions to drive water out of the gas phase and sustain the equilibrium shift throughout the pre-breakthrough period; straightforward regeneration by temperature or pressure adjustment without structural damage; negligible activity toward undesired side reactions; and adequate structural and chemical stability over a large number of adsorption-regeneration cycles.

Selectivity for water over CO_2 deserves particular attention. If the sorbent adsorbs CO_2 alongside water, two penalties arise simultaneously: the pore volume occupied by CO_2 is no longer available for water uptake, which reduces the effective bed capacity and shortens the pre-breakthrough period; and CO_2 retained on the sorbent is removed from the gas phase and therefore cannot react.

A critical but largely overlooked property of zeolite-based sorbents in the SEM context is their mechanical stability under conditions involving solid movement or circulation. Sonnleitner et al. reported that

standalone zeolite 13X loses approximately 4 % of its mass after only 32 hours of bubbling fluidisation, with visible fines formation occurring after as little as 5 hours [42]. This data concerns the base zeolite material without any metal impregnation; whether nickel and cerium incorporation strengthens or further weakens the particle structure has not been investigated. Since nearly all SEM experiments to date have been conducted in fixed-bed reactors where attrition is negligible by construction, this issue has not been encountered experimentally and its significance for continuous reactor configurations remains entirely uncharacterised. This represents a gap that is relevant to any reactor design involving solid transport or circulation.

These requirements narrow the field of viable sorbent materials considerably. Before evaluating candidate zeolites against them, Section 3.4.2 first establishes the two configurations in which catalyst and sorbent can be combined, since the configuration imposes additional physical constraints on which zeolites are structurally eligible.

3.4.2. Bi-functional Compared to Mixed-Bed Configuration

Two strategies exist for combining catalyst and sorbent in a SEM reactor. In a mixed-bed configuration, separate catalyst and sorbent particles are physically combined in the same bed. In a bi-functional material, the active metal is deposited directly into the zeolite support, so that catalytic and adsorption sites reside within the same particle [3, 13]. The fundamental difference between the two approaches is the distance over which water must travel after it is produced at the catalytic site. In a mixed-bed, water produced at the catalyst surface must cross the gas phase to reach a neighbouring sorbent particle, a path on the order of particle diameters, typically in the millimetre range. In a bi-functional material, this path is reduced to the intraparticle distance between active metal sites and adsorption sites, on the order of nanometres to micrometres. Terreni et al. showed that this shorter path produces a more efficient equilibrium shift per unit of sorbent inventory compared to a physical mixture, and identified close proximity of catalytic and adsorption sites as a prerequisite for effective process intensification in SEM [43]. Using neutron imaging, Terreni et al. further demonstrated that for the integrated bifunctional Ni/13X material the overall process rate is limited by the catalytic surface reaction rather than by intraparticle water transport, and that spatial separation of catalyst and sorbent in a mixed setup markedly reduces the sorption enhancement because water accumulates around the active sites instead of being captured immediately [43]. Borgschulte et al. demonstrated this experimentally in the original proof-of-concept study, showing that a bi-functional configuration outperforms a physically mixed bed of the same materials under otherwise identical conditions [12]. This work therefore adopts the bi-functional configuration as established in Chapter 1, and the two practical constraints specific to this configuration are noted below.

The first is a pore size constraint. Borgschulte et al. found that pore openings larger than 5 Å ((1 Å = 0.1 nm)) are required, since reactants must reach the active metal sites inside the pore and the methane product must be able to exit. At pore openings of 3 Å or smaller, methane cannot readily leave the pore interior, which shifts selectivity toward carbon monoxide [3, 44]. Within the zeolite families used in SEM, this restricts viable bi-functional supports to zeolite 5A and zeolite 13X, and excludes zeolite 4A despite its strong performance in mixed-bed configurations. The second is a gradual reduction in the water diffusion coefficient through the zeolite support over repeated cycling, identified by Delmelle et al. for bi-functional Ni/5A, who observed a decrease in the water diffusion coefficient of approximately 40 % after 39 cycles [41]. Since water diffusion through the zeolite is the rate-limiting transport step during regeneration, this can progressively extend regeneration time over the lifetime of the material.

A further distinction relevant to reactor type selection concerns configurations that use two separate vessels: one for reaction and one for sorbent regeneration, with solids continuously transferred between them. In such a configuration, only the spent sorbent should be transferred to the regenerator, while the catalyst remains in the reactor. This selective transfer is achieved by exploiting a size or density difference between the two solids and therefore requires two physically distinct solid phases [45]. In a bi-functional material, catalytic and adsorption functions reside in the same particle, so any particle transferred to the regenerator simultaneously removes both reaction and adsorption capacity from the reactor. Selective circulation is therefore not possible with a bi-functional material.

3.4.3. Zeolite Classes and Candidate Sorbents

Zeolites are the only sorbent class with sufficient experimental backing to satisfy all requirements identified in Section 3.4.1 at methanation temperatures. Non-zeolite alternatives all fail on at least one critical requirement. Calcium oxide is inexpensive and retains water uptake capacity above 200,°C, but its capacity declines rapidly in the presence of CO_2 due to irreversible carbonation of the active material: Agirre et al. observed a 60% reduction in uptake capacity within three cycles [46], and Coppola et al. confirmed a depressing effect of CO_2 on steam capture [45]. Silica gel, absorbing polymers, and salts fail the temperature requirement outright, with working temperatures well below 200,°C [3]. Metal-organic frameworks have high water uptake capacity at room temperature but lose most of their capacity above 200,°C and have not been tested under SEM conditions [3].

Two zeolite structural families dominate the SEM literature. Linde Type A (LTA) zeolites include the 3A, 4A, and 5A variants, which differ in their effective pore opening controlled during manufacture. Faujasite (FAU) zeolites are represented in SEM primarily by zeolite 13X, which has both a larger pore opening and a larger internal cavity than the LTA variants. As established in Section 3.4.2, the pore size constraint for bi-functional use restricts the candidate field for this work to zeolite 5A and zeolite 13X. Zeolites 3A and 4A are discussed below where their behaviour as standalone sorbents informs the comparison, but they are not viable bi-functional supports.

The standalone water uptake capacity of the candidate zeolites was measured by Gavrilović et al. using a thermogravimetric analyser under a 20/80 vol% H_2O/N_2 flow across the SEM-relevant temperature range [40]. At 200,°C, 13X showed the highest capacity at 9.0,$\mathrm{g_{H_2O}}/100,\mathrm{g_{zeolite}}$, followed by 3A at 7.9 and 4A at 7.7,$\mathrm{g_{H_2O}}/100,\mathrm{g_{zeolite}}$. All capacities decreased substantially with temperature: at 250,°C the values dropped to 7.3, 4.9, and 3.8,$\mathrm{g_{H_2O}}/100,\mathrm{g_{zeolite}}$ for 13X, 3A, and 4A respectively, with 3A outperforming 4A at this temperature. These measurements reflect idealised conditions without CO_2 or CH_4 present and therefore represent an upper bound on the capacity available under actual SEM conditions. Long-term cycling reveals an important distinction between zeolite types: over 100 adsorption/desorption cycles, standalone 13X lost approximately 50 % of its initial water uptake capacity, while 4A and 3A showed no measurable degradation [40]. This is consistent with observations from the move-MEGA project, where sorbent capacity of bi-functional Zeolite 13X and Nickel dropped to approximately 80 % of its initial value after a small number of cycles before stabilising, suggesting that the degradation is concentrated in the early cycling period rather than being gradual [33].

The mixed-bed SEM literature provides a useful reference for ranking zeolites as standalone sorbents under actual reaction conditions, even though mixed-bed performance numbers cannot be used as direct design targets for the bi-functional configuration. Gómez et al. conducted the most systematic comparison of all four zeolite variants, using a mixed-bed configuration with a rhodium-based catalyst at 275,°C and atmospheric pressure [47]. In breakthrough experiments under a water/N_2 flow, 4A and 13X achieved pre-breakthrough durations of 17.3 and 16.2,min respectively, approximately double that of 5A and five times that of 3A, confirming 4A and 13X as the strongest candidates at this temperature. Under a stoichiometric H_2/CO_2 feed, zeolite 4A achieved a higher effective water uptake of 0.142,$\mathrm{g_{H_2O}}/\mathrm{g_{zeolite}}$ compared to 0.098,$\mathrm{g_{H_2O}}/\mathrm{g_{zeolite}}$ for 13X, and a water-to-CO_2 selectivity of approximately 84,$\mathrm{g_{H_2O}}/\mathrm{g_{CO_2}}$ compared to 28,$\mathrm{g_{H_2O}}/\mathrm{g_{CO_2}}$ for 13X. The superior selectivity of 4A is attributed to its smaller pore opening, which excludes CO_2 more effectively than the larger pores of 13X. Zeolite 4A showed no detectable structural degradation over 40 adsorption-desorption cycles [47]. In a mixed-bed configuration, 4A would therefore be the preferred sorbent. However, as established in Section 3.4.2, its pore opening of 4A falls below the threshold required for bi-functional use, making 13X the leading candidate for this work by process of elimination within the viable field of 5A and 13X.

The bi-functional performance of Ni-based materials supported on 13X and 5A was systematically studied by Wei [13]. Ni/13X materials achieved substantially higher CO_2 conversion than Ni/5A across the tested temperature range, attributed to sub-nanometre nickel particles penetrating the larger 13X pores and placing catalytic sites in close proximity to adsorption sites. However, Ni/13X materials showed variable and in some cases reduced CH_4 selectivity at lower temperatures, attributed to water accumulating inside the 13X pores around the active sites and inhibiting complete conversion to methane. Ni/5A consistently showed 100 % CH_4 selectivity but at the cost of lower CO_2 conversion. The addition of cerium as a promoter resolves this selectivity limitation for 13X-supported materials. The best-performing formulation, 5 wt% Ni with 2.5 wt% Ce on zeolite 13X, achieved approximately 100

% CO_2 conversion and near-complete CH_4 selectivity over 100 adsorption-regeneration cycles, with only a slight reduction in water breakthrough capacity [13]. This capacity reduction may reflect limited nickel sintering or CO_2 co-adsorption on the zeolite support; the mechanism has not been systematically investigated and remains a gap in literature. This is the only bi-functional SEM formulation for which conversion, selectivity, and long-cycle stability data are simultaneously available. Notably, the bi-functional Ni-Ce/13X material showed considerably less capacity degradation over 100 cycles than the materials including zeolite 13X reported by Gavrilović et al. and Kiefer, Borgschulte, and Bach, despite the standalone zeolite being the same base material [40, 33]. The reason for this discrepancy has not been systematically investigated and represents another gap in literature.

Of the bi-functional SEM formulations reported to date, 5 wt% Ni with 2.5 wt% Ce supported on zeolite 13X, following the formulation of Wei, is the most mature available option [13]. It is the only bi-functional SEM formulation for which conversion, selectivity, and long-cycle stability data are simultaneously available. The capacity degradation observed in standalone 13X is not reflected in this material over 100 cycles.

3.5. Feed Composition

The feed composition directly affects both the achievable CO_2 conversion, the need for a recycle stream, and the tolerance of the catalyst and sorbent to impurities, making it a key operating variable in SEM. The stoichiometric H_2/CO_2 ratio for the Sabatier reaction (Equation 3.1) is 4:1, and this is the operating point adopted across nearly all SEM experiments (see Table 3.1). The ratio has a pronounced effect on both CO_2 conversion and sorbent utilisation.

Gómez et al. systematically evaluated H_2:CO_2 ratios of 3:1, 4:1, and 5:1 at 10 bar and 250 °C using a Ni-based catalyst mixed with zeolite 4A. At 3:1, the higher CO_2 partial pressure increased CO_2 co-adsorption on the zeolite by a factor of eight compared to 5:1, reducing the pore volume available for water uptake and collapsing the pre-breakthrough period from 64 minutes to 3 minutes. The sub-stochiometric feed also increased from 0.02% for stochiometric feed to 0.06 %. At a ratio of 5:1, CO_2 conversion increased from 79.3 % to 94.8%, but CH_4 purity in the product dropped from 100% to 37,vol% due to unconverted H_2 in the outlet flow [48]. Gómez et al. concluded that 4:1 is the preferred operating point, balancing a good CH_4 yield at 84.9% while maintaining high methane purity (84 vol%) and a solid pre-breakthrough time (27 min).

Mercader et al. identified 5:1 as optimal under their specific conditions, where incomplete zeolite regeneration between stages made the lower water production at 5:1 advantageous for preserving residual adsorption capacity in later reaction stages [49]. The apparent disagreement between these recommendations reflects differences in regeneration completeness rather than a fundamental conflict. For a fully regenerated system, 4:1 maximises single-cycle performance; super-stoichiometric operation may offer advantages when regeneration is constrained, at the cost of a H_2 recycle stream.

When biogas is used as feedstock, CH_4 is present in the feed alongside CO_2. Wei showed that CH_4 concentrations from 0 to 81.5,vol% in the feed have no significant effect on CO_2 conversion, as methane does not interact with the catalyst or compete for adsorption sites on the zeolite [13]. This is consistent with findings by Gómez, Martínez, and Murillo, who reported that increasing the CH_4 fraction in the feed from 10 to 40,vol% reduced CO_2 conversion by less than one percentage point [50].

When CO is present in the feed, originating for example from upstream gasification or partial oxidation processes, it reacts readily over Ni-based catalysts via CO methanation. Because CO does not adsorb significantly on the zeolite sorbent, its presence reduces the CO_2 fraction competing for zeolite adsorption sites, which can extend the effective pre-breakthrough period. Gómez et al. demonstrated the sorption-enhanced effect for both pure CO and mixed CO/CO_2 feeds, and Gómez et al. found that adding CO to a CO_2 feed improved overall CO_2 conversion for this reason [14, 48]. The principal process consequence of CO in the feed is a higher thermal load, as CO methanation releases approximately 206,kJ/mol compared to 165,kJ/mol for CO_2 methanation, see equation 3.3, which must be accounted for in the thermal design of the reactor.

The most critical feed impurity for SEM is hydrogen sulphide (H_2S). Ni-based catalysts are highly sensitive to sulphur because they deactiviate Ni active sites due to the formation of NiS. Concentrations

should therefore be as low as 10 - 100 ppb [31]. Upstream desulphurisation is therefore a prerequisite for direct biogas methanation over Ni catalysts. The effect of H_2S and other trace impurities on SEM specifically has not been evaluated experimentally and represents a gap in literature.

3.6. Regeneration

Once the zeolite sorbent approaches its water uptake capacity, the sorption-enhanced effect weakens and the sorbent must be regenerated to restore adsorption performance. In fixed-bed configurations this requires taking the reactor offline, making regeneration a primary determinant of cycle time, energy demand, and the number of parallel vessels required for continuous methane production [35, 17]. A regeneration phase twice as long as the adsorption phase requires at least three parallel reactors [17]. In non-fixed-bed configurations with continuous sorbent circulation, regeneration occurs simultaneously in a separate vessel, decoupling it from the reaction phase, but the energy demand and completeness of desorption remain equally critical design variables. In both configurations, minimising regeneration time and energy demand while achieving complete sorbent restoration is a central engineering objective.

Three strategies for driving desorption have been reported in the SEM literature: temperature swing, pressure swing, and purge gas stripping, which can be applied individually or in combination. A fourth category, electrically driven methods including microwave, radiofrequency dielectric, and acoustic excitation, has not yet been demonstrated under SEM conditions but is of particular interest for power-to-gas applications where renewable electricity is directly available. These are evaluated separately in Section 3.6.3. The thermodynamic basis common to all strategies is established first in Section 3.6.1, before the conventional strategies and their trade-offs are reviewed in Section 3.6.2.

3.6.1. Thermodynamic Basis for Regeneration

The equilibrium water loading of the zeolite sorbent depends on both temperature and the partial pressure of water in the gas phase, a dependence that directly determines when regeneration is required and how it can be driven. This relationship is described by the Dubinin–Astakhov (DA) model, which is used across the SEM modelling literature [15, 35, 51]:

$$W = W_0 \exp\left[-\left(\frac{A}{E_{\mathrm{char}}}\right)^n\right] \tag{3.5}$$

where W is the volumetric adsorption capacity ($\mathrm{m^3\,kg^{-1}}$), W_0 the maximum capacity, E_{char} a characteristic energy of the adsorbent–adsorbate pair ($\mathrm{J\,mol^{-1}}$), and n a structural heterogeneity exponent. The adsorption potential A is defined as:

$$A = \frac{R}{M_{\mathrm{H_2O}}}\, T \ln\left(\frac{p_{\mathrm{sat}}}{p_{\mathrm{H_2O}}}\right) \tag{3.6}$$

where T is temperature, p_{sat} the saturation vapour pressure of water at T, $p_{\mathrm{H_2O}}$ the actual partial pressure of water in the gas phase, R the universal gas constant, and $M_{\mathrm{H_2O}}$ the molar mass of water.

Equations (6.24) and (6.25) together make the thermodynamic basis for all three regeneration strategies explicit. The equilibrium loading W reaches its maximum W_0 when the gas phase is saturated ($p_{\mathrm{H_2O}} = p_{\mathrm{sat}}$), and decreases as the gas moves away from saturation. Regeneration therefore requires reducing the equilibrium loading by increasing the adsorption potential A. Three levers are available. Raising the temperature increases A through two compounding effects: directly, since the temperature appears as a multiplier in Equation (6.25), and indirectly through the increase in the saturation vapour pressure p_{sat}, which follows an approximately exponential dependence on temperature via the Clausius–Clapeyron relation. The indirect effect dominates in practice: p_{sat} increases by roughly a factor of five between 200 °C and 300 °C, while the direct temperature multiplier increases by only a factor of approximately 1.2 over the same range. This is the basis of temperature swing adsorption (TSA). Lowering the water partial pressure $p_{\mathrm{H_2O}}$ through a reduction in total pressure achieves the same result without thermal cycling, and is the basis of pressure swing adsorption (PSA). Combining both levers defines temperature and pressure swing adsorption (TPSA). Sweeping the bed with a dry purge gas achieves the same reduction in $p_{\mathrm{H_2O}}$ through a different mechanism: at constant total pressure, the

purge dilutes and convects away the desorbed water, reducing its mole fraction in the gas phase. The strategies are therefore not fundamentally different mechanisms but different operating choices acting on the same underlying equilibrium expression.

3.6.2. Regeneration Strategies

Water adsorption on zeolites is an exothermic, equilibrium-limited process, so adsorption capacity increases at lower temperatures and desorption is favoured at elevated temperatures or reduced water partial pressures [13]. This establishes a thermodynamic tension between the reaction phase, which requires temperatures above $200\,^{\circ}\mathrm{C}$, and the regeneration phase, which must operate at conditions that meaningfully reduce the equilibrium loading expressed by Equation (6.24).

In temperature swing adsorption (TSA), the bed is heated after the SEM stage, typically to 300–400 °C in an inert or reducing purge stream, until the adsorbed water is desorbed [14]. TSA is the most common mode in lab-scale studies (see Table 3.1) because it imposes few hardware constraints, but heating and subsequent cooling back to reaction temperature lengthen the overall cycle significantly, reducing methane productivity. It is therefore generally considered impractical for larger-scale operation [50]. However, since both the SEM stage and the TSA regeneration operate in overlapping temperature ranges, the heat released by the exothermic methanation reaction can supply the regeneration energy demand, which would substantially reduce the energetic penalty of TSA [13, 15, 33], but this concept has not been experimentally validated for SEM. The reactor-level implementation of this heat integration is discussed in Chapter 4.

Pressure swing adsorption (PSA) reduces the total pressure to near-atmospheric while a purge gas sweeps desorbed water from the bed, with temperature held constant at the reaction value. Gómez et al. demonstrated PSA at a constant 230 °C, achieving stable zeolite water uptake capacity and pre-breakthrough duration over 11 consecutive cycles using a a purge flow of 200 Nℓ/h [18]. Because PSA eliminates the thermal transients of TSA, it substantially shortens cycle time and is the preferred strategy for elevated-pressure SEM [50]. A hard lower bound on the allowable pressure during regeneration was identified by Gómez et al.: vacuum conditions of -0.7 bar destroyed the crystalline framework of zeolite 4A in a single cycle, reducing the pre-breakthrough duration by approximately 80%, while a more moderate vacuum of -0.5 bar halved the pre-breakthrough duration after seven cycles [18]. Since regeneration must remain at or above atmospheric pressure, effective PSA requires that the reaction stage operates at significantly elevated pressure, so that a meaningful pressure differential can be established between the two stages. This adds a process-level argument for elevated operating pressure in SEM, complementing the thermodynamic and adsorption capacity benefits discussed in Section 3.1.

Combined temperature and pressure swing adsorption (TPSA) applies both levers simultaneously, yielding a higher sorbent working capacity than either strategy alone at the cost of increased process complexity [18]. Although TPSA was evaluated first by Gómez et al., the additional heating and cooling steps increased regeneration time and reduced methane productivity compared to PSA, leading the authors to select PSA for cyclic operation at elevated pressure. TPSA becomes preferable over PSA alone when the available pressure differential between the reaction and regeneration stages is insufficient to drive complete sorbent restoration, since the additional temperature swing recovers the working capacity that pressure reduction alone cannot reach at modest operating pressures [18, 22].

Regardless of the thermodynamic strategy, a purge gas is required in practice to convect desorbed water out of the bed and thereby drive regeneration kinetically. In TSA and PSA, the purge gas acts as a carrier: the temperature increase or pressure reduction drives desorption thermodynamically, and the purge removes the released water before it can re-adsorb. In a standalone purge gas strategy, the dry inert itself is the active regeneration mechanism, since diluting the gas phase at constant temperature and pressure reduces the water partial pressure and drives desorption directly. Inert gases, typically nitrogen or dilute hydrogen in nitrogen, are the most effective purge media: they introduce no competing adsorbate and the purge effluent consists almost entirely of inert gas and water, which can be separated and the inert recycled with a limited make-up requirement [18]. The standalone purge approach is the simplest mode to implement at lab scale, requiring no pressure rating above atmospheric and no heating cycle beyond maintaining reaction temperature, which explains its frequent use in experimental studies, see table 3.1

A physically important asymmetry exists between the transport processes during the SEM phase and during regeneration in bi-functional materials. During the reaction phase, water produced at the active metal sites need only diffuse to the nearest adsorption sites within the same particle, a distance of order 10^{-8}–10^{-6} m in bi-functional materials. During regeneration, the same water must diffuse out of the entire pellet before the purge gas can carry it away, corresponding to path lengths of order 10^{-3} m [41]. This several-orders-of-magnitude increase in the relevant diffusion length makes regeneration the transport-limited step in the cycle. Delmelle et al. confirmed this experimentally, showing that a 40% reduction in the effective water diffusion coefficient over 39 cycles had negligible impact on methane yield during the SEM phase but progressively extended the time required for complete regeneration [41]. Regeneration duration is therefore the binding constraint on cycle time. In mixed-bed configurations the regeneration path length is comparable, since water must similarly diffuse out of the sorbent pellet on the same millimetre scale. In the experiments of Gómez et al., the PSA regeneration stage lasted approximately 60 min against a SEM stage of 25 min, a ratio of roughly 2.5, implying that continuous methane production requires a minimum of three to four parallel reactor vessels operating in staggered phases [17, 18].

3.6.3. Alternative Regeneration Concepts

The electrification of the heating step in a TSA cycle has gained attention recently [52] and is of particular interest for power-to-X applications such as the CO_2URAGE programme, where renewable electricity is directly available as the primary energy input. Direct electrical heating of the sorbent offers several potential advantages over conventional heat supply through combustion or heat exchange. Energy is deposited directly into the material rather than transferred through a vessel wall or a carrier gas, which can substantially reduce the time required to reach regeneration temperature and therefore shorten the overall cycle time. Shorter cycles are directly beneficial for dynamic operation under a variable renewable electricity supply, since the system can respond more quickly to changes in feed availability. Direct electrical heating also eliminates the need for a separate combustion-based or heat-exchange-based heat source, reducing both the process footprint and the environmental impact of the regeneration step. Four methods have been studied for zeolite regeneration in related applications: microwave heating, radiofrequency (RF) heating, acoustic excitation, and induction heating. None has been tested under SEM-relevant conditions.

In water-loaded zeolites, adsorbed water is the primary microwave absorber at 2.45 GHz: Ohgushi, Komarneni, and Bhalla showed that hydrated zeolite 4A heats rapidly under microwave irradiation, while the same material with little residual water is difficult to heat at all [53]. This creates a self-limiting behaviour: energy transfer diminishes precisely as the bed approaches full regeneration. A second limitation is non-uniform energy deposition due to the shallow penetration depth at GHz frequencies, which favours fluidized bed geometries over packed beds [54]. Kubota et al. confirmed experimentally that standalone microwave regeneration of a zeolite desiccant bed produced desorption ratios substantially below the theoretical prediction due to non-uniform temperature distribution, and found that a hybrid scheme combining microwave and hot-air heating outperformed either method alone [55]. Polaert et al. further showed that microwave desorption behaviour is strongly dependent on the specific adsorbent-adsorbate pair and reactor geometry, and cannot be generalised across zeolite types [56]. Microwave coupling to a pressurised reactor presents a further practical barrier, as it requires microwave-transparent vessel sections and specialised sealing arrangements that standard steel pressure vessels cannot accommodate.

RF heating operates on the same electromagnetic principle as microwave heating but at substantially lower frequencies; Holzer, Kraus, and Roland used 13.56,MHz compared to the 2.45,GHz typical of microwave systems [57]. The lower frequency penetrates deeper into the packed bed, producing more uniform energy deposition throughout the bulk. A further advantage is that the heating mechanism in zeolites at RF frequencies is driven by the movement of charge-compensating cations within the zeolite framework, rather than by absorption of energy by adsorbed water. Zeolite 13X therefore continues to absorb RF energy as the bed dries, directly addressing the self-limiting behaviour that constrains microwave heating [57]. A reactor vessel rated to 30,bar was designed, built, and tested at atmospheric pressure, achieving a residual moisture at the outflow of only 6 ppmv [57].

Daghooghi-Mobarakeh et al. demonstrated that acoustic excitation reduces energy consumption by 24

% and regeneration time by approximately 24 % when applied at an ultrasound-to-total-power ratio of 0.2–0.25 for zeolite 13X [58]. Acoustic excitation uses ultrasound waves to vibrate adsorbed water molecules loose from the zeolite surface. The benefit diminishes rapidly above this ratio, confirming that acoustic excitation supplements rather than replaces thermal input. Integration of ultrasound transducers into a pressurised gas-phase reactor has not been demonstrated.

Induction heating couples an alternating magnetic field to ferromagnetic or conductive particles within the bed, generating heat through eddy currents and through the reorientation of magnetic domains against the material's coercivity [52]. Nickel is ferromagnetic and couples directly to such a field, meaning that bi-functional catalyst–sorbent materials containing a nickel catalyst could in principle be heated at the particle level without the need for a separate susceptor. The approach is at an early stage of development: for adsorption applications it is reported at a technology readiness level of 3–4, and the largest inventories demonstrated to date are on the order of several grams of composite adsorbent, well below the scale of the other methods reviewed here. No demonstration under sorption-enhanced CO_2 methanation conditions has been reported, and the interaction of an induction field with a nickel catalyst, including the risk of localised overheating or sintering, has not been characterised.

Of the four methods, RF dielectric heating is the most promising for SEM applications, given its deeper field penetration, continued energy absorption as the bed dries, and demonstrated feasibility in a pressure-rated vessel. However, all experiments to date used zeolites under near-ambient drying conditions rather than at the water loadings and temperatures relevant to SEM, and none has been tested in the presence of a nickel catalyst. The potential interaction between nickel and the electromagnetic field, including risks of localised overheating or sintering, is entirely uncharacterised. Material-specific feasibility testing is therefore a prerequisite before any of these methods can be seriously considered for SEM, and this falls outside the scope of the present thesis.

3.7. Literature Overview and Gaps

The experimental literature on SEM has established a consistent operating window and a preferred materials combination, but it remains narrow in scope and is dominated by a single reactor type and a single material configuration. Lab-scale fixed-bed experiments demonstrate that near-complete CO_2 conversion and CH_4 purities above 99 vol% are achievable with a Ni-based catalyst and multiple zeolites (Table 3.1). Of these, only the formulation of 5 wt% Ni with 2.5 wt% Ce on zeolite 13X reported by Wei provides simultaneously high conversion, near-complete selectivity, and 100-cycle stability data [13]. Operating temperatures in the 200–400 °C range have been reported, with the 200–300 °C window representing the preferred operating point balancing sufficient reaction rate against adequate sorbent capacity. A stoichiometric H_2/CO_2 feed ratio of 4:1 is adopted across nearly all studies. Higher operating pressure consistently extends the pre-breakthrough period and improves product purity, though most experiments are conducted at atmospheric pressure due to hardware constraints. Where pressure swing regeneration has been evaluated alongside temperature swing, it has been preferred at elevated pressure due to shorter cycle times. The collective record, summarised in Table 3.1, supports a coherent picture of SEM feasibility at lab scale, but is dominated by mixed-bed configurations in fixed-bed tubes with low pressures and cycle counts that fall well short of industrially relevant durations. The bi-functional configuration, which is the primary target of the present thesis and of the CO_2URAGE programme, is represented by only six studies in the table, all in fixed-bed reactors [12, 13, 15, 38, 59]. The gaps that follow directly from this narrow experimental base and that are most relevant to the design developed in Chapter 4 are identified below.

Non-fixed-bed configurations are essential for two reasons. First, they allow continuous solid circulation between a reactor and a separate regeneration vessel, eliminating the multiplied hardware and inventory cost of the parallel fixed-bed vessels otherwise required for continuous production, and providing a clear path toward industrial scale. Second, they offer better management of the heat released by the strongly exothermic methanation reaction, suppressing the hotspots that fixed-bed configurations struggle to avoid. Despite this, fluidised bed SEM has been demonstrated only once, in a batch apparatus at atmospheric pressure using a mixed-bed configuration, without attrition assessment or extended cycling [20]. No experiment combining any non-fixed-bed reactor with a bi-functional material has been performed at any scale, and no engineering-level model of a bi-functional SEM reactor in a non-fixed-bed geometry has been reported. Crucially, no SEM setup with continuous solid circulation

between a reactor and a dedicated regeneration vessel has been demonstrated, which is the configuration required for continuous methane production in a non-fixed-bed geometry. The closest precedent is the work of Coppola et al., who demonstrated solid circulation in a dual fluidised bed system, but using sorbent particles only and without carrying out the methanation reaction itself [45]. The engineering consequence is that the kinetic and adsorption parameters available for any non-fixed-bed SEM model are derived exclusively from fixed-bed experiments, and their validity in a geometry with continuous solid movement has not been independently verified. The reactor design and engineering-level model developed in this thesis provide the conceptual and analytical foundation that must exist before such experiments can be executed.

As noted in Section 3.4.1, the only available mechanical stability data for zeolite 13X under dynamic conditions is that of Sonnleitner et al., and this concerns the base zeolite without metal impregnation. No attrition data exists for any bi-functional SEM material under transport or circulation conditions. Since virtually all SEM experiments have been conducted in fixed-bed reactors where attrition is negligible, the issue has not been encountered experimentally and has therefore not been reported. The engineering consequence is that the attrition resistance of the bi-functional material in any configuration involving solid transport must currently be inferred from standalone zeolite data that may not be representative. If metal incorporation reduces the structural integrity of the zeolite framework, the actual attrition rate could be substantially higher than the Sonnleitner data suggests. This uncertainty must therefore be carried as an explicit assumption in any reactor configuration that involves solid transport or circulation, since the only available attrition data concerns the standalone zeolite rather than the bi-functional material specifically.

Heat integration between the reaction and regeneration stages has been proposed as a route to reducing the electrical energy demand of SEM, but no working implementation exists at any scale. Mancusi et al. modelled the full SEM cycle including regeneration in an adiabatic fixed-bed reactor using a bi-functional Ni/13X material and showed that the heat released during the reaction stage can in principle be used to drive sorbent drying without an external heat input, but this was a modelling result and was not translated into a process design [17]. At a larger scale, the move-MEGA pilot confirmed that the thermal flows between a first-stage conventional methanation reactor and a second-stage SEM reactor were sufficient in principle to drive temperature swing regeneration, but the actual heat routing was again deliberately not implemented [33]. All existing evidence on heat integration in SEM therefore remains either at the modelling level or unimplemented, and is exclusively based on fixed-bed geometries. The engineering consequence is that the extent to which heat recovery from the reaction stages can reduce the net electrical demand of a separate regeneration vessel in a non-fixed-bed configuration remains entirely unquantified. Quantifying this is one of the objectives of the process simulation developed in this thesis.

Most studies in Table 3.1 perform fewer than 15 cycles, which is insufficient to characterise gradual degradation mechanisms. Delmelle et al. reported a 40 % reduction in the effective water diffusion coefficient through the zeolite framework after 39 cycles for a bi-functional Ni/5A material [41]. Since water diffusion is the rate-limiting transport step during regeneration, this progressive reduction would extend the time required for complete sorbent restoration with each successive cycle, directly reducing the productive fraction of each cycle. Whether a comparable reduction occurs in the Ni–Ce/13X formulation has not been investigated, as Wei characterised conversion and selectivity over 100 cycles but did not measure the diffusion coefficient [13]. A further unresolved question is whether CO_2 co-adsorption on the zeolite is a transient effect that stabilises after initial cycling or compounds progressively over hundreds of cycles, gradually reducing the effective water uptake capacity. Both mechanisms are relevant parameters to monitor in the experimental campaign that follows from this thesis.

Another gap in the literature concerns sulphur tolerance, which has been studied only for the catalyst despite the fact that sorbents in a real biogas-fed process would be exposed to the same impurity stream. The desulphurisation requirement established in Section 6.2.4 is therefore justified from the perspective of catalyst protection alone, and its adequacy for sorbent protection remains unconfirmed.

Finally, no alternative electric regeneration concept as discussed in Section 3.6.3 has been tested under SEM-relevant conditions, leaving the feasibility of RF dielectric, microwave, acoustic regeneration or induction as open experimental questions.

Table 3.1: Overview of catalysts, sorbents, configurations (bi-functional catalyst and sorbent or physically mixed), CH_4 yields, CO_2 conversions, feed ratios, space velocities regeneration types and numbers of cycles performed in SEM experiments.

Ref.	Catalyst	Sorbent	Config.	CH_4 yield / CO_2 conv.	H_2/CO_2 ratio	T / P	Space velocity	Cycles / Regen.
[60] (2026)	Commercial Ni/Al_2O_3, Ni5A, Ni/SiAl	Zeolite 3A	Mixed	100 % CO_2 conv.	4:1	150–250 °C 1 bar	110–2750 $Nml_{CO_2}/g_{cat}/h$	5 / H_2/He purge
[50] (2026)	Commercial Ni-based (Haldor Topsoe)	Zeolite 4A	Mixed	100 % CH_4 yield 100% CO_2 conv	4:1	225 °C 9.5 bar	1.0 $kg_C/kg_{cat}/h$	11 / TPSA + H_2/N_2 purge
[40] (2025)	Ni (0–5 wt% CeO_2)/γ-Al_2O_3	Zeolite 13X, 4A, 3A	Mixed	92% CO_2 conv.	4:1; 8:1	250–320 °C 1 bar	n.r.	10 / H_2 + N_2 purge 100 cyc. sorbent only
[20] (2025)	Commercial Ni/Al_2O_3	Zeolite 3A, 4A	Mixed Fluidized bed	—	4:1	200–250 °C 1 bar	n.r.	— / N_2 purge
[49] (2025)	7.5 wt% Ni + 2.5 wt% Fe/γ-Al_2O_3	Zeolite 5A	Mixed	—	2:1 - 5:1	250 °C - 400 °C 1 bar	7.5×10^3 - 1.2×10^5 $ml/g_{cat}/h$	- / TSA + purge
[38] (2024)	Ni	Zeolite 13X	Bifunct.	99 % CH_4 purity 100% CO_2 conv.	5:1	300 °C 10 bar	-	- / TPSA
[39] (2024)	Ni, Rh and Ru-based	Zeolite 4A, 13X	Mixed	99 % CH_4 99.8% CO_2 conv.	4:1	200 - 275 °C 10 bar	509 $Nml/g_{cat}/h$	3 / TSA
[48] (2024)	Commercial Ni-based (Haldor Topsoe)	Zeolite 4A	Mixed	94.8 % CH_4 100% CO_2 conv.	3:1 - 5:1	200–230 °C 10 bar	0.8–1.4 $kg_C/kg_{cat}/h$	– / PSA
[47] (2023)	Commercial Rh-based (Johnson Matthey)	Zeolite 3A, 4A, 5A, 13X	Mixed	85 vol% CH_4 98.7% CO_2 conv.	3:1 - 5:1	200 - 300 °C 1 bar	230 - 300 $Nml/g_{cat}/h$	40 / TSA
[14] 2022	Commercial Rh-based (Johnson Matthey)	Zeolite 5A	Mixed	98.7 % CO_2 85 % CH_4	3:1 - 5:1	200 - 300 °C 1 bar	230 - 300 $Nml_{CO_2}/g_{cat}/h$	10 / TSA
[15] (2022)	5 wt% Ni (lab-synth.)	Zeolite 13X	Bifunct.	Near 100% conv	4:1	200 - 400 °C 5 bar	–	50 / PSA + air purge
[13] (2021)	5 wt% Ni; 5 wt% Ni + 1-10 wt% Ce	Zeolite 5A, 13X	Bifunct.	100 % conv. 100% yield	4:1	180–320 °C 1 bar	923 $ml/g_{cat}/h$	100 / TSA (H_2 purge)
[46] (2021)	12.6 wt% Ni/CaO–Al_2O_3 (Katalco 57-4Q, JM)	Zeolite 4A, La_2O_3, CaO	Mixed	92–94 % H_2 conv.	3.3:1	290 - 345 °C 10–15 bar	n.r.	3 per cat. / TSA
[45] (2021)	— (sorbent only)	CaO, Zeolite 3A	fluidized bed	—	—	200–300 °C 1 bar	n.r.	10 / TSA
[41] (2018)	5 wt% Ni (lab-synth.)	Zeolite 5A, 13X	Bifunct.	—	4:1	320-500 °C 1 bar	n.r.	39 / TSA
[59] (2016)	5 wt% Ni (lab-synth.)	Zeolite 5A	Bifunct.	100% conv.	4:1	270–320 °C 1 bar	100% conv.	– / H2 purge
[12] (2013)	5 wt% Ni (lab-synth.)	Zeolite 5A	Bifunct.	100% yield	8:1	250 °C 1.2 bar	1000 h^{-1} bed vol	-

The TRL-5 fixed-bed pilot of Kiefer, Borgschulte, and Bach [33] at the Empa institute is not included as no peer-reviewed performance data are available.

4

Basis of Design for a Lab-Scale Unit

4.1. Design Strategy

The design of the lab-scale sorption-enhanced methanation unit follows a top-down funnel structure, moving from external constraints to internal design choices. The chapter opens by establishing the boundary conditions that the process must satisfy regardless of any design decision: the product specification imposed by the Dutch gas grid, the programme objectives of CO_2URAGE, and the physical constraints of the TU Delft laboratory.

Within these constraints, the bi-functional material is selected first, since the physical and chemical properties of the solid inventory directly determine which reactor configurations are mechanically feasible and which are not. The reactor type is then selected through a multi-criteria assessment that evaluates six candidate configurations against the requirements established in Section 4.2, with the moving packed bed emerging as the preferred configuration due to its wide operational bandwidth, continuous sorbent circulation, low attrition risk, and counter-current contact pattern. The selected reactor concept is subsequently embedded in a complete process design covering feed conditioning, a first-stage conventional methanation reactor, a condenser, and the moving packed bed second-stage reactor with a separate electrically heated regenerator, presented as a block flow diagram in Section 4.5.

The central research gap motivating this chapter is the absence of any SEM demonstration or model combining a non-fixed-bed reactor with a bi-functional catalyst-sorbent material at any scale, as established in Section 3.7. The literature review presented in Chapters 2 and 3, together with the basis of design developed in this chapter, closes sub-questions 1 and 2: methane is selected as the target product, the moving packed bed is selected as the reactor type, and the bi-functional 5 wt% Ni with 2.5 wt% Ce on zeolite 13X is selected as the solid inventory. The remaining sub-questions 3 through 5, covering reactor performance, integrated process performance and energy balance, and the mechanical design of the full lab-scale process, are addressed in the subsequent thesis chapters.

4.2. Process Requirements and Design Objectives

The design of the lab-scale sorption-enhanced methanation unit is bounded by four sets of requirements: the product specification imposed by the Dutch gas grid injection standard, the programme objectives set by CO_2URAGE, the facility constraints of the TU Delft Process and Energy laboratory, and a set of design objectives against which the final concept will be evaluated. These are discussed in turn below and summarised in Table 4.1.

4.2.1. Product Specification

The target application is injection into the Dutch gas grid. The binding limits are defined by Gasunie Transport Services [34] and the Regeling Gaskwaliteit [61], and are summarised in Table 4.1. The conversion requirements follow directly from two specifications. The most stringent entry point permits a maximum CO_2 content of 1.5 mol%, requiring at least 96% single-pass CO_2 conversion when

using a stoichiometric feed. More demanding is the hydrogen limit. Current regulation permits a maximum of 0.5 mol% H_2 only in the Maasvlakte system, with the remainder of the Dutch network limited to 0.02 mol% [61]. However, the Dutch government is actively exploring raising the hydrogen limit in broader parts of the network, and the 0.5 mol% value is adopted here as a realistic near-term design target, requiring approximately 99.9% H_2 conversion for a stoichiometric feed. The Wobbe index specification is less binding than the explicit CO_2 limit and does not introduce additional conversion requirements. The water dewpoint limit of -8 °C at 72 bar(e) corresponds to a maximum water content of approximately 430 ppm(v) in the compressed product, requiring a dedicated drying step downstream of the reactor, discussed in Section 4.5. Upstream desulphurisation of any biogas-derived feedstock is required to meet the sulphur specifications and to protect the catalyst, as established in Section 6.2.4.

4.2.2. Programme Requirements

The reactor is developed within the CO_2URAGE programme, which sets four requirements that directly shape the design space. First, the target maturity level at the conclusion of this thesis is TRL 4: the reactor concept must be demonstrated under conditions representative of its intended operating environment, not merely as a proof-of-concept. Second, the programme focuses specifically on bi-functional catalyst-sorbent materials combined with electrified regeneration; the reactor must therefore be compatible with a single-particle bi-functional bed and accommodate an electrically driven regeneration step. Resistive heating is used here as a well-understood baseline; particle-level electrical heating methods such as microwave or inductive heating fall outside the scope of this thesis. Third, the reactor must provide a clear path toward scale-up, consistent with the programme's industrial deployment objective. Fourth, the solid inventory must remain within a particle count tractable for particle-resolved CFD validation carried out in parallel within the programme, targeted at below approximately 500 000 particles.

4.2.3. Facility Constraints

Operation within the TU Delft Process and Energy laboratory bounds the design space through constraints on physical space and a pressure-volume-temperature classification threshold. Atmospheric pressure is selected to satisfy these classification constraints while simplifying the setup (see Table 4.1). This precludes pressure swing regeneration, making temperature swing regeneration the only viable option, with a maximum electrical supply capacity of 10 kW. Process gases including H_2, CH_4, and potentially CO are flammable or toxic and must be handled in accordance with laboratory safety protocols.

Table 4.1: Process requirements for the lab-scale SEM unit. Conversions are based on stoichiometric feed.

Parameter	Requirement	Source
Product specification		
CO_2 in product	$<$ 1.5 mol% ($>$96% CO_2 conversion)	[34]
H_2 in product	$<$ 0.5 mol% ($>$99.9% H_2 conversion)	[61]
O_2 in product	$<$ 5 ppm	[34]
Total sulphur in product	$<$ 20 mg m^{-3}(n)	[34]
Water dewpoint	-8 °C at 72 bar(e) ($<$430 ppm(v))	[34]
Hydrocarbon liquid content	$<$ 5 mg m^{-3}(n) at $T \geq -3$ °C	[34]
Wobbe index	13.83–15.00 kWh m^{-3}(n)	[34]
Programme requirements		
Technology readiness	TRL 4	CO_2URAGE
Material configuration	Bi-functional catalyst-sorbent	CO_2URAGE
Regeneration mode	Electrically driven TSA	CO_2URAGE
CFD tractability	$<$ 500 000 particles	CO_2URAGE
Facility constraints		
Maximum operating pressure	$\approx$1.1 bar	†
Electrical supply	10 kW	†

† Based on discussions with the project supervisor.

4.2.4. Design Objectives

Beyond the hard constraints above, the design is evaluated against four objectives that directly map onto the reactor selection criteria developed in Section 4.4.4.

The first design objective is dynamic operational flexibility, requiring that the reactor maintains acceptable performance across a wide range of feed flow rates and can reach its target operating condition and return to a safe idle state within a short time. This objective is motivated directly by the intermittent nature of the renewable hydrogen supply described in Chapter 1, and is operationalised through the operational bandwidth and start-up time criteria in the reactor comparison.

Capital and operating cost favours a compact, low-complexity design that minimises both investment and running costs. Attrition resistance and heat management determine how quickly particles degrade or sinter, driving material replacement frequency and therefore operating expenditure. Sorbent capacity usage determines the required circulation rate and solid inventory, which set equipment size and capital expenditure. All three map onto corresponding criteria in the reactor comparison.

Scale-up representativeness requires that the geometry, operating conditions, and instrumentation of the lab-scale concept are representative of a larger unit, avoiding shortcuts that would not translate to industrial scale. Continuous sorbent circulation is the primary feature distinguishing a scalable concept from a batch lab demonstrator, and is evaluated through the continuity of production criterion.

Energy efficiency requires that the integrated process minimises electrical energy input per unit of methane produced. This objective depends on the full process configuration rather than the reactor type alone, and is evaluated in Section 4.5.

4.3. Bi-functional Material Selection

The bi-functional configuration, in which the catalytic and water-adsorbing functions are integrated within a single particle, is adopted as the material configuration for this work. Of the bi-functional SEM formulations reported to date, the material developed by Wei [13] is selected as the baseline for this work. It is the only bi-functional SEM formulation for which conversion, selectivity, and long-cycle stability data are simultaneously available: 5 wt% Ni with 2.5 wt% Ce supported on zeolite 13X maintained around 100% CO_2 conversion and CH_4 selectivity over 100 adsorption-regeneration cycles totalling 223 hours on stream, with only a slight reduction in water uptake capacity [13]. The material also provides kinetic and adsorption data that can be used to validate the reactor model. The key material properties are summarised in Table 6.1.

Based on a particle diameter range of 0.5–1 mm and a particle density of approximately 689 kg/m^3 for the zeolite 13X support [58], the bi-functional material falls within the Geldart B classification, a system that groups solid particles by their fluidisation behaviour based on size and density [62, p. 78]. This confirms that bubbling fluidised bed operation is feasible with the material as synthesised without any size reduction. Geldart B solids are characterised by vigorous bubbling immediately above the minimum fluidisation velocity, which is beneficial for solid mixing and heat management but increases gas bypass through the bubble phase compared to finer Geldart A particles [62, p. 68]. Entering the fast fluidisation regime required for riser operation demands particles roughly below 300 µm [62, p. 78]; whether milling the material to this size is achievable without degrading the zeolite pore structure, the nickel dispersion, or the water adsorption capacity has not been investigated in literature and represents an open engineering question.

Table 4.2: Key properties of the bi-functional 5 wt% Ni 2.5 wt% Ce / zeolite 13X material used in this work. All values from Wei [13] unless stated otherwise.

Property	Value	Unit
Particle diameter range[1]	0.5–1	mm
BET surface area (reduced)	611	m^2/g_{cat}
Total pore volume (reduced)	0.21	cm^3/g_{cat}
Water uptake capacity	$\approx$1.3	$mmol/g_{cat}$
Particle density (zeolite 13X only) [35]	1400	kg/m^3

4.4. Reactor Type Selection

For the reactor type selection, three categories are considered: fixed bed reactors, in which the solid inventory is stationary; fluidised bed reactors, in which solids are suspended and can be circulated; and the moving packed bed reactor, in which a packed bed of solids descend continuously under gravity. Parallel fixed bed and parallel bubbling fluidised bed configurations are included in the comparison as reference configurations representing the current state of practice in SEM, against which continuous configurations can be benchmarked. Within each category, relevant configurations are discussed and subsequently evaluated.

A staged reactor configuration partially addresses both the thermal and sorbent inventory constraints that apply to all reactor types discussed in this section. In this arrangement, a first reactor stage without sorbent achieves approximately 80 % conversion, after which the product stream is cooled and condensed water is removed before entering a second, sorption-enhanced stage [13, 15, 33]. This split reduces the total sorbent inventory requiring regeneration and allows the first stage to operate at higher temperatures where reaction kinetics are faster, without penalising adsorption capacity in the second stage. In principle, the heat released in the first stage can supply the energy demand of sorbent regeneration in the second stage, improving the overall energy balance. This concept was adopted as the basis for the most advanced SEM demonstration to date, the TRL-5 move-MEGA pilot at the Empa institute, a 50 kW two-stage fixed-bed plant operating at 50 bar, where the thermal flows between stages were measured and confirmed to be sufficient in principle to drive regeneration, though the actual heat routing was not implemented in the demonstration plant [33]. The staged configuration is therefore assumed throughout the reactor comparisons that follow.

4.4.1. Fixed Bed Reactors

The large majority of SEM experiments reported in literature have been performed in fixed bed reactors, primarily due to their simplicity and low construction cost at lab scale [3]. In a fixed bed, the catalyst and sorbent are packed as a stationary solid inventory through which the feed gas passes axially. The well-defined flow path, predictable concentration profile, and straightforward instrumentation make the fixed bed the natural starting point for proof-of-concept studies.

The principal limitation of the fixed bed in SEM is its inherently batch operation: once the sorbent approaches saturation, the reactor must be taken offline for regeneration. Therefore, continuous methane production requires at least two reactors operated in parallel, alternating between adsorption and regeneration phases [3, 35]. Mancusi et al. showed through simulation that a minimum of three parallel reactors is required when using temperature swing to regenerate zeolite based sorbent, since the regeneration takes longer that the reaction [17].

Heat management presents a further challenge. Under adiabatic conditions, axial temperature gradients of up to 100 K have been reported in simulation near the inlet [35] due to the exothermicity of the Sabatier reaction . Isothermal operation suppresses these excursions but requires mechanically complex cooling arrangements such as multi-tubular heat exchangers. Borgschulte et al. proposed a plate reactor concept in which alternating reaction and regeneration plates are placed adjacent to one another, so that the heat released by the exothermic reaction drives regeneration of the neighbouring plate directly [59]. This concept could in principle resolve both the heat management and the regeneration energy challenges simultaneously and offers a scalable geometry, but no experimental demonstration has been reported to date.

4.4.2. Fluidised Bed Reactors

Fluidised bed reactors suspend the solid inventory in an upward-flowing gas stream, enabling intensive solid mixing that produces near-isothermal operating conditions and suppresses the hot spots that are a persistent concern in fixed-bed SEM. Furthermore, they offer the possibility of continuous sorbent circulation between separate reaction and regeneration vessels [3, 45].These advantages come with inherent limitations, including gas bypass through bubbles, back-mixing of solids, and particle attrition risks [62, p.11]. Four fluidised bed configurations are considered below, followed by a brief discussion of reactor types that were considered but discarded at an early stage.

The fluidised bed configuration for SEM has received considerably less experimental attention than the

fixed bed. Coppola et al. experimentally evaluated CaO and zeolite 3A as sorbents in a dual fluidised bed apparatus under simulated SEM conditions, though without carrying out the methanation reaction itself [45]. Massa, Coppola, and Scala subsequently provided the first proof-of-concept demonstration of full SEM in a fluidised bed reactor using zeolite sorbents, reporting methane production enhancements of 22–66 % relative to conventional methanation depending on temperature and sorbent type [20]. Beyond these two studies, the available evidence consists primarily of computational work [63, 64] and analogous proof-of-concept experiments for sorption-enhanced methanol synthesis [21, 19].

Circulating Fluidised Bed with Riser

In a circulating fluidised bed with riser, the feed gas entrains solid bed material upward through a riser at high velocity in the fast fluidisation regime. Solids are separated from the product gas in a cyclone and conveyed by gravity to a regeneration vessel, from which they are returned base of the riser. A key practical advantage of this configuration is that the riser itself provides the upward solid transport, eliminating the need for a separate mechanical conveying system between reactor and regenerator. The continuous high-velocity particle-wall and inter-particle impacts in both the riser and the cyclone make this the most mechanically aggressive environment of all configurations considered, posing a severe attrition risk for the bi-functional Ni–Ce/13X material given the fragility of zeolite 13X under fluidisation conditions reported by Sonnleitner et al. [42].

The more fundamental limitation is the sorbent inventory available in the riser at any moment. In the fast fluidisation regime, the solid volume fraction in the dilute upper section of the riser is only approximately 4 % [62, p. 87, Ch. 8], meaning that the instantaneous water capture capacity of the riser is inherently low regardless of the solid circulation rate. A quantitative assessment detailed in Appendix B shows that the required solid mass flux to match the water production rate of the reaction exceeds the achievable value by more than two orders of magnitude, even when a two-stage reactor configuration is assumed in which a conventional first stage has already converted approximately 80 % of the CO_2 feed. This gap cannot be closed by any practical combination of gas velocity and particle size within the fast fluidisation regime. The riser is therefore not a viable reactor configuration for SEM with the bi-functional Ni–Ce/13X material and is not considered further.

Circulating Bubbling Bed

A circulating bubbling fluidised bed combines a bubbling fluidised bed (BFB) reactor for the sorption-enhanced reaction phase with a separate regeneration vessel, connected by continuous solid transfer, as shown in the left reactor in Figure 4.1. The vigorous solid mixing in the bubbling regime is the defining advantage of this configuration: heat released by the exothermic methanation reaction is rapidly distributed throughout the entire bed, producing near-isothermal conditions and suppressing the formation of hot spots that risk sintering the catalyst or reducing the local sorbent capacity. This makes it the strongest configuration for heat management of all those considered. The mean solid residence time in the reactor is controlled independently by the withdrawal rate at the bottom of the vessel, though the gas velocity must remain above the minimum fluidisation velocity to sustain the bubbling regime, which sets an upper bound on the achievable gas–solid contact time and a corresponding lower bound on the minimum feed flow rate. The bi-functional Ni–Ce/13X material falls in the Geldart B classification, as established in Section 4.3, confirming that bubbling fluidisation is achievable with the material as synthesised without size reduction.

Particle attrition is a significant concern for this configuration. As established in Section 3.4, zeolite 13X shows significant fines formation after only a few hours of bubbling fluidisation, and no attrition data exists for the bi-functional material specifically. Although the collision velocities are substantially lower than in the riser, the continuous low-energy inter-particle collisions inherent to the bubbling regime cause progressive mechanical degradation of brittle particles over many operating cycles [62, p. 9]. The solid transport lines connecting the reactor and regenerator add further mechanical load.

Parallel Bubbling Fluidised Beds

A configuration of two or more bubbling fluidised beds operated in parallel, alternating between methanation and regeneration, avoids the need for continuous solid transfer between vessels. Since catalyst and sorbent remain in the same vessel throughout the cycle, no mechanical solid transport is required, which reduces system complexity and eliminates the associated attrition risks. The principal limitation

is the same as for the parallel fixed bed: batch operation requires multiple alternating vessels and the total solid inventory is much larger.

Trickle Flow Reactor

In a gas-solid trickle flow reactor, freshly regenerated particles are fed continuously at the top of a vertical vessel and descend individually through an upward-flowing gas stream at a velocity close to their terminal settling velocity, as shown in Figure 4.1 Internal structures, typically perforated trays, can be placed at intervals along the column height to slow the descending particles, increasing their residence time and consequently the solid holdup.

An advantage for the trickle flow reactor is its counter-current contact pattern between gas and solid, which is absent in both the riser and the bubbling bed. Freshly regenerated sorbent enters at the top where the gas, having already reacted along most of its path, is most depleted in the target species. At the gas inlet at the bottom, where the reaction rate is highest, the sorbent is most saturated and therefore least reactive, which distributes the heat release more evenly along the reactor axis than co-current configurations. The mechanical forces on individual particles remain low due to the slow descent velocity, giving the configuration inherently low attrition inside the vessel itself [65].

The principal limitations are a narrow operational bandwidth and sensitivity to particle size. Gas velocity must remain close to the terminal settling velocity throughout the bed: excess velocity entrains particles upward while insufficient velocity causes uncontrolled descent, both reduce contact time whichcan lead to incomplete conversion. Dynamic operation with varying feed conditions is therefore inherently difficult. Particle size is a further constraint, since particles must be large enough that the upward gas flow does not elutriate them, setting a lower bound on diameter that depends on superficial gas velocity and solid density. If attrition in the transport lines or regenerator reduces particle size below this threshold, particles are elutriated from the top of the reactor rather than descending through it, reducing solid inventory and capture efficiency.

Other Configurations Considered

Two further reactor types were considered and discarded. Slurry bed reactors suspend the catalyst in a liquid phase, offering excellent heat transfer, and are used in conventional three-phase methanation [66], but the liquid fills the zeolite pores and destroys the sorption enhancement, while regeneration would require either removing the solid from the liquid phase or heating the entire liquid inventory to temperature swing conditions. Vortex reactors use centrifugal forces to separate gas and solid phases at high rotational velocities, providing very short gas–solid contact times that are incompatible with the residence times required for near-complete single-pass CO_2 conversion in SEM. Neither configuration is considered further.

4.4.3. Moving Packed Bed Reactor

In a moving packed bed reactor, shown in the right of Figure 4.1, solids descend as a continuous touching column under gravity while gas flows upward through the interstitial voids, producing a steady-state concentration and temperature profile in both phases [67]. The solid descent rate is set entirely by a withdrawal device at the bottom of the bed, independent of the gas velocity, which is kept below the minimum fluidisation velocity to maintain packed bed conditions. This decoupling gives the process two independent control handles: the gas velocity can be reduced to very low values to increase contact time and approach complete conversion, while the solid withdrawal rate is adjusted separately to control the mean sorbent loading. The result is an exceptionally wide operational bandwidth that is not constrained by a minimum fluidisation requirement.

The counter-current contact pattern introduced in Section 4.4.2 for the trickle flow reactor applies equally here. Particles approach their breakthrough adsorption capacity progressively as they descend, since they always encounter gas with a higher water content than their current equilibrium loading. The space-time yield benefit of this arrangement is substantial: Munera Parra et al. demonstrated through modelling that a moving packed bed yields a space-time yield approximately one order of magnitude higher than an optimised fixed bed for the same reactor volume in sorption-enhanced reverse water-gas shift [67]. Despite the moderating effect of counter-current solid flow on the temperature peak at the gas inlet, axial temperature gradients still develop in the gas phase and heat management remains an engineering challenge, particularly at larger reactor diameters where radial heat transport becomes

insufficient to prevent localised overheating. Attrition within the reactor is low, as the slow descent velocity produces only compressive contact forces at the base of the column with no bubble-induced collisions. Solid transport between the reactor and the regeneration vessel introduces additional mechanical loading at bends and valves, which remains a risk that must be managed through careful pipe routing, but it is substantially lower than in any fluidised transport configuration. A withdrawal device such as a rotary valve or screw conveyor is required to control the solid descent rate [68].

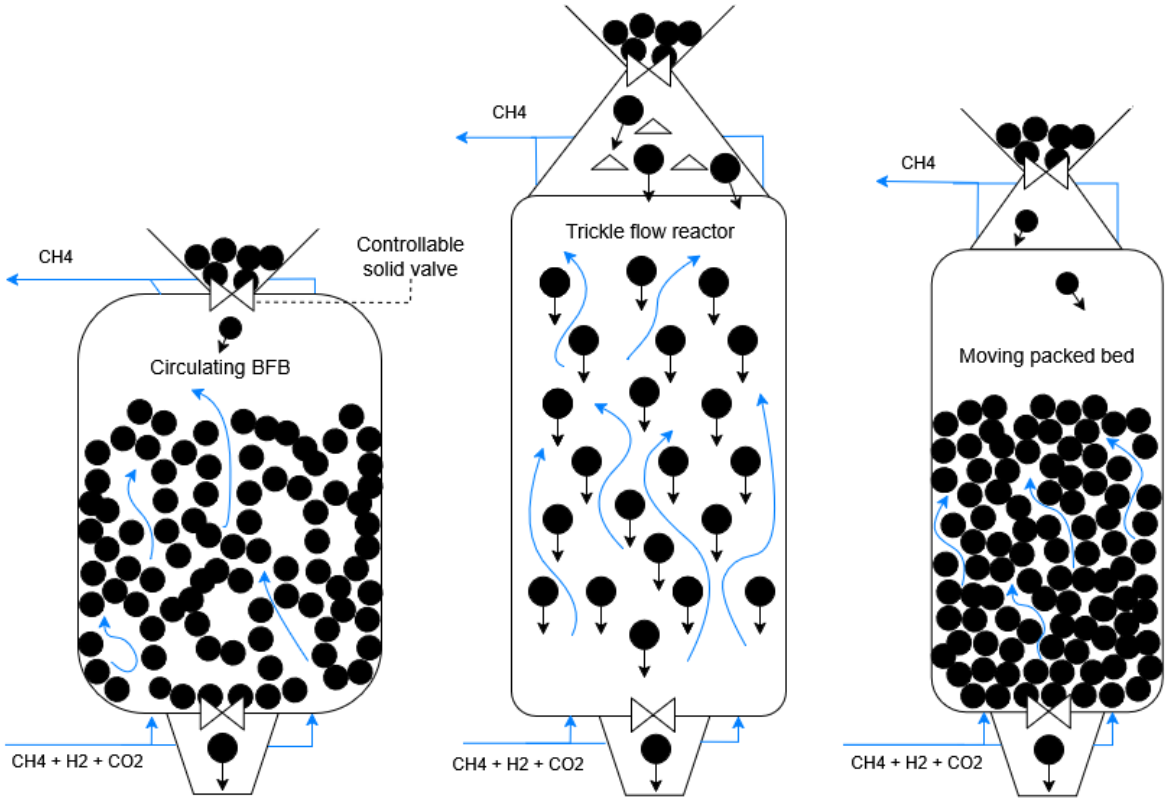


Figure 4.1: Schematic representation of the three continuous reactor configurations considered for sorption-enhanced methanation: circulating bubbling fluidised bed (left), trickle flow reactor (centre), and moving packed bed (right). Black spheres with arrows indicate the solid flow; blue arrows indicate gas flow. All configurations are shown as second-stage reactors receiving the partially converted outlet of a first-stage conventional methanation reactor, with CH_4 + H_2 + CO_2 as the inlet stream and CH_4 as the product stream at the top.

4.4.4. Reactor Selection

This section discusses the selection of the reactor type through a structured multi-criteria comparison of six candidate configurations. Table 4.3 summarises the results, with each criterion defined and its weight justified below, after which all six configurations are scored simultaneously. The section concludes by identifying and justifying the selected reactor concept.

Operational Bandwidth

The operational bandwidth criterion is assigned the highest weight because the reactor must remain functional across a wide range of feed flow rates, reflecting the intermittent and variable nature of a renewable electricity supply. Two requirements are captured jointly. First, the ability to reduce production to a low fraction of the design capacity without losing stable reactor operation: since the system is sized for full-load operation and a reduction in hydrogen supply must not force a complete shutdown, low-end turn-down is the more critical requirement across all configurations. The second requirement is the ability to achieve a sufficiently low superficial gas velocity to approach complete CO_2 conversion in a single pass: lower velocities extend the contact time between the gas phase and the solid bed, increasing the conversion per pass through the reactor.

The parallel fixed bed scores highest, as it can operate at any superficial gas velocity above zero and

is not subject to a minimum or maximum fluidisation requirement. The moving packed bed also scores highest: gas velocity can be reduced to arbitrarily low values since the minimum fluidisation velocity is a ceiling rather than a floor, and packed bed conditions are preserved across the full turn-down range. The parallel BFB and circulating BFB both require a minimum superficial gas velocity to maintain the bubbling regime; below this threshold the bed defluidises and hydrodynamic uniformity is lost. This sets a lower bound on the achievable partial load range, unless an inert diluent is used to maintain fluidisation at reduced feed rates, which adds process complexity. Both score moderately. The trickle flow reactor is more tightly constrained: the gas velocity must remain close to the terminal settling velocity of the particles throughout the bed height, since excess velocity entrains particles upward and insufficient velocity causes uncontrolled descent or settling. This narrow operating band restricts both the load range and the achievable contact time. The riser scores lowest, requiring continuous fast-fluidisation velocities that produce the shortest possible gas–solid contact time and a limited operational bandwidth.

Continuity of Production

The continuity of production criterion is assigned a high weight because batch operation not only interrupts methane delivery but also forces a larger total solid inventory to compensate for the time the reactor spends offline during regeneration, both of which are direct obstacles to scalable fuel production. The circulating BFB, riser, moving packed bed, and trickle flow reactor all enable steady-state methane production through continuous sorbent circulation to a separate regeneration vessel and score highest, while the parallel fixed bed and parallel BFB require periodic switching between adsorption and regeneration phases and score lowest, with a minimum of three parallel units required when regeneration takes longer than the adsorption phase [17].

Attrition Resistance

Attrition resistance is assigned a high weight because progressive particle size reduction would alter bed hydrodynamics, reduce the active solid inventory over time, and ultimately require more frequent material replacement. As established in Section 3.4, the only available mechanical stability data for zeolite 13X under fluidisation conditions shows a 4% mass loss after 32 hours of bubbling, with visible fines formation after just 5 hours [42]. No attrition data exists for the bi-functional material in any configuration involving solid transport, therefore the attrition of the bi-functional material is assumed to be similar.

The parallel fixed bed scores highest, as the stationary bed is subject only to compressive forces from the weight of particles above, with no inter-particle collisions or particle-wall impacts driven by gas flow. The moving packed bed and trickle flow reactor score equally: in the moving packed bed, gravity-driven solid descent produces low compressive contact forces at the base of the column but no bubble-induced collisions; in the trickle flow reactor, individual particles descend slowly under near-terminal-velocity conditions with minimal contact forces between them. Both configurations score lower than the parallel fixed bed because the solid transport lines connecting the reactor to the regeneration vessel introduce additional mechanical loading. The parallel BFB scores lower than these three configurations because the bubbling regime produces continuous low-energy inter-particle collisions throughout the bed height, which accumulate over many operating cycles even though no solid transport is involved. The circulating BFB adds the mechanical loading of pneumatic or mechanical solid transport between two vessels, increasing the attrition relative to the parallel BFB. The riser scores lowest, since it has the highest gas velocities of all configurations with repeated particle-wall and inter-particle impacts.

Heat Management

This criterion captures the ability of the reactor to avoid local temperature excursions that could drive sintering of the active metal or reduce the water adsorption capacity of the zeolite. It is assigned a moderate weight because the consequences of poor heat management, while serious, can be partially mitigated through good engineering at the design stage: for example, the exothermic heat released during the methanation reaction can be routed to supply part of the regeneration energy demand, reducing the temperature swing required in the reactor itself, as discussed in Section 3.6.2.

The parallel BFB and circulating BFB score highest: vigorous solid mixing in the bubbling regime distributes heat rapidly throughout the entire bed, suppressing the formation of hot spots and producing near-isothermal conditions. The trickle flow reactor scores second, benefiting from two independent

heat management advantages simultaneously: its dilute solid loading limits the absolute heat release per unit reactor length, and its counter-current solid flow distributes that heat release evenly along the axis. In counter-current configurations, freshly regenerated sorbent enters the reactor where the gas has already reacted and carries little remaining water, while nearly saturated sorbent exits where fresh feed gas enters and the reaction rate is highest. The most reactive solid therefore never contacts the most reactive gas, moderating the peak temperature compared to configurations where fresh solid and fresh feed gas meet at the same location. The moving packed bed benefits from the same counter-current effect but scores lower than the trickle flow reactor because its dense solid inventory produces a higher heat release per unit reactor length. The riser scores equally to the moving packed bed: its dilute loading limits the absolute heat release per unit length, but its co-current gas–solid flow concentrates that heat release at the inlet where both fresh gas and fresh solid enter simultaneously. The parallel fixed bed scores lowest, as its stationary solid inventory causes the reaction front and associated temperature peak to remain at a fixed axial position throughout the adsorption phase, with no mechanism for redistribution.

Sorbent Utilisation

The sorbent utilisation criterion captures how efficiently the available adsorption capacity of the solid inventory is utilised before regeneration. Withdrawing sorbent before it approaches saturation wastes solid inventory and requires a higher solid circulation rate, increasing both equipment size and attrition risk. The criterion is assigned a low weight because partially loaded sorbent carries less water into the regenerator, so the regeneration energy per cycle scales with the actual water load rather than the theoretical maximum capacity, partially offsetting the cost of incomplete adsorpbion capacity utilisation.

The riser scores lowest: as established in Section 4.4.2, the solid mass in the reactor has insufficient sorption capacity to match the water production rate of the reaction. The moving packed bed and parallel fixed bed score highest: in both configurations the solid inventory is retained until it approaches saturation before regeneration begins. The moving packed bed gains an additional advantage from its counter-current contact pattern: at the reactor outlet where the gas has nearly reached full conversion and its water content is lowest, the solid is freshly regenerated, ensuring that a high adsorption driving force is maintained even where the thermodynamic potential for further reaction is smallest. The BFB configurations and trickle flow reactor score moderately. In BFB configurations, the well-mixed inventory means the solid must be withdrawn at an intermediate mean saturation state, well below breakthrough, to maintain an adequate driving force across the bed. The trickle flow reactor shares the counter-current driving force advantage of the moving packed bed but cannot guarantee particles approach breakthrough capacity within a single pass through the vessel when reactor length is limited.

Start-up Time

Start-up time captures the time required to bring the reactor from a cold idle state to the operating temperature window of 200–300 °C, which is directly relevant when the feed has been interrupted, for example during periods without wind or solar generation, and must be resumed quickly. This criterion is assigned a low weight because a wide operational bandwidth reduces the frequency of required full shutdowns, meaning the reactor rarely needs to be brought from cold to operating temperature. Since a buffer of freshly regenerated sorbent can be held in reserve, meaning the regenerator does not need to be started simultaneously with the reactor, the bottleneck during start-up is the time required to heat the reactor vessel and its solid inventory to the required operating temperature.

The parallel BFB and circulating BFB score highest: vigorous solid mixing distributes the heat input uniformly and the compact, dense bed geometry minimises the thermal mass per unit of production capacity. The parallel fixed bed and moving packed bed score moderately: both are densely packed, but the absence of solid mixing means a temperature front must propagate axially through the bed before uniform operating conditions are established. The trickle flow reactor and riser score lowest: both require large vessel volumes to accommodate their dilute solid loadings, increasing the total thermal mass to be heated before stable operation can begin.

Reactor Selection

Table 4.3 shows that the moving packed bed achieves the highest weighted total of 4.30, driven by its top scores on the two highest-weighted criteria. On contact time flexibility, its decoupled gas velocity and

solid withdrawal rate provide a wide operational bandwidth without a minimum fluidisation requirement, directly supporting dynamic operation under a variable hydrogen supply. On continuity of methane production, continuous sorbent circulation eliminates the batch switching that forces the parallel fixed bed and parallel BFB to operate multiple vessels in parallel. The moving packed bed further benefits from its counter-current contact pattern, which moderates peak temperatures and maximises sorbent utilisation simultaneously, and from its low attrition risk relative to the fluidised configurations.

The moving packed bed is followed by the circulating BFB at 3.70, which scores well on continuity and heat management but is penalised by its higher attrition risk. The trickle flow reactor totals 3.40, the parallel fixed bed 3.30, and the parallel BFB 3.10. The riser scores lowest at 2.25.

The result is robust with respect to the choice of weights: assigning equal weights of 16.7 % to all six criteria, the moving packed bed retains its lead with a score of 3.83 against 3.67 for the circulating BFB, confirming that the selection is not driven by the weighting choices. The moving packed bed is therefore selected as the reactor configuration for the lab-scale SEM unit. The following section presents the complete process in which it is embedded, covering all units from feed conditioning to product outlet.

Table 4.3: Multi-criteria comparison of reactor configurations for sorption-enhanced methanation. Scores are on a scale of 1 to 5, where 5 is most favourable.

Criterion	Wt.	Par. Fixed	Par. BFB	Circ. BFB	Riser	Mov. Packed	Trickle Fl.
Operational bandwidth	25 %	5	3	3	1	5	2
Continuity of production	20 %	1	1	5	5	5	5
Attrition resistance	20 %	5	3	2	1	4	4
Heat management	15 %	1	5	5	3	3	4
Sorbent utilisation	10 %	4	3	3	1	5	3
Start-up time	10 %	3	5	5	1	3	2
Weighted total	100 %	**3.30**	**3.10**	**3.70**	**2.10**	**4.30**	**3.40**

4.5. Block Flow Diagram

The conceptual process, shown in Figure 4.2, comprises six functional sections. A stoichiometric H_2/CO_2 feed at a molar ratio of 4:1 is metered, mixed, and preheated before entering the first-stage reactor, which operates at approximately 300–400,°C and achieves around 80 % CO_2 conversion under conventional methanation conditions without sorbent. If biogas is used as feedstock, upstream desulphurisation is required at this point to protect the nickel catalyst, as established in Section 6.2.4. The partially converted stream is then cooled in a condenser, where bulk liquid water is removed, reducing the water load on the second stage and extending the pre-breakthrough period of the sorbent. The cooled stream enters the moving packed bed reactor from the bottom, while freshly regenerated catalyst-sorbent is introduced continuously from the top, producing a counter-current contact pattern in which the solid approaches saturation progressively as it descends; near-complete CO_2 conversion is targeted during this stage at 250–300 °C. Spent catalyst-sorbent withdrawn continuously from the reactor bottom is transferred to the electrically heated regenerator, where temperature swing at 300–400 °C in a dry purge gas, most likely nitrogen, drives off the accumulated water; the regenerated solid is then returned to the reactor top to close the solid loop. Heat released by the first-stage reaction is available to partly offset the electrical demand of the regenerator. The methane-rich stream leaving the reactor top may still contain residual water vapour above the dewpoint limit specified in Table 4.1; a downstream drying step, such as a packed bed of silica gel or zeolite at near-ambient temperature, may therefore be required before the product meets grid injection quality.

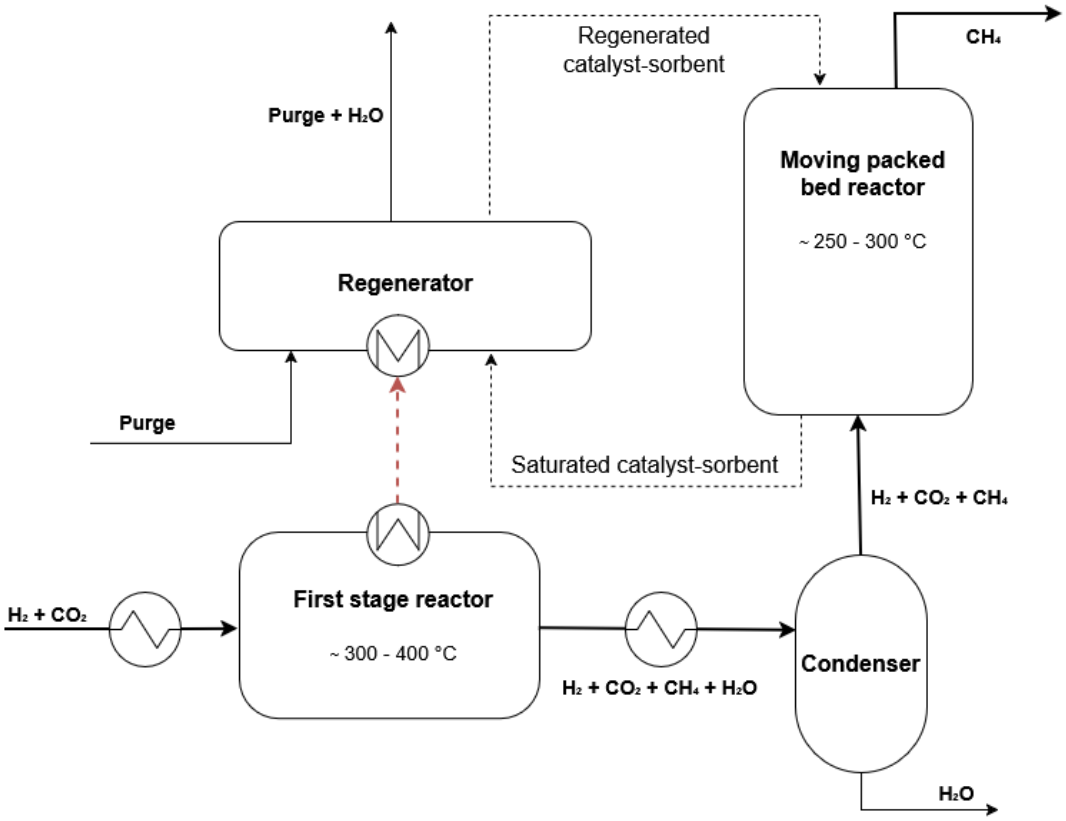


Figure 4.2: Block flow diagram of the two-stage sorption-enhanced methanation process. Dashed arrows indicate solid circulation; the red dashed arrow indicates heat transfer from the first stage to the regenerator.

5

Conclusions and Research Plan

The literature review and basis of design together close the first two sub-questions of the central research question. Chapter 2 selected methane as the target product through a weighted multi-criteria assessment of methane, dimethyl ether, methanol, and carbon monoxide. The decisive advantage is downstream process complexity: sorption enhancement applied to methanation delivers a stream that is already close to the Dutch gas grid injection specification with only water removal required, while the three alternatives all require multi-step downstream separation regardless of the single-pass conversion achieved. Methane is also directly compatible with the existing gas grid.

Chapter 3 confirmed the bi-functional configuration as the most promising material concept and identified the catalyst-sorbent developed by Wei, consisting of 5 wt% Ni and 2.5 wt% Ce supported on zeolite 13X, as the most mature option, with full conversion shown over 100 cycles. The review also revealed a structural gap in the experimental record: the literature is dominated by fixed-bed reactors, even though non-fixed-bed configurations are essential for scaling beyond batch operation and for managing the heat released by the strongly exothermic methanation reaction. No demonstration or model combining a non-fixed-bed reactor with a bi-functional material has been reported, and heat integration between the exothermic reaction stage and the regeneration stage remains unquantified outside fixed-bed geometries.

Chapter 4 selected the moving packed bed reactor through a weighted comparison of six reactor configurations. The moving packed bed supports continuous solid circulation, minimises attrition risk, and provides a counter-current contact pattern that simultaneously moderates the peak temperature and maximises sorbent utilisation. The reactor is embedded in a two-stage process configuration with a conventional first-stage reactor, intermediate cooling and water removal, the moving packed bed second-stage reactor, and a separate regenerator.

Three sub-questions remain open and are addressed in the subsequent thesis chapters. Sub-question 3, the predicted reactor performance under varying operating conditions, is addressed through an engineering-level model of the moving packed bed reactor implemented in Python. The model couples the reaction and adsorption behaviour and predicts the CO_2 conversion, methane purity, temperature profile, required solid circulation rate, and reactor mass and energy balance. These outputs provide the basis for sizing the lab-scale reactor.

Sub-question 4, the configuration and energy balance of the surrounding process, is addressed through a process simulation in Aspen Plus. Heat integration between the exothermic reaction stages and the regeneration stage has been proposed in literature but has not been implemented or quantified at lab scale for any SEM configuration, making it an open question of direct relevance to the energy efficiency of the process.

Sub-question 5, the mechanical design of the lab-scale process, is expressed in basic construction drawings. The model and mechanical design together form the basis for the TRL 4 experimental unit to be built within the CO_2URAGE programme.

6

Reactor Model

6.1. Scope and Objectives

This chapter develops the engineering-level reactor model that answers the third research sub-question: the predicted performance of the moving packed bed reactor under varying operating conditions. The deliverable is a model used to (i) size the lab-scale reactor, (ii) characterise its feasible operating envelope across the dynamic load range, (iii) describe the characteristic conversion, composition, and temperature profiles along the reactor, and (iv) produce the reactor mass and energy balance.

The model is intentionally restricted to engineering-level fidelity: a one-dimensional, steady-state, pseudo-homogeneous description of the coupled reaction, adsorption, and heat transport along the reactor axis. This resolution is chosen to be inexpensive enough to scan a wide range of operating points while still capturing the phenomena that govern the design. Three-dimensional CFD of the reactor hydrodynamics and particle-level modelling of intraparticle transport are carried out in parallel by other partners within the CO2URAGE programme and fall outside the scope of this thesis.

Because no moving packed bed SEM experiment exists in the literature, the model cannot be validated end-to-end against data. Validation is therefore carried out at the level of the constitutive sub-models: the kinetic and adsorption parameters are validated against the fixed-bed experiments of Wei [13] and then carried unchanged into the moving bed model. The assembled reactor model is not claimed to be validated; it is instead verified for internal consistency (mass and energy closure, limiting-case behaviour). Full experimental validation of the moving bed is the future work that this design enables.

The remainder of the chapter is structured as follows:

1. the design basis and modelling scope, fixing reactor geometry, temperature, material properties and feed composition (Section 6.2);
2. the assumptions made in the model (Section 6.3);
3. the model formulation, governing equations, and the numerical solution method (Section 6.4);
4. validation of the kinetic and adsorption sub-models (Section 6.5).

6.2. Design basis and modelling scope

This section establishes the reactor's basic design features and operating envelope, providing the fixed inputs for the model developed in the remainder of this chapter: the operating temperature (Section 6.2.1), the cooling strategy (Section 6.2.2), the reactor geometry (Section 6.2.3), the feed composition (Section 6.2.4), and the reactor and material properties (Section 6.2.5).

6.2.1. Temperature

Inlet temperature is set at 280 °C, a reasonable trade-off between the reaction kinetics, which favour high temperature, and the sorbent capacity and equilibrium conversion, both of which favour low tem-

perature. This is somewhat higher than most literature conditions (see Table 3.1; at TRL 4, prioritising faster kinetics for the sake of scaling is a reasonable choice for a first demonstration. Mercader et al. [49] found 300 °C to be optimal under their conditions; however, because of the temperature rise that occurs within the reactor, the inlet temperature set here is lower than the average bed temperature, which in practice settles closer to this reported optimum.

6.2.2. Cooling

The reactor must be held close to its design temperature. Too high a temperature reduces the water uptake capacity of the sorbent and risks sintering or destruction of the catalyst, while too low a temperature slows the kinetics below a useful rate. Because the methanation reaction is strongly exothermic and water adsorption on the zeolite adds a second exothermic contribution, active cooling is required; without it, axial temperatures can rise by around 100 K [35] in a fixed bed reactor of similar size. In the moving packed bed this released heat is offset by three sinks: cooling at the wall, heat carried out by the solid flux and heat carried out by the gas.

Two simplifications represent the cooling in the model. The reactor wall is held at the inlet temperature of 280 °C, and the overall bed-to-coolant heat transfer coefficient is taken as a constant $h = 25\ \mathrm{W\,m^{-2}\,K^{-1}}$. The controlling resistance is the transfer between the packed bed and the wall: the metal wall itself has negligible thermal resistance and its outer surface is actively cooled with thermal oil, so the wall can reasonably be treated as isothermal at the coolant temperature. The coefficient is set deliberately low. It is a conservative estimate that avoids overstating the achievable cooling, and a modest value also preserves the axial temperature gradients that are the object of study, since a more aggressively cooled reactor would suppress the temperature effects of interest [69].

6.2.3. Reactor Geometry

To ensure sufficient solid–gas contact time in a moving packed bed reactor, the reactor must be sufficiently tall. An initial length of 2 m is adopted here: it fits within the available laboratory space and leaves enough room for axial instrumentation. Erring on the long side at this stage is preferable to erring short, since a vessel built longer than necessary can always be operated with a shorter effective bed length by loading less solid, whereas a vessel built too short cannot be extended without rebuilding it. This initial estimate is used to evaluate the assumption checks in Section 6.3 and to run the model itself; Section 7.2 subsequently uses the model to confirm that 2 m is in fact sufficient across the target operating range, with a margin over the worst-case operating point.

For the diameter, it should be wide enough to make sure wall effects are not dominating. The reactor inner diameter of 50 mm is selected to satisfy the wall effect criterion ($d_t/d_p = 20 > 10$) and simultaneously be easy to work with, since it can easily fit sensors and it is a standard size (DN50), so pre-fabricated components can be utilised.

6.2.4. Feed

The SEM reactor receives the effluent of a two-stage conventional methanation cascade, see Figure 4.2. After the second reactor a flash separator removes condensed water. In practice, gas leaving a flash separator remains saturated with water at the separator's operating temperature, so some residual water would still be present in the feed to the moving packed bed. For simplicity, this residual water is disregarded here, and the feed is taken to consist of 80 % CH_4, 16 % H_2, and 4 % CO_2; Section 7.5 examines the effect of relaxing this assumption.

6.2.5. Reactor and Material Properties

Table 6.1 summarises the reactor dimensions and material properties used throughout the model. The particle diameter and intraparticle porosity are taken from Wei [13], measured directly on the 5 wt% Ni 2.5 wt% Ce/13X formulation selected in Section 4.3. The remaining properties, including bed void fraction, solid heat capacity, and sorbent density, are taken from Bareschino et al. [35]. The particle porosity ε_p is set to 0.242, derived from the total pore volume of the reduced 5 wt% Ni–2.5 wt% Ce/13X material measured by Wei ($V_{total} = 0.21\,\mathrm{cm^3/g}$, Table S.5.5 in [13]) and the particle density of $1400\,\mathrm{kg\,m^{-3}}$. This is a 13 % reduction relative to the fresh 13X support ($V_{total} = 0.24\,\mathrm{cm^3/g}$), reflecting pore volume occupied by deposited Ni and Ce.

The bed is diluted with an inert solid making up 75.7 wt% of the total solid volume. This is a common safety strategy for lab-scale reactors handling strongly exothermic gas-solid reactions, as it lowers the volumetric heat release rate for a given reactor length and moderates the axial temperature gradients. The inert fraction is assumed to share the density and heat capacity of the active material, since its actual composition is not specified [38].

Table 6.1: Key reactor and material properties for the bi-functional 5 wt% Ni 2.5 wt% Ce / zeolite 13X material and model used in this work.

Property	Symbol	Value	Unit	Source
Reactor length	L	2	m	—
Internal diameter	d_t	50	mm	—
Particle diameter	d_p	2.5	mm	[35]
Bed void fraction	ε_b	0.4	–	[35]
Bi-functional material mass	m_{bif}	0.80	kg	this work
Filler mass (75.7 wt% of bed)	m_{fill}	2.49	kg	this work
Intraparticle porosity	ε_p	0.242	–	[13][1]
Tortuosity	τ_p	4.0	–	[51]
Solid heat capacity	$c_{p,s}$	1100	J/(kg·K)	[35]
Sorbent density	ρ_p	1400	kg/m^3	[35]
Reaction enthalpy	ΔH_{r}	-165×10^3	J/mol	[35]
Adsorption enthalpy	ΔH_{ads}	-45×10^3	J/mol	[35]

[1] Calculated from Wei's reported total pore volume and particle density; not reported directly.

6.3. Model Assumptions

The reactor model rests on the assumptions listed below. Rather than presenting these as a single undifferentiated list, they are split into two groups. The first group consists of assumptions whose validity is checked quantitatively against a dimensionless criterion; each is discussed in its own subsection and summarized in Table 6.2. The criteria are computed at reference point (T = 280 °C, $GHSV = 1.0\ \mathrm{m^3_{STP}\,kg^{-1}_{ads}\,h^{-1}}$), The second group consists of simplifications that are made without a dedicated quantitative justification, either because they are standard practice for this class of reactor or because their effect is expected to be secondary; these are listed at the end of this section.

Table 6.2: Summary of quantitatively-justified model assumptions, computed at the design point ($T = 280\,^\circ$C, $GHSV = 1.0\ \mathrm{m^3_{STP}\,kg^{-1}_{ads}\,h^{-1}}$)

Assumption	**Criterion**	**Value**	**Threshold**	**§**
Gas-phase axial dispersion negligible	Bo_{bed}	$\approx 1.5 \times 10^3$	$\gtrsim 100$	6.3.1
Axial heat conduction negligible (gas)	$Pe_{h,bed,gas}$	≈ 514	$\gtrsim 100$	6.3.1
Axial heat conduction negligible (solid)	$Pe_{h,bed,solid}$	$\approx 2.3 \times 10^4$	$\gtrsim 100$	6.3.1
Radial gradients / 1D model	Bi_{wall}	≈ 1.6	$\gtrsim 1$	6.3.2
– resulting error estimate	$\Delta T_{center-avg}$	≈ 17 K	–	6.3.2
Pseudo-homogeneous (gas–solid thermal equilibrium)	NTU	$\approx 7.3 \times 10^3$	$\gg 1$	6.3.3
Pressure drop negligible	$\Delta P/P$	1.08%	$\ll 5\%$	6.3.4
Intrinsic kinetics (no internal mass-transfer limitation)	C_{WP}	0.34	< 1.33	6.3.5

In addition to the assumptions in Table 6.2, the following simplifications are made without a dedicated quantitative derivation:

- The ideal gas law is assumed throughout.
- The model is solved at steady state.

- The reaction rate expression (LHHW) permits no negative reaction rate, only plateauing toward zero as the gas composition approaches equilibrium.
- CH_4 selectivity is assumed to be unity: only the Sabatier reaction (Eq. 3.1) is modelled, and the rWGS and CO methanation side reactions (Eqs. 3.2–3.3) are neglected, consistent with the LHHW rate expression of Koschany, Schlereth, and Hinrichsen [37], in which these side reactions are likewise omitted, and with the near-complete selectivity reported below 400°C at ambient pressure [31] and over 100 cycles up to 320°C for the material used here [13].
- CO_2 adsorption on the solid is neglected and only water is assumed to adsorb; this is supported by Gómez et al. [47], who measured a water-to-CO_2 selectivity of approximately 28 g_{H_2O}/g_{CO_2} for zeolite 13X at 275 °C under a stoichiometric H_2/CO_2 feed, close to the operating temperature used in this work.
- The solid is assumed to be fully regenerated between cycles.
- No catalyst deactivation or carbon deposition is considered [13, 41].
- Solid heat capacity and density are assumed independent of the saturation state.
- Gas mixture properties (viscosity, thermal conductivity, heat capacity) are evaluated from ideal-gas mixing rules at the local temperature and composition.
- All particles are assumed identical in size and shape, with particle diameter and bed voidage held constant along the bed and over time, and no attrition.
- The filler material is assumed to share the same properties (size, density, thermal conductivity and heat capacity) as the bi-functional material.
- The solid moves as a dense, gravity-driven bed well below minimum fluidization velocity, with uniform, near-monodisperse particles, and is assumed to move in plug flow with a uniform velocity profile and no axial back-mixing.

6.3.1. Axial transport

Plug flow is assumed for both phases: velocity and concentration (or loading) profiles are flat across the bed cross-section, with no axial back-mixing. For the gas phase this means neglecting axial dispersion relative to advective transport; the same argument, applied to heat, justifies neglecting axial heat conduction in both phases. Solid-phase plug flow is addressed separately, since no dispersion coefficient applies to a moving bed of solids.

Gas-phase axial mass dispersion

The gas phase plug flow assumption is justified by the bed-scale Bodenstein number,

$$Bo_{bed} = Pe_L \frac{L}{d_p}, \qquad \frac{1}{Pe_L} = \frac{1}{\tau}\frac{\varepsilon_b}{Re_p Sc} + \frac{1}{2}, \tag{6.1}$$

where Pe_L is the particle-level Peclet number for longitudinal dispersion, the standard two-asymptote form for dispersion in packed beds, with $\tau = \sqrt{2}$ and $Pe_L \to 2$ at high $Re_p Sc$ [70], using the unmodified Reynolds number $Re_p = \rho_g u_0 d_p/\mu$. At the design point, ($Re_p \approx 11.8$, $Sc \approx 0.8$, $\varepsilon_b = 0.4$), this gives $Pe_L \approx 1.9$ and $Bo_{bed} \approx 1.5 \times 10^3$, well above the $Bo_{bed} \gtrsim 100$ threshold commonly used to justify plug flow [71]. This is also consistent with Barbaresi et al., who found diffusive transport negligible compared to advective transport in a similar SEM system.

Axial heat transport

Axial heat conduction is neglected relative to convection, justified by the axial heat Peclet number,

$$Pe_{h,bed} = \frac{\rho\, u\, c_p\, L}{k_{eff,ax}}, \tag{6.2}$$

valuated once for the gas phase (ρ_g, u_g, $c_{p,g}$) and once for the solid phase ($\rho_{bed,tot}$, u_s, $c_{p,cat}$) against the effective axial thermal conductivity of the packed bed [51]. The same $\gtrsim 100$ threshold used for axial mass dispersion is applied here. At the design point, $Pe_{h,bed,gas} \approx 514$ and $Pe_{h,bed,solid} \approx 2.3 \times 10^4$

at $u_s = 5$ mm/s, both comfortably above this threshold, so axial heat conduction is neglected in both phases.

6.3.2. Radial gradients

Gradients in the radial direction are neglected since a 1D model is used. This is attractive given the reactor geometry (small diameter, long length), but is not free of cost: the wall Biot number,

$$Bi_{wall} = \frac{h_w R_t}{\lambda_{ar}}, \tag{6.3}$$

indicates whether radial temperature gradients are non-negligible ($Bi_{wall} \gtrsim 1$). Here λ_{ar} is approximated by the intrinsic solid conductivity of zeolite 13X, $k_s = 0.4$ W/(m·K) [51], which is a conservative lower bound on the true effective radial bed conductivity, since gas-phase conduction and flow-induced dispersion can only increase it. With $h_w = 25$ W/(m^2·K) (Section 6.2.2) and $R_t = 0.025$ m, this gives $Bi_{wall} \approx 1.6 > 1$: the 1D assumption introduces a quantifiable, and likely conservatively estimated, error rather than being exact. This is consistent with [35], whose 2D model found significant radial temperature non-uniformity under similar SEM conditions. Barbaresi et al. faced the same trade-off and adopted a 1D lumped-parameter model regardless, for computational tractability, and still obtained reasonable accuracy against steady-state measurements. The same trade-off is made here.

An estimate of the resulting non-uniformity follows from the steady radial heat balance with a uniform volumetric heat source q_v, giving a parabolic profile with center-to-average difference

$$\Delta T_{center-avg} = \frac{q_v R_t^2}{8\lambda_{ar}}. \tag{6.4}$$

Taking $q_v \approx 85$ kW/m^3, representative of the traveling methanation heat-release peak reported by Bareschino et al. for an equivalent-geometry SEM reactor (rather than the larger, transient start-up peak), this gives $\Delta T_{center-avg} \approx 17$ K, or $\approx 3\,\%$ of the absolute operating temperature. This estimate assumes that all locally generated heat is removed radially; the converged reactor model (Section 7.3.5) instead attributes the majority of heat removal to axial convection by the counter-currently flowing solid, so $\Delta T_{center-avg}$ is a conservative upper bound on the true local radial gradient rather than a prediction of it. Even so, this bounded, quantified swing confirms the 1D simplification as a genuine, non-negligible limitation that is accepted here in exchange for a tractable model, rather than an error that can be assumed away.

6.3.3. Pseudo-homogeneous

The pseudo-homogeneous assumption treats the gas and solid phases as being at the same local temperature, i.e. the interphase heat transfer resistance is neglected. This is justified when the number of gas-solid heat transfer units is large:

$$NTU = \frac{h_{gs}\, a_v\, L}{\rho_g\, c_{p,g}\, u_g} \tag{6.5}$$

where h_{gs} is the gas-solid heat transfer coefficient, $a_v = 6(1-\varepsilon_b)/d_p$ is the specific interfacial area, and L is the bed length. This gives $NTU \approx 7.3 \times 10^3 \gg 1$ for the conditions of the present reactor, corresponding to an estimated gas-solid temperature difference of $\Delta T_{gs} \approx 0.36$ K. This is consistent with [35], who reported $T_s - T_g \leq 5\,^\circ$C in their heterogeneous 2D SEM model. The pseudo-homogeneous simplification is therefore retained.

6.3.4. Pressure drop

The pressure drop across the moving packed bed was estimated using the Ergun equation,

$$\frac{\Delta P}{L} = \frac{150\,\mu\, u_0\,(1-\varepsilon_b)^2}{d_p^2\,\varepsilon_b^3} + \frac{1.75\,\rho_g\, u_0^2\,(1-\varepsilon_b)}{d_p\,\varepsilon_b^3}, \tag{6.6}$$

where u_0 is the superficial gas velocity, $d_p = 2.5$ mm, $\varepsilon_b = 0.40$, and μ and ρ_g are the gas viscosity and density evaluated at local conditions. At the design operating point ($T = 280$ °C, $P = 1$ bar, GHSV $= 1.0$ $m^3_{STP}\,kg^{-1}_{ads}\,h^{-1}$), the inlet superficial velocity is $u_0 = 212$ mm s^{-1}. The modified particle Reynolds number,

$$Re_p = \frac{\rho_g\, u_0\, d_p}{\mu\,(1-\varepsilon_b)}, \tag{6.7}$$

evaluates to $Re_p = 18.4$, placing the flow in the mixed viscous–inertial regime, with the viscous (Blake–Kozeny) term contributing about 82% of the total pressure drop. The resulting pressure drop over the $L = 2$ m bed is $\Delta P = 10.8$ mbar, or 1.08% of the operating pressure. Because this estimate uses the inlet velocity, it is a conservative upper bound: the methanation reaction reduces the total molar flow by up to 8% at full conversion, so the true average velocity is somewhat lower. In line with Bareschino et al., who found that at comparable gas velocities the pressure drop remains well below 5% and can therefore be safely neglected, a constant-pressure assumption is retained throughout the reactor model; errors introduced by this simplification are of order $\Delta P/P = 1.08\%$, which is negligible relative to the uncertainty in the kinetic parameters.

6.3.5. Intraparticle mass transfer

The reaction kinetics are assumed intrinsic, i.e. no internal diffusion limitation reduces the observed rate below the kinetic rate (effectiveness factor $\eta = 1$). This is checked with the generalized Weisz–Prater criterion of [72], extended to arbitrary reaction order n,

$$\frac{r_{obs}\,\rho_p\,L^2}{D_{eff}\,C_{CO_2,s}} < \frac{2}{n+1}, \tag{6.8}$$

where $L = V_p/S_p = R_p/3$ is the particle's characteristic length (for a sphere) and $D_{eff} = D_{CO_2-CH_4}\,\varepsilon_p/\tau_p$ is the effective intraparticle diffusivity of CO_2 through the CH_4-dominated background gas, estimated via the Fuller correlation. The observed rate r_{obs} is taken per unit particle mass (rather than per unit active-catalyst mass), since ρ_p and D_{eff} act on the whole pellet; it is therefore rescaled from the peak per-active-mass rate by the catalytically active mass fraction of the bifunctional material.

With $n \approx 0.5$, the approximate reaction order in CO_2 implied by the LHHW rate expression, the threshold is $2/(n{+}1) \approx 1.33$. At the peak local reaction rate, the criterion evaluates to $C_{WP} \approx 0.34$, well below this threshold, so internal diffusion limitation is neglected and the intrinsic kinetics are used directly.

6.4. Model Equations

Building on the assumptions introduced in the previous section, this section presents the governing equations of the reactor model: mass and energy balances for the gas and solid phases, the reaction kinetics, and the adsorption dynamics. The corresponding boundary and initial conditions are given alongside each set of equations.

6.4.1. State Variables and Coordinate System

The model is formulated in terms of molar fluxes F_i [$\mathrm{mol\,m_{bed}^{-2}\,s^{-1}}$], defined as the molar flow rate of species i per unit bed cross-sectional area. This choice absorbs the superficial gas velocity and bed void fraction into a single variable, so neither appears explicitly in the species balances.

Two spatial coordinates are used. The axial coordinate z runs from 0 (gas inlet, bottom) to L (gas outlet, top), in the direction of gas flow. The solid coordinate $\zeta = L - z$ runs in the opposite direction, from 0 (solid inlet, top) to L (solid outlet, bottom), consistent with the counter-current contact pattern in which freshly regenerated solid enters at the top while feed gas enters at the bottom.

All fixed model parameters are collected in Table 6.1.

6.4.2. Species Balances

The steady-state species balances are written as:

$$\frac{dF_{CO_2}(z)}{dz} = -\rho_{bed,cat}\, r_{rxn}(z) \tag{6.9}$$

$$\frac{dF_{H_2}(z)}{dz} = -4\,\rho_{bed,cat}\, r_{rxn}(z) \tag{6.10}$$

$$\frac{dF_{CH_4}(z)}{dz} = +\rho_{bed,cat}\, r_{rxn}(z) \tag{6.11}$$

$$\frac{dF_{H_2O}(z)}{dz} = +2\,\rho_{bed,cat}\, r_{rxn}(z) - \rho_{bed,ads}\, r_{ads}(z) \tag{6.12}$$

The source terms are volumetric: the reaction rate r [mol kg_{cat}^{-1} s^{-1}] is multiplied by the catalyst bulk density $\rho_{bed,cat}$ [kg_{cat} m_{bed}^{-3}] to convert from per-mass to per-volume, and similarly for adsorption.

The stoichiometric coefficients (1, 4, 1, 2) follow directly from $CO_2 + 4\,H_2 \longrightarrow CH_4 + 2\,H_2O$. The H_2O flux has an additional sink $\rho_{bed,ads}\, K_{LDF}(q^* - q)$ [$\mathrm{mol\,m_{bed}^{-3}\,s^{-1}}$], where K_{LDF} [s^{-1}] is the linear driving force coefficient defined in Section 6.4.6, q^* [$\mathrm{mol\,kg_{ads}^{-1}}$] is the equilibrium water loading, q [$\mathrm{mol\,kg_{ads}^{-1}}$] is the current solid loading, and $\rho_{bed,ads}$ [$\mathrm{kg_{ads}\,m_{bed}^{-3}}$] is the sorbent bulk density.

Partial pressures are obtained from gas molar fluxes assuming ideal gas behaviour at constant total pressure P:

$$p_i(z) = \frac{F_i(z)}{\sum_j F_j(z)}\, P \tag{6.13}$$

The boundary condition at the gas inlet is:

$$F_i(z = 0) = F_{i,\mathrm{in}} \tag{6.14}$$

6.4.3. Solid Loading Balance

The solid descends counter-currently in the ζ coordinate. At steady state, the water loading q [$\mathrm{mol\,kg^{-1}}$] evolves as:

The solid moves counter-currently from $z = L$ to $z = 0$. In the solid coordinate $\zeta = L - z$, the loading balance is:

$$u_s\, \frac{dq(\zeta)}{d\zeta} = K_{LDF}(\zeta)\,(q^*(\zeta) - q(\zeta)) \tag{6.15}$$

where u_s [m s^{-1}] is the solid velocity, q [mol kg_{ads}^{-1}] is the current H_2O loading, q^* is the equilibrium loading, and K_{LDF} [s^{-1}] is the linear driving force coefficient, defined in Section 6.4.6. The boundary condition imposes fully regenerated solid at the top:

$$q(\zeta = 0) = 0 \tag{6.16}$$

The adsorption rate appearing in Equation 6.12 is:

$$r_{ads}(z) = K_{LDF}(z)\,(q^*(z) - q(z)) \tag{6.17}$$

In a fixed-bed reactor, water uptake evolves in time; in the moving packed bed, the steady solid flow converts this time axis into the spatial coordinate ζ, so the same driving force describes a spatially varying loading profile rather than a transient one.

6.4.4. Energy Balance

The energy balance is pseudo-homogeneous, treating gas and solid as being at the same local temperature. The balance reads:

$$\frac{dT(z)}{dz} = \frac{(-\Delta H_r)\,\rho_{bed,cat}\,r_{rxn}(z) + (-\Delta H_{ads})\,\rho_{bed,ads}\,K_{LDF}(z)\big(q^*(z) - q(z)\big) - U_a\big(T(z) - T_{wall}\big)}{\sum_i F_i(z)\,C_{p,i} - u_s\,\rho_{bed}\,C_{p,s}} \tag{6.18}$$

where $\sum_i F_i C_{p,i}$ [$\mathrm{W\,m^{-2}\,K^{-1}}$] is the gas-phase thermal flux, $u_s\,\rho_{bed}\,C_{p,s}$ the solid-phase thermal flux in the opposing direction, ΔH_r [$\mathrm{J\,mol^{-1}}$] the reaction enthalpy, ΔH_{ads} [$\mathrm{J\,mol^{-1}}$] the adsorption enthalpy, and U_a [$\mathrm{W\,m^{-3}\,K^{-1}}$] the volumetric wall heat transfer coefficient. All quantities on the right-hand side are evaluated locally at position z.

The energy balance (Eq. 6.18) is a statement of convective heat transport by two counter-flowing streams. Written as a single first-order ODE for T, the coefficient multiplying dT/dz is the *difference* of the two streams' thermal capacity fluxes, $\sum_i F_i c_{p,i} - u_s \rho_{\mathrm{bed}} c_{p,\mathrm{cat}}$: the gas contribution advecting in $+z$ and the solid contribution advecting against it. When this difference is integrated in z with the coefficient approaching zero, dT/dz diverges — not because the physics is singular, but because the chosen integration direction stops being the one along which temperature information actually propagates. The critical velocity u_s^* is exactly the point where the two fluxes are equal and net advective transport reverses direction: for $u_s < u_s^*$ the gas flux dominates and the temperature front is carried in $+z$, so integrating in z from the gas inlet is well-posed; for $u_s > u_s^*$ the solid flux dominates and the front is carried in $+\zeta$ (i.e. from the solid inlet), so the equation must instead be integrated in ζ. Switching coordinate at u_s^* therefore keeps the denominator bounded away from zero and positive in both cases, rather than merely avoiding a formal division by zero.

The operating envelope considered in this thesis lies entirely in the solid-dominated regime ($u_s > u_s^*$); the gas-dominated branch is retained in the model for robustness and completeness, so that the solver remains applicable should operating conditions outside this envelope become relevant in future work.

6.4.5. Reaction Kinetics

The reaction rate is described by the LHHW expression of Koschany, Schlereth, and Hinrichsen, selected in Section 6.4.5 over competing kinetic models, whose form and parameters are adopted without modification.

$$r_{rxn} = \frac{k\,(p_{CO_2}\,p_{H_2})^{0.5}\,(1-\beta)}{DEN^2} \tag{6.19}$$

where β is the approach-to-equilibrium factor:

$$\beta = \frac{p_{CH_4}\,p_{H_2O}^2}{K_{eq}\,p_{CO_2}\,p_{H_2}^4} \tag{6.20}$$

and DEN accounts for competitive adsorption of surface species:

$$DEN = 1 + K_{OH}\frac{p_{H_2O}}{p_{H_2}^{0.5}} + K_{H_2}\,p_{H_2}^{0.5} + K_{mix}\,p_{CO_2}^{0.5} \tag{6.21}$$

The rate constant k and adsorption equilibrium constants K_j follow Arrhenius and van't Hoff temperature dependences relative to a reference temperature $T_{ref} = 555\,\mathrm{K}$:

$$k = k_{ref}\exp\left[-\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right], \qquad K_j = A_j\exp\left[-\frac{\Delta H_j}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right] \tag{6.22}$$

Parameter values are taken directly from Koschany et al. Section 6.5.2 tests whether these independently regressed parameters reproduce Wei's sorption-enhanced breakthrough behaviour. The thermodynamic equilibrium constant is:

$$K_{eq} = 137\,T^{-3.994}\exp\left(\frac{158700}{RT}\right) \tag{6.23}$$

The LHHW form is preferred over the power law fitted by Wei to his own data, which omits the equilibrium term and consequently overestimates the rate near equilibrium, which is precisely the regime that dominates in the moving packed bed where the gas approaches full conversion.

6.4.6. Adsorption Isotherm and Linear Driving Force Coefficient

The equilibrium water loading q^* on the zeolite is described by the Dubinin–Astakhov (DA) isotherm:

$$q^* = \frac{\rho_w(T)}{M_{H_2O}} W_0 \exp\left[-\left(\frac{A}{E}\right)^n\right] \tag{6.24}$$

where W_0 [$\mathrm{m^3\,kg^{-1}}$] is the limiting micropore volume, E [$\mathrm{J\,kg^{-1}}$] is the characteristic adsorption energy, n [-] is the structural heterogeneity exponent, and $\rho_w(T)$ [$\mathrm{kg\,m^{-3}}$] is the temperature-dependent density of liquid water. The adsorption potential A [$\mathrm{J\,kg^{-1}}$] is:

$$A = \frac{R}{M_{H_2O}} T \ln\frac{p_{sat}(T)}{p_{H_2O}} \tag{6.25}$$

where p_{sat} is the saturation vapour pressure of water at temperature T.

The DA isotherm parameters follow the calibration described in Section 6.5.1: only W_0 is fitted, against Wei's breakthrough capacities, while E and n are held at the values of Mette et al. The fitted values are given in Table 6.3.

Table 6.3: Fitted Dubinin–Astakhov parameters for 5 wt% Ni–2.5 wt% Ce/13X, regressed at a regeneration temperature of 300 °C.

Parameter	Value	Unit
W_0	1.90×10^{-4}	$\mathrm{m^3\,kg^{-1}}$
E	1.19×10^{6}	$\mathrm{J\,kg^{-1}}$
n	1.55	–

The rate at which the equilibrium loading is approached is described by the linear driving force (LDF) model, with coefficient K_{LDF} derived from the Glueckauf approximation assuming pore diffusion control:

$$K_{LDF} = \frac{15\,\varepsilon_p\,D_M}{r_p^2\,\tau_p\,\rho_p\,R\,T\,\dfrac{dq^*}{dp_{H_2O}}} \tag{6.26}$$

where ε_p [-] is the particle porosity, r_p [m] the particle radius, τ_p [-] the tortuosity, ρ_p [$\mathrm{kg\,m^{-3}}$] the particle density, and dq^*/dp_{H_2O} the local isotherm slope evaluated by central finite difference. The molecular diffusivity of water vapour in the gas phase D_M [$\mathrm{m^2\,s^{-1}}$] is estimated for the H_2O–CH_4 binary at $P = 1$ bar, which is the dominant carrier at reactor conditions (80 mol % CH_4) [35]:

$$D_M = 7.28 \times 10^{-5}\left(\frac{T}{300}\right)^{1.75} \tag{6.27}$$

The pressure dependence is neglected since the model operates at atmospheric pressure.

6.4.7. Boundary Conditions

The model has boundary conditions at both ends of the reactor, making it a split boundary value problem: gas conditions are imposed at $z = 0$ and the solid condition at $z = L$.

$$F_i(z = 0) = F_{i,in} \tag{6.28}$$

$$q(\zeta = 0) = q(z = L) = 0 \tag{6.29}$$

$$T(\zeta = 0) = T_{in} \tag{6.30}$$

Equation 6.30 applies in the solid-dominated regime ($u_s > u_s^*$), where the temperature boundary condition is imposed at the solid inlet. In the gas-dominated regime ($u_s < u_s^*$), the temperature boundary condition is instead imposed at the gas inlet:

$$T(z = 0) = T_{in} \tag{6.31}$$

The reactor wall is assumed isothermal at the gas inlet temperature, $T_{wall} = T_{in}$, and the volumetric heat transfer coefficient U_a is obtained from the bed-to-wall heat transfer coefficient h [$\mathrm{W\,m^{-2}\,K^{-1}}$] (introduced in Section 6.2.2) via

$$U_a = h \cdot \frac{4}{d_t}, \tag{6.32}$$

where the factor $4/d_t$ [$\mathrm{m^{-1}}$] is the specific wall surface area of the cylindrical bed (wall surface area per unit reactor volume), converting h from a per-unit-wall-area basis to the per-unit-bed-volume basis used in the energy balance.

6.4.8. Numerical solution strategy

The counter-current geometry makes the model a split boundary-value problem: gas conditions are imposed at $z = 0$ while the solid loading condition is imposed at the opposite end, $z = L$. No single initial-value integration from one end to the other is therefore possible, and an iterative decoupling strategy is used instead. The gas and solid equations are coupled in both directions. The gas-phase water balance (Eq. 6.12) contains an adsorption sink that depends on the solid loading $q(z)$, which is itself governed by the solid loading balance (Eq. 6.15) whose driving force $q^*(p_{H_2O}) - q$ depends on the gas-phase water partial pressure $p_{H_2O}(z)$. The energy balance couples both phases further: the temperature profile $T(z)$ sets the reaction rate, the equilibrium loading q^*, and the LDF coefficient K_{LDF}, all of which feed back into both balances simultaneously. Because gas and solid flow in opposite directions, neither phase can be integrated independently from inlet to outlet with the other held fixed at its final converged state: the gas enters at $z = 0$ without knowledge of what the solid loading will be downstream, while the solid enters at $z = L$ without knowledge of the water partial pressure it will encounter as it descends.

The coupling is resolved by a Gauss–Seidel iteration, illustrated in Fig. 6.1, in which the gas and solid sub-problems are solved alternately, each treating the other phase as known from the previous sweep. The temperature profile is initialised as uniform at the inlet temperature, $T^{(0)}(z) = T_{\mathrm{in}}$, while the solid loading profile is warm-started from a physics-based estimate obtained from an isothermal, no-adsorption gas pass for the first case in a scan, and from the profile of the neighbouring case thereafter to speed up the iteration. Which phase carries the energy balance during the iteration depends on whether the solid or the gas thermal flux dominates, $u_s \rho_{\mathrm{bed}} c_{p,\mathrm{cat}} \lessgtr \sum_i F_i c_{p,i}$, with the crossover denoted u_s^*; the results reported in this thesis lie almost entirely in the solid-dominated regime ($u_s \gg u_s^*$), and the description below is written with focus on that case.

In the first step of each iteration, the solid loading profile $q(z)$ and temperature profile $T(z)$ from the previous iteration are held fixed, and the gas-phase species balances (Eqs. 6.9–6.12) are integrated from $z = 0$ to $z = L$ using the gas inlet conditions as initial values. This yields updated molar flux profiles $F_i(z)$ and, from these, updated partial pressure profiles $p_i(z)$. (In the gas-dominated regime,

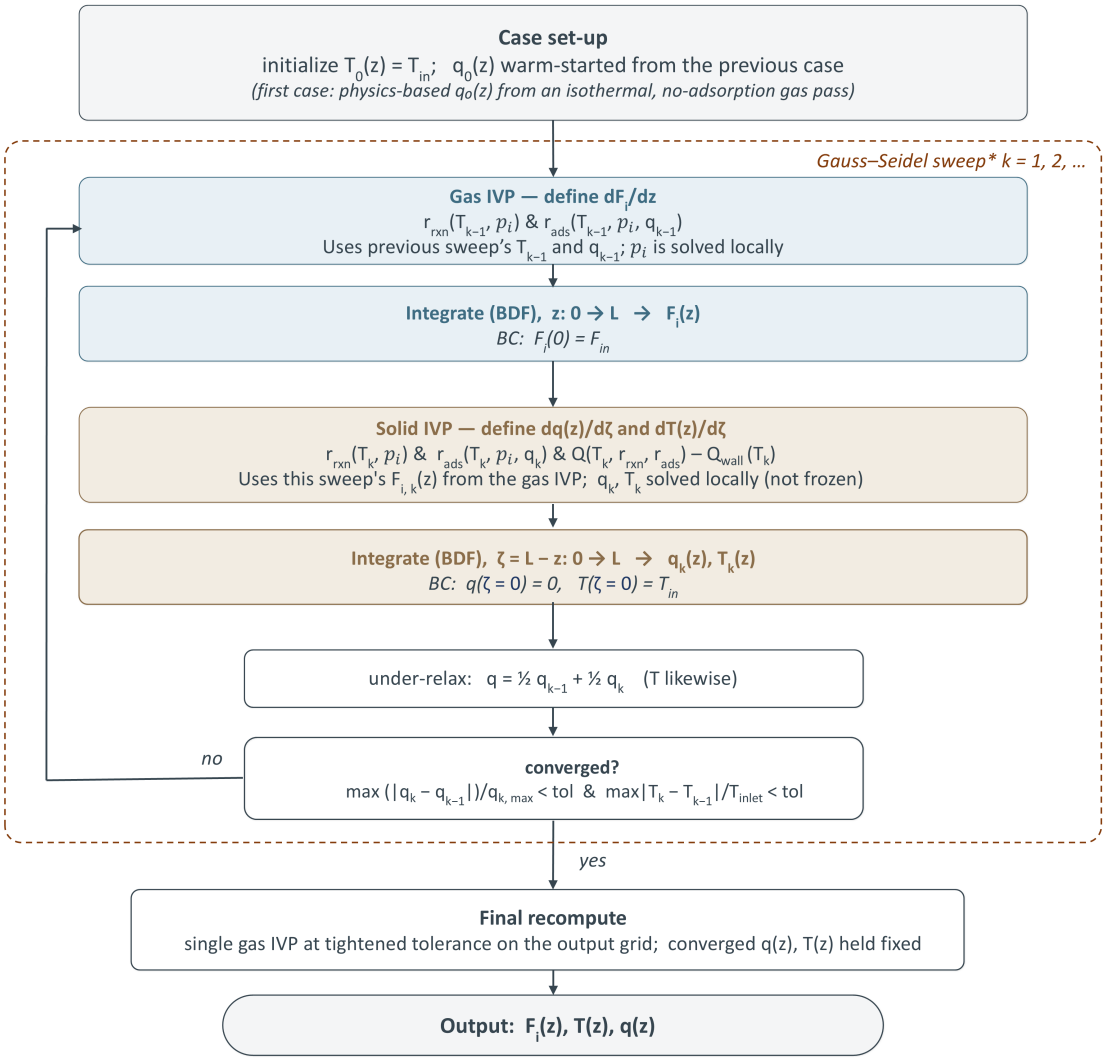


Figure 6.1: Solution algorithm for the steady-state, counter-current moving packed-bed (MPB) reactor model in molar-flux form. Each case is solved by a decoupled Gauss–Seidel iteration that alternates between a gas initial-value problem (IVP), integrated in the gas flow direction ($z\colon\ 0 \to L$) for the species fluxes $F_i(z)$, and a solid IVP, integrated in the solid flow direction ($\zeta = L - z\colon\ 0 \to L$) for the sorbent loading $q(z)$ and temperature $T(z)$. The two problems are coupled on a shared axial grid: the gas IVP uses the previous sweep's T_{k-1} and q_{k-1}, while the solid IVP uses the current sweep's freshly solved fluxes $F_{i,k}(z)$. Profiles are under-relaxed and the sweep is repeated until the maximum relative change in both q and T falls below the tolerance, after which a single gas IVP is recomputed at tightened tolerance on the output grid. Each new case is warm-started from the previous one; the first case uses a physics-based $q_0(z)$ from an isothermal, no-adsorption gas pass. *The diagram shows the solid-dominated regime, in which the solid thermal flux exceeds the gas thermal flux. When the gas flux dominates, the temperature is instead solved as a fifth state in the gas IVP, the solid IVP reduces to q alone, and convergence is checked on q only.

$u_s < u_s^*$, the energy balance, Eq. 6.18, is integrated simultaneously as a fifth equation in this same IVP, with $T(z = 0) = T_{\text{in}}$; in the solid-dominated regime this step advances species only, and $T(z)$ is instead updated in the solid step below.)

In the second step, the updated partial pressure profile $p_{H_2O}(z)$ and temperature profile $T(z)$ from the gas step are held fixed, and the solid loading balance (Eq. 6.15) is integrated from $\zeta = 0$ to $\zeta = L$ with the boundary condition $q(\zeta = 0) = 0$. In the solid-dominated regime, the energy balance is included as a second state variable in this step and integrated jointly with q, with $T(\zeta = 0) = T_{\text{in}}$. (In the gas-dominated regime, T is not solved here, having already been updated in the gas step above.)

Both IVPs are solved with the Backward Differentiation Formula (BDF) method, an implicit linear multistep scheme well suited to stiff ordinary differential equations, as implemented in SciPy's `solve_ivp`. Stiffness arises here from the strong nonlinearity of the LHHW rate expression: an explicit method would require prohibitively small step sizes in these regions, whereas BDF solves an implicit corrector equation at each step, allowing the integrator to take large steps where the solution is smooth and automatically reducing the step size only where gradients are steep.

To suppress oscillatory behaviour that can arise when source terms are stiff, the updated profiles are blended with the previous iteration using a relaxation factor of one half,

$$q^{(k+1)} = \tfrac{1}{2}\, q^{(k)} + \tfrac{1}{2}\, q^{(k)}_{\text{new}} \tag{6.33}$$

and analogously for T in the solid-dominated regime. Convergence is assessed after each iteration by the maximum relative change in the solid loading profile,

$$\varepsilon^{(k)} = \frac{\max_z \left| q^{(k+1)}(z) - q^{(k)}(z) \right|}{\max_z q^{(k+1)}(z)} \tag{6.34}$$

in the gas-dominated regime, and by the stricter of $\varepsilon^{(k)}$ and the analogously normalised maximum change in temperature, $\max_z |T^{(k+1)} - T^{(k)}|/T_{\text{in}}$, in the solid-dominated regime. The iteration is considered converged once this criterion falls below $\text{tol} = 10^{-5}$; cases that do not reach this tolerance within the maximum number of iterations are reported as unconverged and excluded from the results. Upon convergence, a single gas-phase IVP is recomputed at tightened tolerance on the output grid, with the converged $q(z)$ and $T(z)$ held fixed, to produce the final reported profiles.

6.4.9. Numerical tolerances

The moving-packed-bed model is solved by decoupling the counter-current gas and solid energy/species balances into two initial value problems (IVPs), which are integrated in an outer Gauss–Seidel loop until the sorbent loading and temperature profiles no longer change between sweeps. Three numerical parameters control the accuracy and computation time of this procedure and are easily conflated:

The axial grid z_{grid} (with N points) is not merely used for output: at every Gauss–Seidel sweep, the previous iteration's profiles $q(z)$ and $T(z)$ are linearly interpolated on this grid to couple the two IVPs. A coarser N therefore introduces a genuine coupling error between the gas- and solid-phase solutions, distinct from the accuracy of either IVP integration itself. After 400 grid points, the differences become less than 0.1%.

Within a single IVP solve, the adaptive BDF integrator (`scipy.integrate.solve_ivp`) chooses its own internal step size, evaluating the right-hand side far more often than N times regardless of the requested output grid. The accuracy of this continuous integration is governed entirely by the specified relative and absolute tolerances, not by N. During the Gauss–Seidel iterations, $\text{rtol} = 10^{-4}$ is used throughout, with $\text{atol} = 10^{-8}$ for the molar fluxes (tight, since fluxes are on the order of 10^{-3} to 10^{-1} $\text{mol}/(\text{m}^2\,\text{s})$) and a looser $\text{atol} = 10^{-1}$ K for temperature (looser because T is around 500–600 K, so a much larger absolute tolerance still corresponds to a tight relative one). After the outer loop converges, one final gas-side solve is repeated on a fixed 300-point grid with tightened tolerances ($\text{rtol} = 10^{-6}$, $\text{atol} = 10^{-10}$ for fluxes and 10^{-3} K for temperature), affordable since it is only run once.

The Gauss–Seidel iteration terminates once the relative change in $q(z)$ (and, where solved, $T(z)$) between successive sweeps falls below a convergence tolerance $\mathrm{tol} = 10^{-5}$. This controls how well the two decoupled IVPs have converged to a mutually consistent solution, independent of both N and the internal solver tolerances.

6.5. Model validation

The reactor model is assembled from three constitutive sub-models: the water adsorption equilibrium (Dubinin–Astakhov isotherm), the water mass-transfer coefficient (K_{LDF}), and the reaction kinetics (LHHW). To validate these against fixed-bed data, a dedicated one-dimensional transient model was implemented, independent of the moving-bed code: the coupled species and solid-loading balances are discretised on a fixed grid ($N = 100$, first-order upwind) and integrated in time as an initial-value problem at the exact geometry and operating conditions of Wei's Fig. 5.10 lab column, with a quasi-steady total-molar-flux closure that lets the local gas velocity fall along the bed as the mole-reducing reaction and water adsorption shrink the carrier gas [13]. Of all the parameters entering the three sub-models, only two quantities are calibrated to Wei's experiments on the 5%Ni 2.5%Ce/13X material for which the reactor is designed: the sorbent water capacity (through the isotherm parameter W_0) and the mass fraction within the bifunctional material that acts a catalyst Crucially, these two are fitted to separate, non-overlapping datasets (W_0 to the measured breakthrough capacities, the active fraction to the reaction-only conversion) so that any experiment depending on both at once constitutes an independent cross-check rather than a circular fit. The remaining ingredients are taken over unchanged: the LHHW rate expression and its parameters from Koschany, Schlereth, and Hinrichsen, and the water mass-transfer coefficient from Mette et al. Because the underlying materials are similar but not identical to Wei's, exact agreement is not expected; the two calibrations bring the model close enough for the purposes of this design-stage study, though not to exact quantitative agreement.

Since no moving-bed SEM experiment exists in the literature, the assembled moving packed bed model cannot be validated against data directly. Validation is therefore carried out at the level of the fixed-bed sub-models, and the same validated sub-models are carried unchanged into the moving-bed model. The section proceeds from the two independent calibrations (Sections 6.5.1 and 6.5.2) to their coupled verification in a single breakthrough experiment (Section 6.5.3), whose three features probe the sub-models to differing degrees: the plateau conversion re-confirms the active fraction, the breakthrough timing re-tests the sorbent capacity, and the front steepness offers the only possible check (albeit non-exhaustive) on K_{LDF}, which is carried in unchanged and untouched by either calibration.

6.5.1. Adsorption capacity

The equilibrium water uptake of the zeolite is described by the Dubinin–Astakhov (DA) isotherm (Eq. 6.24). Since the sorbent-catalyst material used here is not identical to that studied by Mette et al., the micropore volume W_0 is regressed against the breakthrough capacities measured by Wei on the same 5 wt% Ni–2.5 wt% Ce/13X material, making the calibration a single-parameter rescaling of the capacity magnitude onto the present material. The shape parameters E and n are instead taken unchanged from Mette et al., whose isotherm measurements on zeolite 13X remain the most extensive in the literature; Kiefer et al. adopts the same value of n, but uses a different E. Because Mette et al.'s measurements extend only to 250 °C, some extrapolation is involved to cover the operating temperature range of the reactor.

As Figure 6.2 shows, the parameter set of Mette et al. overestimates and that of Kiefer et al. underestimates Wei's measured capacities, whereas fitting W_0 alone, with E and n held at Mette et al.'s values, tracks the data well across thenoperating window. The fitted parameter is $W_0 = 1.90 \times 10^{-4}\ \mathrm{m^3\,kg^{-1}}$, with $E = 1.192 \times 10^6\ \mathrm{J\,kg^{-1}}$ and $n = 1.55$ taken from Mette et al.; these values are collected in Table 6.3. The fitted W_0 lies 44 % below Mette's value, likely reflecting material differences: the present material carries 7.5 wt% deposited Ni and Ce, which add to the sorbent mass basis without adsorbing water while also occupying pore volume.

This capacity calibration is the first of the two fitted quantities; it is exercised again, independently, by the breakthrough timing in Section 6.5.3.

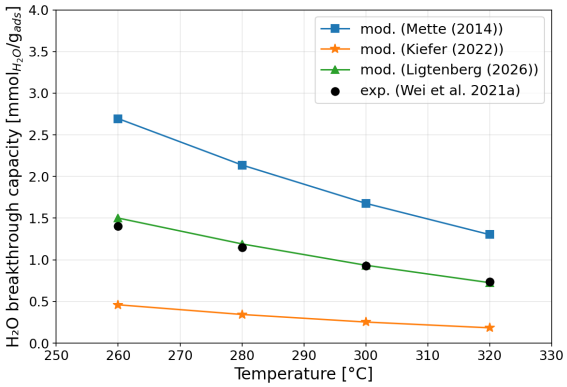


Figure 6.2: Water breakthrough capacity of zeolite 13X vs. temperature: three Dubinin–Astakhov parameter sets compared against Wei [15, 51]. The fitted (green) set matches the data well.

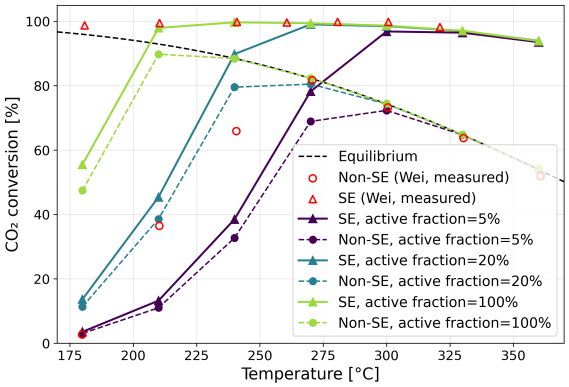


Figure 6.3: CO_2 conversion vs. temperature for 5%Ni2.5%Ce/13X: SEM column model at three active catalyst fractions (5%, 20%, 100%) vs. Wei (2022, Fig. 5.3). Circles: reaction-only; triangles: sorption-enhanced (fresh bed); dashed: equilibrium. GHSV $=$ 923 mL/(g·h), feed: 2.5% CO_2/10% H_2/81.5% CH_4/6% N_2.

6.5.2. Reaction kinetics and the active fraction

The reaction rate is described by the LHHW expression of Koschany, Schlereth, and Hinrichsen, whose form and parameters are adopted without modification (see Section 6.4.5). The single kinetic quantity fitted in this work is the active fraction: the mass fraction of the bifunctional material that is taken to be catalytically active in the rate expression. This matters because the LHHW rate is expressed per kg of catalyst per second, so the catalyst mass assumed enters the predicted rate linearly, and this fitted fraction is needed to reconcile a mass-basis ambiguity in the rate expression. Existing SEM modelling literature takes the catalyst weight in the rate law to be the 5 wt% Ni loading of the bifunctional material, thereby neglecting the zeolite mass entirely [35, 17, 15]. This is defensible in that the zeolite is present primarily as an adsorbent, not as catalyst. However, it is also not fully justified: the zeolite additionally acts as catalyst support, and Koschany, Schlereth, and Hinrichsen himself includes the mass of his (alumina) support in the corresponding weight basis. Neither taking 5 wt% (support neglected) nor 100 wt% (all mass treated as active) is therefore correct, and an intermediate active fraction is introduced to bridge the two conventions.

The active fraction is fixed against the non-sorption-enhanced conversion, which isolates the kinetics: with adsorption inactive, the outlet depends on the rate law alone and not on the capacity calibrated in Section 6.5.1. As Figure 6.3 shows, 5 wt% underestimates and 100 wt% overestimates the measured non-SE conversion, while 20 wt% reproduces it reasonably well across the temperature range. This value is then held fixed.

The sorption-enhanced conversion in the same figure is not used to fit anything; it is a coupled check, because the SE branch depends jointly on the active fraction fixed here and on W_0 fitted to the unrelated capacity data of Section 6.5.1. The model reproduces the SE conversion across most of the window, with the sorbent-enhanced conversions being underestimated below 270 °C. That the two independently calibrated quantities together track this curve reasonably well, without either being fitted to it, offers support to the model, though the underestimation below 270 °C shows the agreement is not exact.

6.5.3. Coupled breakthrough dynamics

The two calibrations are exercised together, and the mass-transfer sub-model is tested for the first time, against a single dynamic SEM breakthrough experiment. A breakthrough curve is informative precisely because its features are governed by different sub-models: the pre-breakthrough plateau conversion is set by the kinetics (active fraction), the breakthrough timing by the sorbent capacity (W_0), and the steepness of the breakthrough front by the water mass-transfer coefficient K_{LDF}. Of these, K_{LDF} is the one quantity never fitted to Wei (it is taken from Mette et al.) so the front shape is the closest thing to an unconditioned validation the available data allow.

It should be stressed that a single breakthrough experiment is a weak constraint on the intrinsic rate. During the pre-breakthrough period the reactor operates close to the water-removal-enhanced equilibrium at near-complete conversion, so the outlet composition is relatively insensitive to the exact rate constant, provided the kinetics are fast enough to approach equilibrium within the bed. The comparison in Figure 6.4 should therefore be read as confirming that the assembled reaction–adsorption–transport model reproduces the overall breakthrough behaviour, not as an independent determination of the rate expression.

Before the comparison is drawn, one operating condition must be made explicit. The breakthrough experiment of Wei was carried out at 240 °C, whereas the model is evaluated here at 260 °C. This 20 °C offset is not an independent adjustment fitted to the present curve, but a direct consequence of the low-temperature conversion underestimate already documented in Section 6.5.2, where the sorption-enhanced conversion was underpredicted below 270 °C. At 240 °C the model would not reach the near-complete conversion that sets the water production rate, so the water-transport comparison would be distorted by a conversion error rather than testing transport itself. Raising the modelled temperature by 20 °C restores the correct plateau conversion and water loading rate, making the front-steepness comparison meaningful; it is a known limitation of the kinetic sub-model at low temperature, not a parameter tuned to this curve, and the front-shape validation below is insensitive to the absolute temperature once the plateau conversion is matched.

On that basis the agreement is good. The pre-breakthrough plateau CO_2 conversion is reproduced almost exactly: 99.5 % in the model against 100.0 % measured. This confirms that the active fraction fixed in Section 6.5.2 carries over to the dynamic experiment. This is corroborated by the H_2 curve, whose pre-breakthrough equilibrium value and transient slope both agree well with the model. The CO_2 comparison is less informative than it appears, however: post-breakthrough, as conversion falls off (visible in the measured H_2 concentration rising again) the measured CO_2 concentration remains pinned at zero, which is inconsistent with the mass balance of the feed. Given the stoichiometric feed ratio (H_2:CO_2 = 4:1), the same drop in conversion that raises H_2 should also produce a proportionally smaller but still nonzero rise in CO_2, which is not observed. This post-breakthrough discrepancy in the measured CO_2 trace is therefore disregarded, and the H_2 curve is taken as the more reliable observable of conversion behaviour.

Notably, the modelled H_2O front is steeper than the measured one. This points to K_{LDF} as the likely source of the mismatch. Two, not mutually exclusive, readings follow. Either K_{LDF}, taken unchanged from Mette et al.'s zeolite-13X system, does not transfer exactly to the sorbent-catalyst composite used here. This is plausible, since the pore structure and tortuosity of a bifunctional pellet need not match a pure zeolite one, or the mass transfer is in reality fast enough that the front shape is no longer governed by K_{LDF} alone but also by axial dispersion, a mechanism the present model does not include. In the latter case a steeper-than-measured front would persist regardless of the value of K_{LDF}, and the front-slope comparison would confirm only that mass transfer is fast relative to the residence time, without being able to pin down K_{LDF} itself any more precisely. Distinguishing between these two explanations is not possible from a single breakthrough curve, and is left as a recommended refinement rather than resolved here. Some caution in over-interpreting the mismatch is in any case warranted given the measurement inconsistency already noted above, and the reference curve should not be treated as an exact target. Because this uncertainty in K_{LDF} cannot be resolved from the available validation data, a sensitivity analysis on K_{LDF} is carried out in Section 7.4 to assess how strongly the reactor design depends on this parameter.

A further discrepancy is that the model underestimates the breakthrough time by approximately 4 minutes. This is a capacity effect rather than a kinetic or transport one: breakthrough timing is governed by the sorbent water capacity, and Wei himself attributes such run-to-run variation to differences between production batches of the sorbent-catalyst material. The discrepancy over the first 10 minutes is, separately, an initial-condition artefact: the experimental bed started filled with regeneration gas whereas the model initialises with feed, and is likewise unrelated to the kinetics.

Taken together, the fixed-bed comparisons give solid support to the model: the adsorption capacity and reaction kinetics are each validated against independent data, and the coupled breakthrough experiment confirms that both calibrations carry over correctly, reproducing the plateau conversion almost exactly and matching the breakthrough timing to within batch-to-batch variation. The water mass-transfer

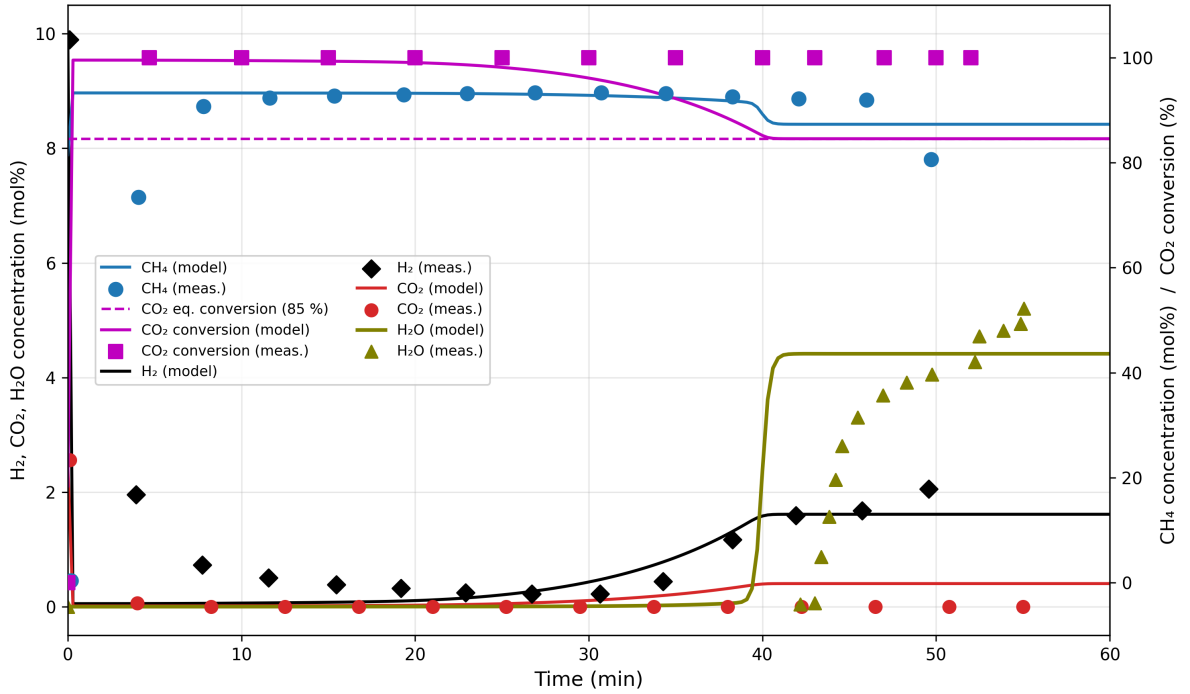


Figure 6.4: Model results validated against Wei (Fig. 5.10). Catalyst: 5%Ni2.5%Ce/13X; feed: 10 ml/min H_2, 2.5 ml/min CO_2, 81.5 ml/min CH_4, 6 ml/min N_2; GHSV = 923 ml g_{cat}^{-1} h^{-1}; $P = 1$ bar. The experiment was carried out at 240 °C; the model is evaluated at 260 °C to compensate for the low-temperature conversion underestimate documented in Section 6.5.2, so that the front-steepness comparison is made at the correct plateau conversion (see text). The experimental bed was regenerated with a 10% H_2 in N_2 mixture at 300 °C and was therefore filled with this gas, rather than feed, at $t = 0$; the model initialises with feed concentrations and reacts immediately, which accounts for the discrepancy over the first 10 minutes. Thereafter the model underestimates the breakthrough time by approximately 4 minutes, consistent with batch-to-batch variation in the material's adsorption capacity. The model assumes 20 wt% active catalyst.

coefficient is the one sub-model not directly fitted here, and it holds up reasonably well too: the modelled front shape is close, if somewhat steeper than measured, most plausibly because K_{LDF} does not transfer perfectly from Mette et al.'s pure-zeolite system to the bifunctional composite, or because axial dispersion (not included in the model) also shapes the front. Either way, this is a minor discrepancy against an otherwise strong set of comparisons. To assess how much these remaining uncertainties matter for the reactor design, Chapter 7 closes with a sensitivity analysis on all three calibrated and carried-over parameter sets: the kinetic parameters, K_{LDF}, and W_0. The moving-bed configuration itself cannot yet be validated directly, since no moving-bed SEM data exists in the literature, but this makes the fixed-bed validation carried out here all the more valuable: it gives good confidence in each of the sub-models individually before they are combined into a genuinely first-of-its-kind moving-bed model. Dedicated experiments isolating water mass transfer for this specific sorbent-catalyst composite remain a natural next step, but the present results already provide a solid foundation for the moving-bed model that follows. With the design basis fixed (Section 6.2), the model assumptions and governing equations established (Sections 6.3–6.4), and the constitutive sub-models validated here, Chapter 7 applies the assembled moving-bed model to size the reactor and characterise its performance across the operating envelope.

7

Results and Discussion

This chapter applies the reactor model developed in Chapter 6 to establish the operating envelope and design point of the lab-scale moving packed bed, and to examine the internal behaviour and robustness of the resulting design. Section 7.1 introduces the capacity margin Λ, a dimensionless criterion that bounds the solid-to-gas throughput ratio for which sorption enhancement remains physically meaningful and the model numerically tractable; this criterion, together with a solid-velocity floor, is then used in Section 7.2 to size the reactor and define its operating window across the target GHSV range. A single design point within this envelope is selected in Section 7.3 to characterise the reactor's internal conversion, composition, solid-loading, and temperature profiles in detail, closing with the overall mass and energy balance. Finally, Section 7.4 tests how sensitive this design is to the uncertainty inherent in the three constitutive sub-models (adsorption kinetics, reaction kinetics, and sorbent capacity) since these are drawn from materials and experiments related to, but not identical with, the Ni-Ce/13X composite used in this work.

7.1. Capacity margin

The moving-bed configuration relies on a continuous supply of unsaturated sorbent to remove the water produced by the reaction as it descends; if this supply cannot keep pace with the rate of water generation, the entire premise of sorption enhancement collapses locally. This balance is captured by a dimensionless capacity margin,

$$\Lambda = \frac{u_s \, \rho_{\mathrm{bed,ads}} \, q^*_{\max}}{2 \, F_{\mathrm{CO_2,in}}}, \tag{7.1}$$

the ratio of the maximum rate at which the solid can, in principle, remove water (set by the solid throughput u_s, the sorbent bulk density $\rho_{\mathrm{bed,ads}}$, and the equilibrium capacity ceiling $q^*_{\max}$ reachable at the maximum water partial pressure the reactor stoichiometry permits) to the rate at which water is generated by the reaction, $2F_{\mathrm{CO_2,in}}$. Note that the denominator assumes complete conversion of the incoming CO_2, so Λ represents a conservative (worst-case) margin: at any conversion below 100%, less water is actually produced than Λ accounts for, and the true margin is correspondingly larger. As the sorbent approaches saturation, the local driving force $(q^* - q)$ collapses, causing water to accumulate in the gas phase; this raises the local water partial pressure, which itself lowers q^* through the adsorption isotherm, further eroding the remaining uptake capacity and reinforcing the accumulation. The resulting rise in $p_{\mathrm{H_2O}}$ simultaneously suppresses the reaction rate through the $K_{\mathrm{OH}} p_{\mathrm{H_2O}}$ inhibition term in the LHHW kinetics and pulls the system back toward its unenhanced equilibrium conversion, since water is a reaction product. In this regime the reactor no longer benefits from sorption enhancement at all: it degrades toward the performance of a conventional fixed bed at exactly the location where sorbent capacity is exhausted.

In practice, the sorbent never retains its full thermodynamic capacity $q^*_{\max}$ as net carry-out, which means the true operability boundary lies well above the nominal $\Lambda = 1$ threshold. This is a direct consequence of the counter-current contact pattern: as the solid descends, it becomes progressively loaded with

water at locations where $p_{\mathrm{H_2O}}$ is high since the methanation reaction has already produced significant amounts of water. As this loaded solid continues toward $z = 0$, where $p_{\mathrm{H_2O}}$ is considerably lower, the local equilibrium loading drops, stopping the adsorption and even creating a driving force for desorption; the solid releases part of its adsorbed water back to the gas phase, effectively a localized, in-situ regeneration step. The extent of this effect is governed by the intraparticle mass-transfer resistance (K_{LDF}) to water diffusion within the zeolite particle: only the fraction of water that can diffuse out fast enough, given the available residence time, is actually released. At higher solid velocities this partial regeneration becomes negligible for two compounding reasons: the shorter residence time means the solid never reaches as high a water loading in the first place, so the driving force for desorption near $z = 0$ is smaller, and even where some driving force exists, the residence time is too short for water to diffuse out of the particle to any significant extent.

Figure 7.1 illustrates this directly for six representative solid velocities at fixed GHSV. At $u_s = 8$ mm/s the solid loading remains well below the equilibrium ceiling $q^*_{\mathrm{max}} = 1.2$ mol/kg throughout the bed, leaving ample driving force for adsorption everywhere. As u_s decreases, the same water production rate must be carried out by progressively less solid, and the loading rises accordingly. This is accompanied by a growing partial-desorption effect near the solid inlet, most pronounced at $u_s = 3.4$ mm/s, where sorbent loaded near the wet end of the bed gives water back as it descends into leaner gas, in turn raising the local water partial pressure. Both effects are undesirable: elevated $p_{\mathrm{H_2O}}$ suppresses the reaction and works against the purpose of sorption enhancement, which is why Λ should be kept in a range where this behaviour remains minimal, in practice above approximately 2.5. As Table 7.1 shows, the peak gas-phase water partial pressure $p^{\mathrm{max}}_{H_2O}$ rise as Λ falls, confirming the physical degradation described above: as the solid approaches saturation, more water is left in the gas phase, suppressing the reaction and eroding the benefit of sorption enhancement. This is an undesirable operating regime and it is the primary reason $\Lambda \approx 2.5$ is adopted as the lower bound of the operating envelope in this work. A consequence of adopting this conservative margin is that the sorbent capacity is used only $\sim$40%, requiring $2.5\times$ more solid circulation than the capacity balance alone would need. This is a genuine disadvantage of operating well above $\Lambda = 1$, since it increases the solid circulation rate and the associated solids-handling duty. However, because the sorbent leaves the reactor only partially loaded, the downstream regeneration step has correspondingly less water to remove per unit mass of solid, reducing its energy demand.

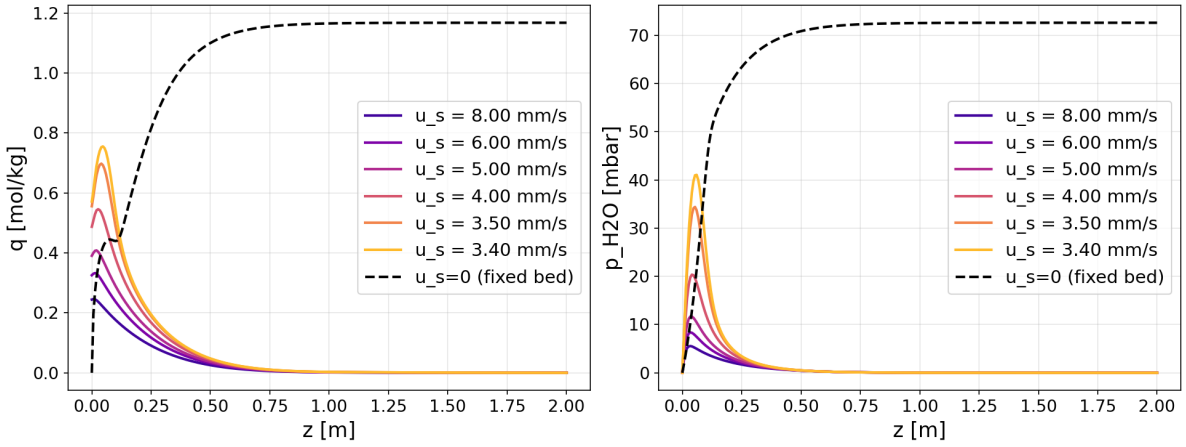


Figure 7.1: Solid water loading $q(z)$ for various solid velocities at GHSV = 1 $\mathrm{m^3_{STP}/(kg_{ads}\,h)}$, T_{in} = 280 °C. Lower u_s visibly pushes the loading profile closer to saturation and produces the steep uptake/partial-desorption fronts associated with low Λ. 99.9% conversion is reached in all cases.

Beyond this physical degradation, operating near or below this margin also renders the problem numerically stiff. Two steep fronts develop simultaneously along the solid path: at the solid inlet, fresh, unloaded zeolite meets gas carrying the highest water partial pressure in the reactor, producing a sharp adsorption front as uptake proceeds rapidly; at the solid outlet, the same particles now meet the leanest gas in the reactor, close to the feed's inlet water content, driving the partial regeneration described above and a second, oppositely-directed front. Comparing Λ against the solver's convergence

behaviour across a representative solid-velocity sweep (Table 7.1) confirms that these compounding gradients are a source of the observed numerical difficulty: iteration counts stay comparatively low through the middle of the sweep ($\Lambda = 3.07$–3.68, 31–35 iterations) and rise sharply as Λ approaches 2.5 from above, reaching 149 and 233 iterations at $\Lambda = 2.15$ and 2.08 respectively, consistent with the increasing sorbent utilisation and steepening profiles shown in Figure 7.1. The highest solid velocity tested ($u_s = 8$ mm/s, $\Lambda = 4.91$) is the exception to this trend, requiring 152 iterations despite its large margin; since this point is run first in the sweep, it converges from a comparatively poor initial guess rather than from the previous point's converged solution, making this a solver initialisation artefact rather than a sign of physical difficulty. Operating points below the $\Lambda \approx 2.5$ threshold are therefore excluded from consideration in this work, both because sorption enhancement provides little physical benefit there and because the resulting steep, competing fronts fall outside what the current solution strategy can robustly resolve. It is unfortunate that the sharp conversion decline this physical argument predicts cannot be demonstrated directly below $\Lambda \approx 2.5$: precisely because the adsorption and desorption fronts steepen in this regime, the solver also loses convergence there.

This constraint is not redundant with Λ: at low GHSV, Λ can remain comfortably above 2.5 even as u_s drops below 3 mm/s, since Λ scales inversely with GHSV at fixed u_s. Solving such a case (GHSV ≈ 0.5 m^3_{STP}/(kg_{ads} h), $u_s < 3$ mm/s, $\Lambda > 3$) still fails to converge, confirming that a low u_s introduces a separate source of numerical difficulty, independent of the capacity balance. This is attributed to the solid residence time L/u_s: at low u_s, the solid remains in the reactor considerably longer, strengthening the coupling between the counter-current gas- and solid-phase balances at any given position, since the local solid state now depends on conditions integrated over a much larger part of the domain. The iterative solution scheme, which alternates between the two phases, consequently requires substantially more sweeps to reach a self-consistent solution, regardless of how favourable Λ is. Therefore another constraint is added: the solid velocity must be above 3 mm/s. This floor is also physically sensible in its own right: maintaining a continuous, uniform solids flow becomes increasingly difficult at very low velocities, where flow can become intermittent or prone to stagnant zones rather than the steady plug flow assumed in the model.

Table 7.1: Capacity margin, sorbent utilisation, and solver behaviour across the solid-velocity sweep at GHSV = 1 m^3_{STP}/(kg_{ads} h), T_{in} = 280 °C.

u_s [mm/s]	Λ [-]	$q(0)$ [mol/kg]	Utilisation $q(0)/q^*_{max}$ [-]	$p^{max}_{H_2O}$ [mbar]	Iterations
8.00	4.91	0.244	0.203	5.60	152
6.00	3.68	0.326	0.272	8.47	31
5.00	3.07	0.390	0.325	11.86	35
4.00	2.45	0.487	0.406	20.77	63
3.50	2.15	0.556	0.463	35.20	149
3.40	2.08	0.572	0.477	42.05	**233**

7.2. Operating envelope and reactor sizing

For each GHSV, u_s is set to hold the capacity margin $\Lambda = 2.5$ (Section 7.2), subject to the $u_s \geq 3$ mm/s floor; only at the lowest GHSV (0.5 m^3_{STP}/(kg_{ads} h)) does the floor bind instead, giving $\Lambda = 3.85$. A target conversion of 99% is used throughout: comfortably above typical grid-injection CH_4 purity requirements established in Section 4.2.1 [34, 61], while stopping short of the near-complete conversion that would demand disproportionate bed length or sorbent for diminishing returns in unreacted H_2.

Figure 7.1 shows the resulting conversion profiles. As GHSV increases, the 99% crossing (red X) moves further into the bed; at GHSV ≥ 2.5 m^3_{STP}/(kg_{ads} h) it is not reached at all, setting the upper bound of the operating window.

The worst case within the target window (GHSV = 2.0 m^3_{STP}/(kg_{ads} h)) reaches 99% conversion at $z = 1.72$ m. A reactor length of $L = 2.0$ m is selected, giving a ~16% margin over this worst case and defining the operating window as GHSV $\in [0, 2]$ m^3_{STP}/(kg_{ads} h), corresponding to a solid velocity range of $u_s \in [3, 7.7]$ mm/s. As Figure 7.2 shows, conversion rises steeply near the inlet and then approaches its asymptote, so beyond the crossing point additional length buys diminishing returns in

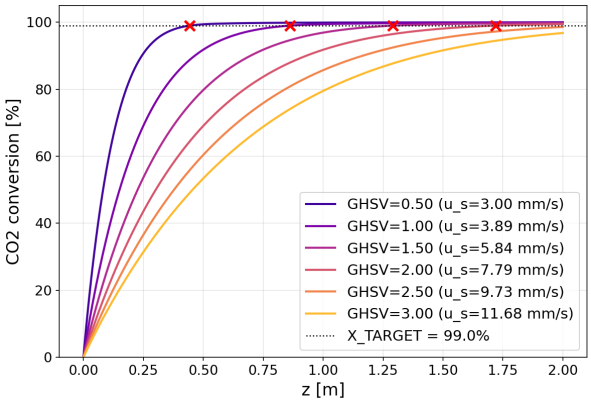


Figure 7.2: Conversion profiles at various GHSV, with u_s set to hold a capacity margin $\Lambda = 2.5$ (floored at $u_s \geq 3$ mm/s). Red X marks the axial position where 99% conversion is reached. Reactor length of 2 m selected on this basis.

conversion at the cost of further catalyst and sorbent inventory; extending the bed well past 2 m to also accommodate GHSV $>$ 2 is therefore not an efficient trade. The resulting catalyst and sorbent inventory is summarized in Table 6.1.

This envelope is directly relevant to the dynamic operability requirement established in Chapter 1, since it defines a feedforward control law rather than a single design point: for any feed rate within the target window, the solid withdrawal rate can be set as

$$u_s(\text{GHSV}) = \max\left(3\ \text{mm/s},\ \frac{2\,\text{GHSV}\cdot F_{\text{CO}_2,\text{in}}}{2.5\,\rho_{\text{bed,ads}}\,q^*_{\max}}\right), \tag{7.2}$$

holding the capacity margin at $\Lambda = 2.5$ (Section 7.1) subject to the $u_s \geq 3$ mm/s floor, so that near-complete conversion is maintained without adjusting reactor geometry. It should be stressed that this is a steady-state envelope rather than a demonstrated transient response: the model characterises a family of achievable operating points, not the trajectory the reactor follows between them when the feed rate changes.

7.3. Design-point performance

To illustrate the reactor's internal behaviour, this section considers a single operating point within the envelope established in Section 7.2: GHSV $= 1.5$ $\text{m}^3_{\text{STP}}/(\text{kg}_{\text{ads}}\,\text{h})$, i.e. 75% of the upper bound of the operating window. This point is chosen to sit comfortably inside the envelope while still exhibiting the relevant effects discussed below, rather than to represent the worst case that governs the reactor length (Section 7.2). The solid velocity follows from the capacity-margin rule of Section 7.1, holding $\Lambda = 2.5$ at this GHSV, which gives $u_s = 5.8$ mm/s. This design point is used throughout the remainder of this section to characterise the reactor's internal behaviour (Sections 7.3.1–7.3.4) and its overall mass and energy balance (Section 7.3.5).

7.3.1. Conversion profile

Figure 7.3 shows the CO_2 conversion X_{CO_2} and the reaction rate as a function of bed position z for the MPB, for a fixed bed without sorption enhancement, and for the thermodynamic equilibrium limit, all at the design point.

The equilibrium ceiling of 84.5% is clearly exceeded: the in-situ removal of water shifts the reaction beyond its unenhanced thermodynamic limit, and the MPB reaches a conversion of $\sim$99.7% at $z = 2$ m. Up to $z \sim 0.55$ m, however, the MPB actually trails the non-enhanced fixed bed. This is a consequence of the lower resulting temperature in the MPB compared to the fixed bed (see 7.3.4).

The reaction rate profile is spread out along the bed rather than sharply peaked near the inlet, reaching a maximum of about 21 mmol/(kg_{cat} s). This more distributed rate profile, in contrast to the sharp front typical of the non-enhanced fixed bed, reflects how sorption enhancement sustains reactivity further into the bed instead of concentrating it near the entrance.

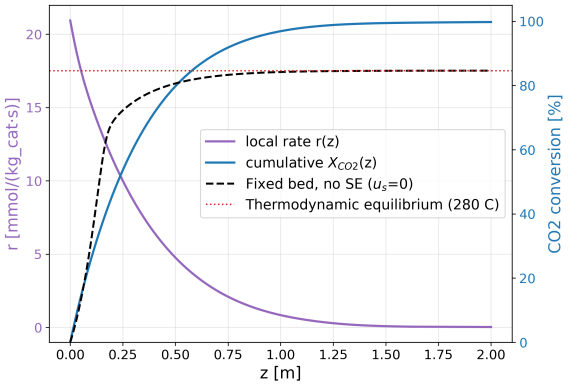


Figure 7.3: CO_2 conversion versus bed position for the moving packed bed, a fixed bed without sorption enhancement, and the thermodynamic equilibrium limit, at the design point $T_{in} = 280\text{C}$, GHSV $= 1.5\ \mathrm{m^3_{STP}/(kg_{ads}\ h)}$, $u_s = 5.8$ mm/s. The MPB reaction rate peaks at ∼21 mmol/(kg_{cat} s).

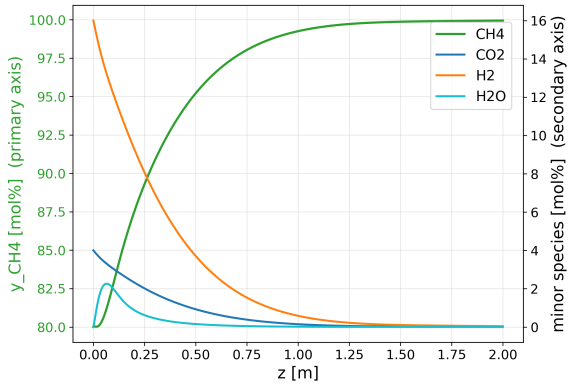


Figure 7.4: Gas-phase composition along the bed at the design point ($T_{in} = 280\text{C}$, GHSV $= 1.5\ \mathrm{m^3_{STP}/(kg_{ads}\ h)}$, $u_s = 5.8$ mm/s). CH_4 on the primary axis, minor species (CO_2, H_2, H_2O) on the secondary axis.

7.3.2. Gas-phase composition

Figure 7.4 shows the mole fractions of CH_4, CO_2, H_2, and H_2O along the bed at the design point, plotted on a dual axis to accommodate the disparate scales of the major and minor species.

The outlet composition can be framed against the grid-injection specification introduced in Chapter 4, which it meets comfortably. A small H_2O bump appears at $z \sim 0.07$ m, where water produced by the reaction briefly outpaces the local uptake by the sorbent, but this is quickly suppressed as sorption enhancement takes hold further into the bed. Over the remainder of the reactor, the partial pressure of H_2O remains very low, confirming that the sorption enhancement is functioning effectively.

7.3.3. Solid loading and adsorption driving force

Figure 7.5 shows the actual solid water loading $q(z)$ against the Dubinin–Astakhov equilibrium loading $q^*(z)$, with the shaded region marking the local driving force $q^* - q$. Where $q^* > q$, adsorption proceeds as expected; where the curves cross and q^* falls below q, the shading flips sign and marks an unwanted desorption driving force, with the solid releasing water back to the gas phase. This crossover occurs near the gas inlet ($z = 0$), where $q(0) \approx 0.5$ mol/kg against $q^* \approx 0$. Since $z = 0$ is simultaneously the gas inlet and the solid outlet, the sorbent arriving there is already loaded from its passage through the bed, while the fresh feed gas carries no water, so $p_{H_2O} \approx 0$ right at this point and the solid is forced to desorb. The effect is short-lived: p_{H_2O} climbs rapidly with distance from the inlet, driven both by the reaction and by the desorbed water itself, so the regeneration zone only extends over a short length of the bed and has limited impact.

7.3.4. Temperature profile and local energy budget

Figure 7.6 merges two panels on a shared z-axis: the temperature profile $T(z)$ on top, and the local volumetric heat generation and removal terms below.

The temperature rises by only $+12$ K to a peak at $z \sim 0.02$ m, decaying back towards the inlet value by $z \sim 1.25$ m. This modest rise reflects the solid's role as the dominant heat carrier: entering cold at $z = L$, it warms as it moves counter-currently through the bed, absorbing heat that would otherwise accumulate in the gas and catalyst. The large sensible heat removed by the solid stream (quantified in Section 7.3.5) keeps the reactor comfortably below the 100 K+ rises reported for fixed-bed sorption-enhanced methanation at similar GHSV [35, 17]. This lower temperature is precisely what causes the MPB to trail the fixed bed in conversion over the first ∼55 cm of the reactor (Section 7.3.1): reaction kinetics are slower at lower temperature. Nonetheless, the trade-off is favourable overall, since the lower temperature also shifts the equilibrium conversion upward and reduces the risk of catalyst sintering.

The lower panel decomposes the local energy budget into reaction heat, adsorption heat, and wall heat removal. Reaction heat dominates, peaking at ∼140 kW/m^3, with wall removal a comparatively minor

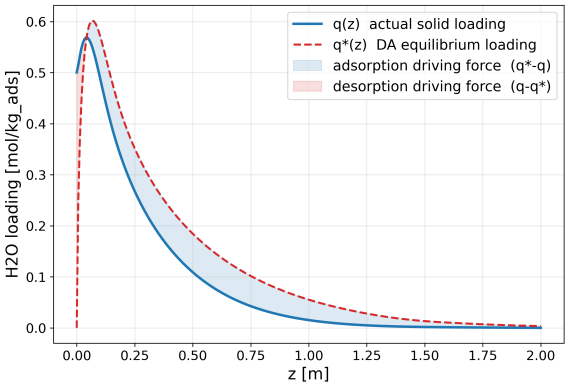


Figure 7.5: Solid H_2O loading $q(z)$ against the Dubinin–Astakhov equilibrium loading $q^*(z)$ at the design point ($T_{in} = 280\text{C}$, GHSV $= 1.5\ \mathrm{m^3_{STP}/(kg_{ads}\,h)}$, $u_s = 5.8$ mm/s). The shaded region show the local adsorption or desorption driving force.

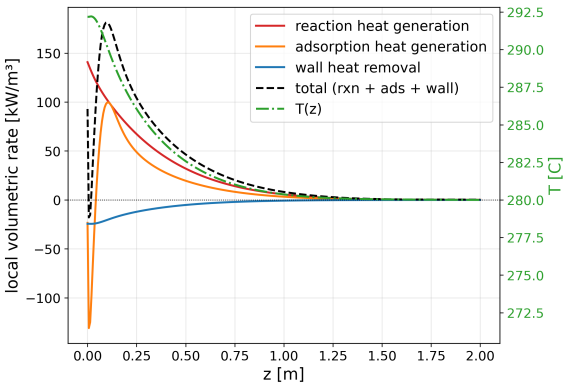


Figure 7.6: Local volumetric heat generation and removal terms along the bed at the design point ($T_{in} = 280\text{C}$, GHSV $= 1.5\ \mathrm{m^3_{STP}/(kg_{ads}\,h)}$, $u_s = 5.8$ mm/s): reaction heat, adsorption heat, wall heat removal, and their sum. The temperature profile is also shown.

term at $\sim$25 kW/m^3. The adsorption term is negative near $z = 0$, reaching ~ -130 kW/m^3: this is the same desorption event identified in Section 7.3.3, confirmed here by heat absorption coinciding with the trailing conversion, the H_2O bump, and the region where $q > q^*$. Overall, the sum of the terms produces a smooth curve free of hotspots, in line with the counter-current design intent, and markedly smoother than the sharp reaction front reported for fixed-bed operation [17].

7.3.5. Overall mass and energy balance

Table 7.2 summarizes the species and water balance at the reference design point. CO_2 conversion reaches $X_{CO_2} = 99.77\,\%$, consistent with the reactor length selected in Section 7.3. The corresponding CH_4 production of 275.94 mmol m^{-2} s^{-1} matches the stoichiometric water production of 551.88 mmol m^{-2} s^{-1} (2 mol H_2O per mol CO_2 converted), confirming that the species and water balances are internally consistent. Of this produced water, 550.79 mmol m^{-2} s^{-1} (99.8 %) leaves the reactor adsorbed on the solid, while only a negligible 0.01 mmol m^{-2} s^{-1} remains in the gas phase at the outlet, confirming that the sorbent keeps the gas-phase water content low throughout the bed. The small residual gap between the water produced and the water accounted for at the outlet ($\sim$1 mmol m^{-2} s^{-1}, 0.2 %) reflects numerical tolerance in the solution rather than a physical loss.

Table 7.2: Species and water mass balance at the reference design point.

Species	Inlet flux [mmol m^{-2} s^{-1}]	Outlet flux [mmol m^{-2} s^{-1}]
CO_2	276.59	0.65
H_2	1106.37	2.59
CH_4	5531.84	5807.78
H_2O, gas	0.00	0.01
H_2O, adsorbed	–	550.79
CO_2 conversion X_{CO_2}	99.77%	
H_2O produced (stoichiometric)	551.88	

Table 7.3 summarizes the energy balance at the reference design point. Reaction and adsorption heat generation together amount to 138.50 W, balanced by 15.84 W removed at the wall and 122.23 W carried out by the solid stream, closing to within 0.3 %. This large solid-side removal is consistent with the conclusion in Section 7.3.4, where it was identified as the reason the reactor temperature to stay an order of magnitude below the 100 K+ rises modelled for fixed-bed sorption-enhanced methanation [35, 17]. Sensible heat removed by the gas is negligible: since the gas enters at $T_{\mathrm{in}} = 280\,^{\circ}$C and the resolved outlet gas temperature is likewise $280\,^{\circ}$C, the gas carries no net sensible heat across the

reactor boundary, and essentially all sensible heat removal is attributed to the solid stream.

This result is consistent with the pseudo-homogeneous closure discussed in Chapter 6. Because the number of gas–solid heat-transfer units is large ($NTU \approx 1.3 \times 10^4$, Section 6.3), the physical thermal entrance length over which the gas equilibrates to the local solid/bed temperature is already very short, of order $L/NTU \approx 0.1$–0.2 mm, which is below the resolution of the axial grid. The energy equation retains the gas-phase heat capacity through its contribution to the combined thermal capacity flux $\sum_i F_i c_{p,i}$, but the two phases share a single temperature field rather than being resolved as separate states. One consequence of this closure is that the resolved internal profile $T(z)$ compresses this equilibration to a single axial point, with the gas entering the bed already at the local solid temperature, rather than being resolved over the short but finite physical entrance length.

Because this equilibration is internal to the entrance layer, the heat exchanged there is drawn from, and returned to, the adjacent solid rather than supplied across an external system boundary. Feed gas physically enters the reactor at $T_{\mathrm{in}} = 280\,^{\circ}\mathrm{C}$ (at $z = 0$, the gas inlet), heats up to $292\,^{\circ}\mathrm{C}$ over this short length, and then cools back down to $280\,^{\circ}\mathrm{C}$ as it travels through the reactor. The gas therefore contributes zero net sensible heat across the reactor's external boundary, which is why it is reported as such in Table 7.3.

Table 7.3: Energy balance at the reference design point.

Quantity	Rate [W]	% of total
Reaction heat generated	89.40	64.55
Adsorption heat generated	49.09	35.45
Total heat generated	138.50	100
Wall heat removed	15.84	11.47
Sensible heat, gas	0.00	0.00
Sensible heat, solid	122.23	88.53
Total heat removed	138.07	100

7.4. Sensitivity analysis of the parameters from the submodels

The reactor model draws its three constitutive sub-models (the LHHW reaction kinetics, the water mass-transfer coefficient, and the Dubinin–Astakhov capacity parameters) from materials and experiments that are similar to, but not identical with, the Ni-Ce/13X sorbent-catalyst composite used in this work. The reaction kinetics are fitted to a Ni-Al_2O_3 catalyst, K_{LDF} is taken unchanged from zeolite-13X data without the deposited Ni and Ce active phase, and W_0, while regressed against Wei's measurements on the actual material, still carries some uncertainty from the synthesis route and the combined effect of the zeolite support with the deposited Ni and Ce. Differences in synthesis method, particle morphology, or the interaction between the catalytic and sorptive functions could therefore shift any of these three parameters away from their assumed values. To assess how much this matters for the reactor's performance, each parameter is varied independently around the reference design point (GHSV $= 1.5$, $u_s = 5.8$ mm/s unless stated otherwise), and the resulting conversion, temperature, and mass-transfer behaviour are compared in Table 7.4. As the following subsections show, the three parameters differ considerably in how strongly they constrain the design: K_{LDF} has only a marginal effect on reactor performance even under a substantial overestimate, whereas the reaction kinetics and W_0 each require a roughly proportional compensating adjustment to remain fully accounted for.

7.4.1. Adsorption kinetics (K_{LDF})

The rate at which the sorbent approaches equilibrium water loading is governed by the linear driving force (LDF) coefficient K_{LDF}, which lumps intraparticle pore-diffusion resistance into a single mass-transfer coefficient (Eq. 6.15). Because it was taken unchanged from Mette et al.'s zeolite-13X data rather than fitted to the sorbent-catalyst composite used here, K_{LDF} is the most uncertain of the three sub-model parameters.

As noted in the validation (Section 6.5), the measured breakthrough front was less steep than the modelled one, suggesting the model may overestimate K_{LDF}. Table 7.4 tests this by reducing K_{LDF}

to $0.2\times$ and $0.04\times$ its base value, i.e. a five- and twenty-five-fold overestimate.

The reactor design is remarkably insensitive to this parameter. At $0.2 \times K_{LDF}$, $X_{\mathrm{CO_2}}$ falls only from 99.77 % to 99.37 %, and even at $0.04 \times K_{LDF}$ it remains above 97.7 %. The extra length needed for 95 % conversion is also modest: a five-fold reduction in K_{LDF} requires only 2 cm of additional bed length.

This robustness stems from an asymmetry between adsorption and desorption. Adsorption takes place over most of the reactor length, so even a much lower K_{LDF} still leaves enough residence time for the sorbent to approach equilibrium loading. Desorption, by contrast, is confined to a short stretch near the inlet, where the water partial pressure in the feed is essentially zero and the driving force is correspondingly large. A lower K_{LDF} throttles this fast but spatially limited desorption far more than it affects the slow but spatially extensive adsorption. This is visible in the maximum desorption rate, which drops by more than an order of magnitude (from 15.44 to 0.54 mmol/(kg·s)) while q_{out} barely moves. The reactor design therefore does not depend critically on the exact value of K_{LDF}.

7.4.2. Reaction kinetics

The reaction kinetics are taken from Koschany, Schlereth, and Hinrichsen, fitted to a Ni-Al_2O_3 catalyst rather than the Ni-Ce/13X composite used here, so some uncertainty in the rate constant is to be expected; the active fraction fitted in Section **??** is itself a correction for this mismatch, which makes a sensitivity analysis on k a natural complement to the one on K_{LDF}.

Unlike K_{LDF}, the rate constant has a substantial effect on reactor performance. Halving k drops $X_{\mathrm{CO_2}}$ from 99.77 % to 96.72 %, and roughly doubles the bed length needed to reach 95 % conversion (from 0.878 m to 1.786 m); doubling k correspondingly compresses this length to 0.438 m and pushes conversion up to 99.91 %. The peak reaction rate $r_{\max}$ scales accordingly. Unlike the water mass-transfer coefficient, therefore, the kinetics are a first-order design driver. An overestimation of the kinetics in the model would in practice require compensating with a proportionally lower GHSV or a longer reactor, and vice versa for an underestimation.

7.4.3. Sorbent DA parameters

Since W_0 was tuned rather than taken directly from the literature, a sensitivity analysis is warranted: Mette et al. and Kiefer et al. report different W_0 values for related zeolite 13X materials, and it is useful to know how strongly the reactor's performance depends on this choice [15, 51]. As Eq. **??** shows, W_0 is linearly related to q^*, and therefore, through Eq. **??**, also to the capacity margin Λ. Because Λ must remain above the threshold of 2.5 established in Section **??**, a naive sweep of W_0 at the reference design point ($u_s = 5.84$ mm/s) is not possible: even a slight reduction in W_0 already pushes Λ below this threshold.

To probe a wider range of W_0 while remaining within the model's valid operating regime, the solid velocity is instead raised to $u_s = 11.68$ mm/s (double the reference value) for this sweep, holding GHSV$= 1.5$ fixed. Since $\Lambda \propto u_s W_0$, this raises the capacity margin at the nominal W_0 to $\Lambda = 5.0$, allowing W_0 to be varied by a factor of two in either direction (Table 7.4) while keeping $\Lambda \geq 2.5$ throughout; the $\times 0.5$ case sits exactly at this threshold ($\Lambda = 2.5$), representing the most conservative sorbent capacity that can still be tested without leaving the model's valid regime.

Across this range, $X_{\mathrm{CO_2}}$ and the solid outlet loading $q(0)$ remain essentially unchanged (99.60–99.87% and 0.250–0.251 mol/$\mathrm{kg_{ads}}$, respectively), confirming that reactor performance at the adjusted design point is not sensitive to moderate uncertainty in sorbent capacity. The peak gas-phase water partial pressure $p_{H_2O}^{max}$, however, scales strongly with W_0 (5.78 mbar at baseline, rising to 18.48 mbar at $\times 0.5\, W_0$ and falling to 2.25 mbar at $\times 2\, W_0$), consistent with a smaller sorbent capacity providing less driving force for adsorption and allowing more water to remain in the gas phase. This confirms the expected trend: if the sorbent's true adsorption capacity were lower than estimated, the operating envelope would need to be adjusted proportionally to restore the same margin against water breakthrough, either by increasing u_s or by reducing GHSV by the same factor, and vice versa for a higher capacity.

Across the three submodels tested, the reaction kinetics emerge as the dominant sensitivity: halving k drops conversion from 99.77% to 96.72% and roughly doubles the bed length needed to reach 95% conversion, whereas even a 5-fold overestimate of K_{LDF} leaves conversion above 99.37%, and a

Table 7.4: Sensitivity of reactor performance to model uncertainties. The base case is evaluated at the design point ($u_s = 5.8$ mm/s, GHSV $= 1.5$) unless stated otherwise.

Adsorption kinetics – K_{LDF}

Case	$X_{\mathrm{CO_2}}$	q_{out} [mol/kg$_{\mathrm{ads}}$]	$z(X_{CO_2}{=}95\%)$ [m]	max desorp. [mmol/(kg·s)]
Base case	99.77%	0.500	0.878	15.44
$\times 0.2$	99.37%	0.499	0.898	2.94
$\times 0.04$	97.68%	0.487	1.262	0.54

Kinetics – rate constant k

Case	$X_{\mathrm{CO_2}}$	$r_{\max}$ [mmol/(kg$_{\mathrm{cat}}$ s)]	$z(X_{CO_2}{=}95\%)$ [m]	$\Delta T_{\max}$ [K]
Base case	99.77%	20.93	0.878	12.2
$\times 0.5$	96.72%	10.00	1.786	11.1
$\times 2$	99.91%	42.84	0.438	12.9

Adsorption capacity – DA isotherm parameter W_0

Case	$X_{\mathrm{CO_2}}$	$q(0)$ [mol/kg$_{\mathrm{ads}}$]	$p^{\max}_{\mathrm{H_2O}}$ [mbar]	Λ [-]
Base case	99.76%	0.251	5.78	5.00
$\times 0.5$	99.60%	0.250	18.48	2.50
$\times 2$	99.87%	0.251	2.25	10.00

The adsorption capacity sweep uses $u_s = 11.68$ mm/s (rather than the $u_s = 5.84$ mm/s reference design point) so that the capacity margin Λ remains above the 2.5 threshold (Section 7.1) even at $\times 0.5\, W_0$; GHSV $= 1.5$ is unchanged.

factor-of-two uncertainty in the sorbent capacity W_0 leaves conversion and outlet loading essentially unchanged (Table 7.4). The sorbent capacity nonetheless plays an important role in the operating envelope: since the capacity margin Λ scales linearly with W_0 (Section 7.1), an overestimate of W_0 must be compensated by a proportional increase in u_s (or a corresponding decrease in GHSV) to keep Λ above the 2.5 threshold; once this adjustment is made, the reactor operates as intended.

These sensitivities are not merely a modelling caveat but a practical consideration for a lab-scale unit built around materials and kinetic parameters that are not fitted directly to the sorbent-catalyst composite used here. Several design choices already provide the flexibility needed to accommodate this uncertainty once it is resolved experimentally, without requiring the reactor hardware itself to be rebuilt. The 2 m reactor length is chosen with a $\sim$16% margin over the worst case in the operating envelope (Section 7.2), giving headroom should the kinetics prove slower than modelled. The bed dilution Table 6.1) is a second, more substantial degree of freedom: the bed is currently 75.7 wt% inert filler, so the active catalyst-sorbent loading could in principle be increased up to roughly fourfold by reducing this dilution, densifying the bed to compensate for lower-than-expected kinetics or sorption capacity, or diluted further should the opposite prove true. Finally, since u_s and GHSV are operating variables rather than fixed design parameters, both can be adjusted after commissioning and initial testing to compensate for any discrepancy between the assumed and measured kinetics or sorbent capacity, restoring the intended capacity margin and conversion target without hardware changes. Taken together, these degrees of freedom mean the reactor design is not brittle to the uncertainties quantified in this section, but is deliberately built with the flexibility to be tuned once experimental data on the actual material becomes available.

7.5. Other operating scenario's

Feed Water Content
Different First-Stage Conversion
temperature analysis
what happens if you enter colder zeolite?

Adiabatic operation (probably works fine, since solid flux carries out 88% of heat currently)
little to no dilution in bed (now 75%)
not fully regenerated zeolite at inlet

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A

GANTT chart

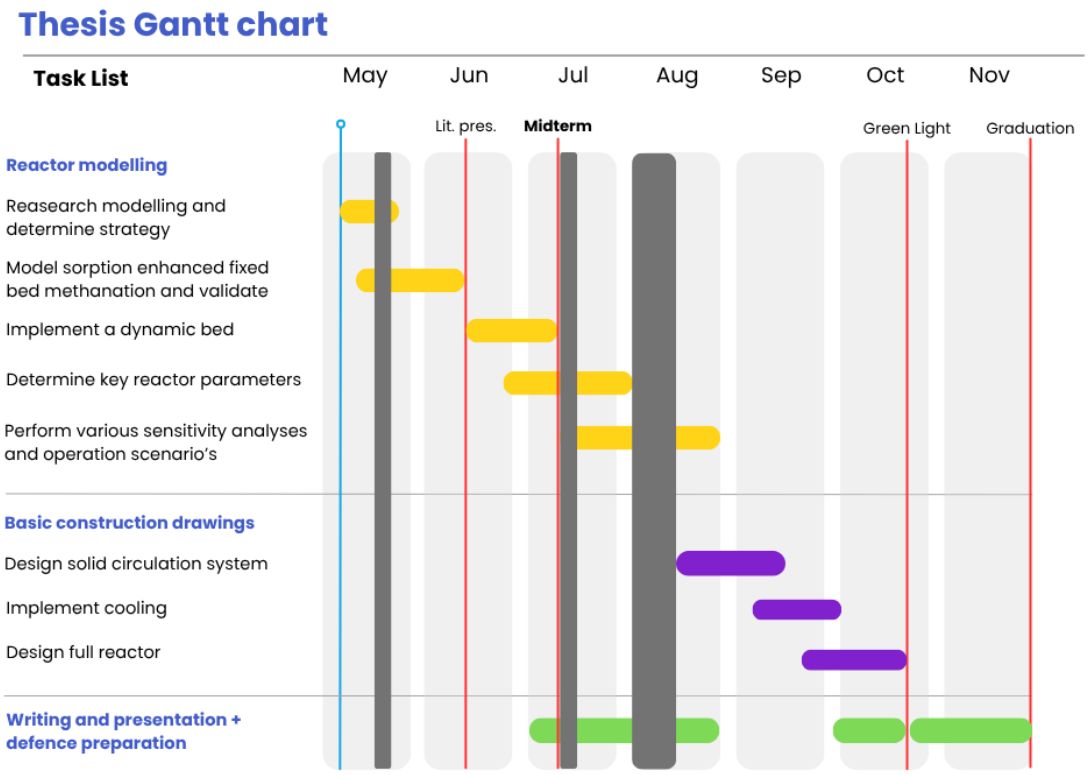


Figure A.1: Project planning overview for the MSc thesis, structured into three main work packages: reactor modelling, basic construction drawings, and writing and defence preparation. Key milestones including the literature presentation (June), midterm review (July), green light (October), and graduation (November). Coloured bars indicate planned task durations, and grey columns denote milestone weeks.

B

Riser calculation

The residence time per pass through the riser is of the order of seconds, but since the rate limiting step is the catalytic reaction, that should not pose a problem, see 6.4.5. The more fundamental concern for SEM is whether sufficient sorbent inventory is present in the riser at any moment to capture the water produced. The dilute solid holdup characteristic of fast fluidisation means the total sorbent mass in the riser is low, making the solid circulation rate the critical design variable: the sorbent must cycle rapidly enough that its cumulative water uptake capacity per unit time matches the water production rate of the reaction. The solid vraction is approximately 4 % in the dilute part at the top of the riser and 15 - 30 % at the bottom [62, p. 87].

To quantify this, the hydrodynamics of a CFB riser of 5 cm diameter and 2 m total height were characterised using the correlations of Kunii and Levenspiel. The sorbent is assumed to be milled to a particle diameter of $d_p = 0.1$ mm. The Archimedes number was computed from the particle and gas properties (gas density $\rho_g = 0.27$ kg/m^3, dynamic viscosity $\mu = 1.75 \times 10^{-5}$ Pa·s, solid density $\rho_s = 689$ kg/m^3), yielding $Ar = 6.97$. The solid density is assumed to be constant, even though it realistically increases 2 - 3 % when fully saturated. The gas density was taken as $\rho_g = 0.27$ kg/m^3, calculated via the ideal gas law at 300 °C and 1 bar for a molar composition of {CO_2 : H_2 : CH_4} = {0.2 : 0.8 : 0.8}, corresponding to the outlet of the first-stage reactor where 80% of the CO_2 feed has been converted to methane. The dimensionless particle diameter $d^* = Ar^{1/3} = 1.91$ places the system in the intermediate flow regime, giving a terminal velocity of $u_t = 0.187$ m/s. At a superficial gas velocity of $u_o = 1$ m/s the gas travels approximately five times faster than the terminal velocity of the solid. The steady state solid mass flux follows is limited by the outflow of solid at the dilute top of the riser [62, Ch. 8, Eq. (4)]:

$$G_s = \rho_s \cdot \varepsilon_{se} \cdot (u_0 - u_t) = 689 \cdot 0.04 \cdot (1.0 - 0.19) = 22.3, \text{kg/m}^2\text{/s} \tag{B.1}$$

where $\varepsilon_{se} = 0.04$ where $\varepsilon_{se} = 0.04$ is extrapolated from Figure 9(b) of [62] at $u_0 = 1$ m/s; as the operating point lies outside the plotted range, this value represents an optimistic estimate that, if anything, strengthens the conclusion that the riser cannot meet the required solid throughput. The steady-state water capture rate of the riser is then the solid mass flux multiplied by the cross-sectional area and the specific adsorption capacity from Table 6.1:

$$\dot{n}_{H_2O,\text{cap}} = G_s \cdot A_t \cdot c_{\text{ads}} = 22.3 \cdot 19.6 \cdot 10^{-4} \cdot 1.3 = 0.057 \,\text{mmol/s} \tag{B.2}$$

The water production rate, assuming full conversion of the remaining 4 vol% CO_2 in the SEM stage feed, is $\dot{n}_{H_2O} = 9.2 \times 10^{-3}$ mol/s, representing a worst-case upper bound on sorbent demand. The required solid mass flux to match this production rate is:

$$G_{s,\text{req}} = \frac{\dot{n}_{H_2O}}{A_t \cdot c_{\text{ads}}} = \frac{9.2 \cdot 10^{-3}}{19.6 \cdot 10^{-4} \cdot 1.3 \cdot 10^{-3}} = 3600 \,\text{kg/m}^2\text{/s} \tag{B.3}$$

This value exceeds $G_s = 22.3\,\mathrm{kg/m^2/s}$ by more than two orders of magnitude. Even after milling the bi-functional material to $100\,\mu$m, the riser cannot supply fresh sorbent rapidly enough to remove the water produced at any gas velocity within the fast fluidisation regime. The CFB riser is therefore not a viable reactor configuration for sorption-enhanced methanation with the bi-functional Ni/Ce–13X material.